Main Category	Category	Terms
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Apoptosis
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Cell cycle
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Cell division
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Chromosome partition
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Mitosis
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Transcription
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Transcription regulation
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Host-virus interaction
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Centromere
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Chromosome
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cytoskeleton
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Microtubule
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Nucleus
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cytoplasm
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Kinetochore
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Protease inhibitor
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Repressor
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Thiol protease inhibitor
Functional Annotations	UP_KW_PTM	Acetylation
Functional Annotations	UP_KW_PTM	Phosphoprotein
Functional Annotations	UP_KW_PTM	Ubl conjugation

Main Category	Category	Terms
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:C->A: Disrupts interaction with histone H3pT3, no effect on interaction with INCENP.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:C->A: Loss of cytoprotection.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:C->R: Disrupts interaction with histone H3pT3, no effect on interaction with INCENP.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:D->A: No change. Loss of interaction with AURKB; when associated with A-70.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:D->A: No change. Loss of interaction with AURKB; when associated with A-71.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:E->A: Almost abolishes RAN-binding. Does not disrupt binding to AURKB or CDCA8. Disrupts mitotic spindle assembly. Does not disrupt nuclear export.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:E->A: Loss of FBXL7 binding.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->A,Q: Mimics acetylation. Localization primarily within the nucleus.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Increases ubiquitination and blocks dissociation from centromeres; when associated with R-23; R-62 and R-78.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Increases ubiquitination and blocks dissociation from centromeres; when associated with R-23; R-62 and R-79.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Increases ubiquitination and blocks dissociation from centromeres; when associated with R-23; R-78 and R-79.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Increases ubiquitination and blocks dissociation from centromeres; when associated with R-62; R-78 and R-79.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Loss of acetylation. Localization primarily within the cytoplasm.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:KK->RR: Loss of FBXL7 mediated polyubiquitination.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:R->A: Disrupts interaction with histone H3pT3, no effect on interaction with INCENP.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:T->A,E: Localizes normally during mitosis but cannot support cell proliferation. Increased affinity for CDCA8/borealin.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:T->A: Loss of LAMTOR5 binding.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:T->A: Prevents phosphorylation by AURKB. Still able to localize correctly but prevents interaction with INCENP.

Main Category	Category	Terms
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:T->E: Higher affinity for LAMTOR5 binding.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:T->E: Mimics phosphorylation. Disrupts subcellular localization during mitosis and prevents interaction with INCENP.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:W->A: Disrupts interaction with histone H3pT3, no effect on interaction with INCENP.
Functional Annotations	UP_SEQ_FEATURE	REPEAT:BIR
Functional Annotations	UP_SEQ_FEATURE	SITE:Interaction with FBXL7
Gene Ontology	GOTERM_BP_DIRECT	G2/M transition of mitotic cell cycle
Gene Ontology	GOTERM_BP_DIRECT	microtubule cytoskeleton organization
Gene Ontology	GOTERM_BP_DIRECT	mitotic cell cycle
Gene Ontology	GOTERM_BP_DIRECT	mitotic cytokinesis
Gene Ontology	GOTERM_BP_DIRECT	apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	mitotic spindle organization
Gene Ontology	GOTERM_BP_DIRECT	chromosome segregation
Gene Ontology	GOTERM_BP_DIRECT	mitotic spindle assembly checkpoint signaling
Gene Ontology	GOTERM_BP_DIRECT	meiosis I
Gene Ontology	GOTERM_BP_DIRECT	regulation of mitotic cell cycle
Gene Ontology	GOTERM_BP_DIRECT	sensory perception of sound
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of cell population proliferation
Gene Ontology	GOTERM_BP_DIRECT	protein-containing complex localization
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of exit from mitosis
Gene Ontology	GOTERM_BP_DIRECT	regulation of apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of apoptotic process

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of neuron apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of DNA-templated transcription
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of mitotic cell cycle
Gene Ontology	GOTERM_BP_DIRECT	chromosome localization
Gene Ontology	GOTERM_BP_DIRECT	mitotic spindle midzone assembly
Gene Ontology	GOTERM_BP_DIRECT	cell division
Gene Ontology	GOTERM_BP_DIRECT	regulation of insulin secretion involved in cellular response to glucose stimulus
Gene Ontology	GOTERM_BP_DIRECT	regulation of type B pancreatic cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of mitotic cell cycle spindle assembly checkpoint
Gene Ontology	GOTERM_BP_DIRECT	mitotic spindle assembly
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of mitotic sister chromatid separation
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of attachment of mitotic spindle microtubules to kinetochore
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of mitotic cytokinesis
Gene Ontology	GOTERM_CC_DIRECT	nuclear chromosome
Gene Ontology	GOTERM_CC_DIRECT	chromosome, centromeric region
Gene Ontology	GOTERM_CC_DIRECT	kinetochore
Gene Ontology	GOTERM_CC_DIRECT	nucleus
Gene Ontology	GOTERM_CC_DIRECT	nucleoplasm
Gene Ontology	GOTERM_CC_DIRECT	chromosome
Gene Ontology	GOTERM_CC_DIRECT	cytoplasm
Gene Ontology	GOTERM_CC_DIRECT	centriole

Main Category	Category	Terms
Gene Ontology	GOTERM_CC_DIRECT	spindle
Gene Ontology	GOTERM_CC_DIRECT	cytosol
Gene Ontology	GOTERM_CC_DIRECT	cytoskeleton
Gene Ontology	GOTERM_CC_DIRECT	microtubule
Gene Ontology	GOTERM_CC_DIRECT	spindle microtubule
Gene Ontology	GOTERM_CC_DIRECT	cytoplasmic microtubule
Gene Ontology	GOTERM_CC_DIRECT	microtubule cytoskeleton
Gene Ontology	GOTERM_CC_DIRECT	midbody
Gene Ontology	GOTERM_CC_DIRECT	interphase microtubule organizing center
Gene Ontology	GOTERM_CC_DIRECT	chromosome passenger complex
Gene Ontology	GOTERM_CC_DIRECT	protein-containing complex
Gene Ontology	GOTERM_CC_DIRECT	spindle midzone
Gene Ontology	GOTERM_CC_DIRECT	survivin complex
Gene Ontology	GOTERM_MF_DIRECT	cysteine-type endopeptidase inhibitor activity
Gene Ontology	GOTERM_MF_DIRECT	protein binding
Gene Ontology	GOTERM_MF_DIRECT	microtubule binding
Gene Ontology	GOTERM_MF_DIRECT	zinc ion binding
Gene Ontology	GOTERM_MF_DIRECT	tubulin binding
Gene Ontology	GOTERM_MF_DIRECT	enzyme binding
Gene Ontology	GOTERM_MF_DIRECT	peptidase inhibitor activity
Gene Ontology	GOTERM_MF_DIRECT	small GTPase binding

Main Category	Category	Terms
Gene Ontology	GOTERM_MF_DIRECT	identical protein binding
Gene Ontology	GOTERM_MF_DIRECT	protein homodimerization activity
Gene Ontology	GOTERM_MF_DIRECT	cysteine-type endopeptidase inhibitor activity involved in apoptotic process
Gene Ontology	GOTERM_MF_DIRECT	metal ion binding
Gene Ontology	GOTERM_MF_DIRECT	protein heterodimerization activity
Gene Ontology	GOTERM_MF_DIRECT	cobalt ion binding
Gene Ontology	GOTERM_MF_DIRECT	protein-folding chaperone binding
General Annotations	CHROMOSOME	17
General Annotations	CYTOGENETIC_LOCATION	17q25.3
General Annotations	ENTREZ_GENE_SUMMARY	This gene is a member of the inhibitor of apoptosis (IAP) gene family, which encode negative regulatory proteins that prevent apoptotic cell death. IAP family members usually contain multiple baculovirus IAP repeat (BIR) domains, but this gene encodes proteins with only a single BIR domain. The encoded proteins also lack a C-terminus RING finger domain. Gene expression is high during fetal development and in most tumors, yet low in adult tissues. Alternatively spliced transcript variants encoding distinct isoforms have been found for this gene. [provided by RefSeq, Jun 2011]
General Annotations	UP_COMMENT_ALTERNATIVE_PRODUCTS	Event=Alternative splicing; Named isoforms=7; Name=1; Synonyms=Alpha; IsoId=O15392-1; Sequence=Displayed; Name=2; Synonyms=2B, Beta; IsoId=O15392-2; Sequence=VSP_002454; Name=3; Synonyms=DeltaEx3; IsoId=O15392-3; Sequence=VSP_020338; Name=4; Synonyms=3B; IsoId=O15392-4; Sequence=VSP_020342; Name=5; Synonyms=SI; IsoId=O15392-5; Sequence=VSP_020341; Name=6; Synonyms=3 alpha; IsoId=O15392-6; Sequence=VSP_020339; Name=7; Synonyms=2 alpha; IsoId=O15392-7; Sequence=VSP_020340;
General Annotations	UP_COMMENT_DEVELOPMENTAL_STAGE	Expression is cell cycle-dependent and peaks at mitosis. {ECO:0000269|PubMed:18591255}.
General Annotations	UP_COMMENT_DOMAIN	The BIR repeat is necessary and sufficient for LAMTOR5 binding. {ECO:0000269|PubMed:12773388}.
General Annotations	UP_COMMENT_FUNCTION	Multitasking protein that has dual roles in promoting cell proliferation and preventing apoptosis (PubMed:20627126, PubMed:21364656,

Main Category	Category	Terms
		PubMed:25778398, PubMed:28218735, PubMed:9859993). Component of a chromosome passage protein complex (CPC) which is essential for chromosome alignment and segregation during mitosis and cytokinesis (PubMed:16322459). Acts as an important regulator of the localization of this complex; directs CPC movement to different locations from the inner centromere during prometaphase to midbody during cytokinesis and participates in the organization of the center spindle by associating with polymerized microtubules (PubMed:20826784). Involved in the recruitment of CPC to centromeres during early mitosis via association with histone H3 phosphorylated at 'Thr-3' (H3pT3) during mitosis (PubMed:20929775). The complex with RAN plays a role in mitotic spindle formation by serving as a physical scaffold to help deliver the RAN effector molecule TPX2 to microtubules (PubMed:18591255). May counteract a default induction of apoptosis in G2/M phase (PubMed:9859993). The acetylated form represses STAT3 transactivation of target gene promoters (PubMed:20826784). May play a role in neoplasia (PubMed:10626797). Inhibitor of CASP3 and CASP7 (PubMed:21536684). Essential for the maintenance of mitochondrial integrity and function (PubMed:25778398). Isoform 2 and isoform 3 do not appear to play vital roles in mitosis (PubMed:12773388, PubMed:16291752). Isoform 3 shows a marked reduction in its anti- apoptotic effects when compared with the displayed wild-type isoform (PubMed:10626797). {ECO:0000269|PubMed:10626797, ECO:0000269|PubMed:12773388, ECO:0000269|PubMed:16291752, ECO:0000269|PubMed:16322459, ECO:0000269|PubMed:18591255, ECO:0000269|PubMed:20627126, ECO:0000269|PubMed:20826784, ECO:0000269|PubMed:20929775, ECO:0000269|PubMed:21364656, ECO:0000269|PubMed:21536684, ECO:0000269|PubMed:25778398, ECO:0000269|PubMed:28218735, ECO:0000269|PubMed:9859993}.
General Annotations	UP_COMMENT_INDUCTION	Up-regulated by COMP. {ECO:0000269|PubMed:17993464}.
General Annotations	UP_COMMENT_INTERACTION	E7CU85; Q53HL2: CDCA8; NbExp=2; IntAct=EBI-10488580, EBI-979174;
General Annotations	UP_COMMENT_INTERACTION	O15392; Q92870-2: APBB2; NbExp=3; IntAct=EBI-518823, EBI-21535880; O15392; P05067: APP; NbExp=3; IntAct=EBI-518823, EBI-77613; O15392; P05067-2: APP; NbExp=3; IntAct=EBI-518823, EBI-17264467; O15392; O14965: AURKA; NbExp=2; IntAct=EBI-518823, EBI-448680; O15392; Q96GD4: AURKB; NbExp=13; IntAct=EBI-518823, EBI-624291; O15392; Q9UQB9: AURKC; NbExp=10; IntAct=EBI-518823, EBI-3926851; O15392; Q14457: BECN1; NbExp=3; IntAct=EBI-518823, EBI-949378; O15392; O15392: BIRC5; NbExp=2; IntAct=EBI-518823, EBI-518823; O15392; Q15834: CCDC85B; NbExp=3; IntAct=EBI-518823, EBI-739674; O15392; Q53HL2: CDCA8; NbExp=21; IntAct=EBI-518823, EBI-979174;

Main Category	Category	Terms
		O15392; P06493: CDK1; NbExp=6; IntAct=EBI-518823, EBI-444308; O15392; Q14203-5: DCTN1; NbExp=3; IntAct=EBI-518823, EBI-25840379; O15392; Q9NR28: DIABLO; NbExp=2; IntAct=EBI-518823, EBI-517508; O15392; Q86XJ1: GAS2L3; NbExp=4; IntAct=EBI-518823, EBI-9248152; O15392; Q8TB36: GDAP1; NbExp=3; IntAct=EBI-518823, EBI-11110431; O15392; P42858: HTT; NbExp=15; IntAct=EBI-518823, EBI-466029; O15392; Q9NQS7: INCENP; NbExp=11; IntAct=EBI-518823, EBI-307907; O15392; O43504: LAMTOR5; NbExp=3; IntAct=EBI-518823, EBI-713382; O15392; Q96CV9: OPTN; NbExp=3; IntAct=EBI-518823, EBI-748974; O15392; Q7Z412: PEX26; NbExp=3; IntAct=EBI-518823, EBI-752057; O15392; Q9BXM7: PINK1; NbExp=3; IntAct=EBI-518823, EBI-2846068; O15392; P62826: RAN; NbExp=7; IntAct=EBI-518823, EBI-286642; O15392; P37840: SNCA; NbExp=3; IntAct=EBI-518823, EBI-985879; O15392; Q13148: TARDBP; NbExp=6; IntAct=EBI-518823, EBI-372899; O15392; P09936: UCHL1; NbExp=3; IntAct=EBI-518823, EBI-714860; O15392; O14980: XPO1; NbExp=2; IntAct=EBI-518823, EBI-355867; O15392-1; Q96GD4: AURKB; NbExp=2; IntAct=EBI-518838, EBI-624291; O15392-1; O15392-1: BIRC5; NbExp=2; IntAct=EBI-518838, EBI-518838; O15392-1; O15392-2: BIRC5; NbExp=2; IntAct=EBI-518838, EBI-518842; O15392-1; Q53HL2: CDCA8; NbExp=2; IntAct=EBI-518838, EBI-979174; O15392-2; Q96GD4: AURKB; NbExp=2; IntAct=EBI-518842, EBI-624291; O15392-2; Q53HL2: CDCA8; NbExp=2; IntAct=EBI-518842, EBI-979174;
General Annotations	UP_COMMENT_PTM	Acetylation at Lys-129 by CBP results in its homodimerization, while deacetylation promotes the formation of monomers which heterodimerize with XPO1/CRM1 which facilitates its nuclear export. The acetylated form represses STAT3 transactivation. The dynamic equilibrium between its acetylation and deacetylation at Lys-129 determines its interaction with XPO1/CRM1, its subsequent subcellular localization, and its ability to inhibit STAT3 transactivation. {ECO:0000269|PubMed:20826784}.
General Annotations	UP_COMMENT_PTM	In vitro phosphorylation at Thr-117 by AURKB prevents interaction with INCENP and localization to mitotic chromosomes (PubMed:14610074). Phosphorylation at Thr-48 by CK2 is critical for its mitotic and anti-apoptotic activities (PubMed:21252625). Phosphorylation at Thr-34 by CDK15 is critical for its anti-apoptotic activity (PubMed:24866247). Phosphorylation at Ser-20 by AURKC is critical for regulation of proper chromosome alignment and segregation, and possibly cytokinesis. {ECO:0000269|PubMed:11069302, ECO:0000269|PubMed:14610074, ECO:0000269|PubMed:21252625, ECO:0000269|PubMed:24866247, ECO:0000269|PubMed:27332895}.
General Annotations	UP_COMMENT_PTM	Ubiquitinated by the Cul9-RING ubiquitin-protein ligase complex, leading to its degradation. Ubiquitination is required for centrosomal targeting. Deubiquitinated by USP35 or USP38; leading to stabilization

Main Category	Category	Terms
		(PubMed:34438346). {ECO:0000269|PubMed:16322459, ECO:0000269|PubMed:24793696, ECO:0000269|PubMed:34438346}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the IAP family. {ECO:0000256|ARBA:ARBA00006672}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the IAP family. {ECO:0000305}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Chromosome, centromere, kinetochore {ECO:0000256|ARBA:ARBA00004629}. Cytoplasm, cytoskeleton, spindle {ECO:0000256|ARBA:ARBA00004186}. Midbody {ECO:0000256|ARBA:ARBA00004214}. Nucleus {ECO:0000256|ARBA:ARBA00004123}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Cytoplasm {ECO:0000269|PubMed:20627126, ECO:0000269|PubMed:20826784, ECO:0000269|PubMed:21364656}. Nucleus {ECO:0000269|PubMed:20627126, ECO:0000269|PubMed:20826784, ECO:0000269|PubMed:21364656}. Chromosome {ECO:0000269|PubMed:14610074}. Chromosome, centromere {ECO:0000269|PubMed:11084331, ECO:0000269|PubMed:14610074, ECO:0000269|PubMed:16322459}. Cytoplasm, cytoskeleton, spindle {ECO:0000269|PubMed:11084331}. Chromosome, centromere, kinetochore {ECO:0000269|PubMed:11084331}. Midbody {ECO:0000269|PubMed:15665297}. Note=Localizes at the centromeres from prophase to metaphase, at the spindle midzone during anaphase and a the midbody during telophase and cytokinesis. Accumulates in the nucleus upon treatment with leptomycin B (LMB), a XPO1/CRM1 nuclear export inhibitor (By similarity). Localizes on chromosome arms and inner centromeres from prophase through metaphase. Localizes to kinetochores in metaphase, distributes to the midzone microtubules in anaphase and at telophase, localizes exclusively to the midbody (PubMed:11084331). Colocalizes with AURKB at mitotic chromosomes (PubMed:14610074). Acetylation at Lys-129 directs its localization to the nucleus by enhancing homodimerization and thereby inhibiting XPO1/CRM1-mediated nuclear export (PubMed:20826784). {ECO:0000250|UniProtKB:E3SCZ8, ECO:0000269|PubMed:11084331, ECO:0000269|PubMed:14610074, ECO:0000269|PubMed:20826784}.
General Annotations	UP_COMMENT_SUBUNIT	(Microbial infection) Interacts with Epstein-Barr virus (EBV) EBNA1; this interaction is probably important for EBV episome maintenance in Burkitt's lymphoma cells. {ECO:0000269|PubMed:28077791}.
General Annotations	UP_COMMENT_SUBUNIT	Monomer or homodimer. Exists as a homodimer in the apo state and as a monomer in the CPC-bound state. The monomer protects cells against apoptosis more efficiently than the dimer. Only the dimeric form is capable of enhancing tubulin stability in cells. When phosphorylated, interacts with LAMTOR5/HBXIP; the resulting complex binds pro-CASP9, as well as active CASP9, but much less efficiently. Component of the

Main Category	Category	Terms
		chromosomal passenger complex (CPC) composed of at least BIRC5/survivin, CDCA8/borealin, INCENP, AURKB or AURKC; in the complex forms a triple-helix bundle-based subcomplex with INCENP and CDCA8 (PubMed:17956729). Interacts with JTB. Interacts (via BIR domain) with histone H3 phosphorylated at 'Thr-3' (H3pT3). Interacts with EVI5. Interacts with GTP-bound RAN in both the S and M phases of the cell cycle. Interacts with USP9X. Interacts with tubulin. Interacts with BIRC2/c-IAP1. The acetylated form at Lys-129 interacts with STAT3. The monomeric form deacetylated at Lys-129 interacts with XPO1/CRM1. The monomeric form interacts with XIAP/BIRC4. Both the dimeric and monomeric form can interact with DIABLO/SMAC. Interacts with BIRC6/bruce. Interacts with FBXL7; this interaction facilitates the polyubiquitination and subsequent proteasomal degradation of BIRC5 by the SCF(FBXL7) E3 ubiquitin-protein ligase complex (PubMed:25778398, PubMed:28218735). {ECO:0000269|PubMed:12773388, ECO:0000269|PubMed:14610074, ECO:0000269|PubMed:15249581, ECO:0000269|PubMed:15665297, ECO:0000269|PubMed:16239925, ECO:0000269|PubMed:16291752, ECO:0000269|PubMed:16322459, ECO:0000269|PubMed:16427043, ECO:0000269|PubMed:16436504, ECO:0000269|PubMed:16764853, ECO:0000269|PubMed:17956729, ECO:0000269|PubMed:18329369, ECO:0000269|PubMed:18591255, ECO:0000269|PubMed:20826784, ECO:0000269|PubMed:21225229, ECO:0000269|PubMed:21536684, ECO:0000269|PubMed:25778398, ECO:0000269|PubMed:28218735}.
General Annotations	UP_COMMENT_TISSUE_SPECIFICITY	Expressed only in fetal kidney and liver, and to lesser extent, lung and brain (PubMed:10626797). Abundantly expressed in adenocarcinoma (lung, pancreas, colon, breast, and prostate) and in high-grade lymphomas (PubMed:14741722, PubMed:16329164). Also expressed in various renal cell carcinoma cell lines (PubMed:10626797). Expressed in cochlea including the organ of Corti, the lateral wall, the interdental cells of the Limbus as well as in Schwann cells and cells of the cochlear nerve and the spiral ganglions (at protein level). Not expressed in cells of the inner and outer sulcus or the Reissner's membrane (at protein level) (PubMed:20627126, PubMed:21364656). {ECO:0000269|PubMed:10626797, ECO:0000269|PubMed:14741722, ECO:0000269|PubMed:16329164, ECO:0000269|PubMed:20627126, ECO:0000269|PubMed:21364656}.
Identifiers	ENTREZ_GENE_ID	332
Identifiers	ENSEMBL_GENE_ID	ENSG00000089685
Identifiers	UNIPROT_ID	K7EP76_HUMAN
Identifiers	UNIPROT_ID	K7ELG2_HUMAN
Identifiers	UNIPROT_ID	K7EMW2_HUMAN
Identifiers	UNIPROT_ID	H3BLT4_HUMAN

Main Category	Category	Terms
Identifiers	UNIPROT_ID	E7CU85_HUMAN
Identifiers	UNIPROT_ID	BIRC5_HUMAN
Identifiers	UNIPROT_ID	A3E0Z4_HUMAN
Identifiers	UNIPROT_ID	A3E0Z7_HUMAN
Identifiers	UNIPROT_ID	A3E0Z6_HUMAN
Identifiers	UNIPROT_ID	A3E0Z5_HUMAN
Identifiers	UNIPROT_ID	A3E0Z8_HUMAN
Identifiers	UNIPROT_ID	A0A7L8XZM3_HUMAN
Identifiers	UNIPROT_ID	A0A0B4J1S3_HUMAN
Interactions	DRUGBANK	Reserpine
Interactions	DRUGBANK	Berberine
Interactions	DRUGBANK	LY2181308
Interactions	PUBCHEM	N-[[5-chloro-3-[5-cyano-4-[2-iodo-5-(trifluoromethyl)phenyl]-6-oxo-1H-pyridin-2-yl]-2-hydroxyphenyl]methyl]-N-methyl-1-propylpiperidine-4-carboxamide
Interactions	PUBCHEM	6-(5-Bromo-2-hydroxyphenyl)-2-oxo-4-phenyl-1,2-dihydropyridine-3-carbonitrile
Interactions	PUBCHEM	7-(Piperidin-1-ylmethyl)-5-[[4-[4-(trifluoromethyl)phenyl]triazol-2-yl]methyl]quinolin-8-ol
Interactions	PUBCHEM	N-(5-chloro-3-(4-(2-chloro-5-(trifluoromethyl)phenyl)-5-cyano-6-oxo-1,6-dihydropyridin-2-yl)-2-hydroxybenzyl)-1-butyryl-N-methylpiperidine-4-carboxamide
Interactions	PUBCHEM	3-(3-Bromophenyl)quinoline
Interactions	PUBCHEM	3-[(2Z)-2-hydroxyimino-3-[4-[4-(trifluoromethoxy)phenyl]phenoxy]propyl]-5,5-dimethylimidazolidine-2,4-dione
Interactions	PUBCHEM	4-(2,5-Dichloro-phenyl)-6-(4-hydroxy-biphenyl-3-yl)-2-oxo-1,2-dihydro-pyridine-3-carbonitrile
Interactions	PUBCHEM	6-(5-Cyclopentyl-2-hydroxyphenyl)-4-(2,5-dichlorophenyl)-2-oxo-1,2-dihydropyridine-3-carbonitrile
Interactions	PUBCHEM	4-(2,5-Dichlorophenyl)-6-(2-hydroxy-5-isobutylphenyl)-2-oxo-1,2-dihydropyridine-3-carbonitrile
Interactions	PUBCHEM	6-(5-Cyclohexyl-2-hydroxyphenyl)-4-(2,5-dichlorophenyl)-2-oxo-1,2-dihydropyridine-3-carbonitrile
Interactions	PUBCHEM	6-(5-Bromo-2-hydroxyphenyl)-4-(2,4-dimethylphenyl)-2-oxo-1,2-dihydropyridine-3-carbonitrile
Interactions	PUBCHEM	6-(5-Bromo-2-hydroxyphenyl)-4-(2,5-dimethylphenyl)-2-oxo-1,2-dihydropyridine-3-carbonitrile

Main Category	Category	Terms
Interactions	PUBCHEM	6-(5-Bromo-2-hydroxyphenyl)-4-(2,5-dichlorophenyl)-2-oxo-1,2-dihydropyridine-3-carbonitrile
Interactions	PUBCHEM	6-(5-Bromo-2-hydroxyphenyl)-4-(2-chloro-5-(trifluoromethyl)phenyl)-2-oxo-1,2-dihydropyridine-3-carbonitrile
Interactions	PUBCHEM	6-(5-Chloro-2-hydroxyphenyl)-4-(2-chloro-5-(trifluoromethyl)phenyl)-2-oxo-1,2-dihydropyridine-3-carbonitrile
Interactions	PUBCHEM	N-(5-chloro-3-(4-(2-chloro-5-(trifluoromethyl)phenyl)-5-cyano-6-oxo-1,6-dihydropyridin-2-yl)-2-hydroxybenzyl)-N-methylcyclohexanecarboxamide
Interactions	PUBCHEM	N-(5-chloro-3-(4-(2-chloro-5-(trifluoromethyl)phenyl)-5-cyano-6-oxo-1,6-dihydropyridin-2-yl)-2-hydroxybenzyl)cyclohexanecarboxamide
Interactions	PUBCHEM	N-(5-chloro-3-(4-(2-chloro-5-(trifluoromethyl)phenyl)-5-cyano-6-oxo-1,6-dihydropyridin-2-yl)-2-hydroxybenzyl)-1-acetyl-N-methylpiperidine-4-carboxamide
Interactions	PUBCHEM	N-(5-chloro-3-(4-(2-chloro-5-(trifluoromethyl)phenyl)-5-cyano-6-oxo-1,6-dihydropyridin-2-yl)-2-hydroxybenzyl)-1-acetylpiperidine-4-carboxamide
Interactions	PUBCHEM	N-(5-chloro-3-(4-(2-chloro-5-(trifluoromethyl)phenyl)-5-cyano-6-oxo-1,6-dihydropyridin-2-yl)-2-hydroxybenzyl)-N-methyl-3-(pyrrolidin-1-yl)propanamide
Interactions	PUBCHEM	N-(5-chloro-3-(4-(2-chloro-5-(trifluoromethyl)phenyl)-5-cyano-6-oxo-1,6-dihydropyridin-2-yl)-2-hydroxybenzyl)-N-methyl-3-morpholinopropanamide
Interactions	PUBCHEM	N-(5-chloro-3-(4-(2-chloro-5-(trifluoromethyl)phenyl)-5-cyano-6-oxo-1,6-dihydropyridin-2-yl)-2-hydroxybenzyl)-4-methoxy-N-methylcyclohexanecarboxamide
Interactions	PUBCHEM	Tert-butyl 4-((5-chloro-3-(4-(2-chloro-5-(trifluoromethyl)phenyl)-5-cyano-6-oxo-1,6-dihydropyridin-2-yl)-2-hydroxybenzyl)(methyl)carbamoyl)piperidine-1-carboxylate
Interactions	PUBCHEM	N-(5-chloro-3-(4-(2-chloro-5-(trifluoromethyl)phenyl)-5-cyano-6-oxo-1,6-dihydropyridin-2-yl)-2-hydroxybenzyl)-N-methylpiperidine-4-carboxamide
Interactions	PUBCHEM	N-(5-chloro-3-(4-(2-chloro-5-(trifluoromethyl)phenyl)-5-cyano-6-oxo-1,6-dihydropyridin-2-yl)-2-hydroxybenzyl)-1-(2-methoxyacetyl)-N-methylpiperidine-4-carboxamide
Interactions	PUBCHEM	N-(5-chloro-3-(4-(2-chloro-5-(trifluoromethyl)phenyl)-5-cyano-6-oxo-1,6-dihydropyridin-2-yl)-2-hydroxybenzyl)-1-(cyclohexanecarbonyl)-N-methylpiperidine-4-carboxamide
Interactions	PUBCHEM	6-[5-Chloro-2-hydroxy-3-[(1-propyl-4-piperidinyl)carbonyl(methyl)aminomethyl]phenyl]-4-[2-chloro-5-(trifluoromethyl)phenyl]-3-cyanopyridine-2(1H)-one

Main Category	Category	Terms
Interactions	PUBCHEM	N-(5-chloro-3-(4-(2-chloro-5-(trifluoromethyl)phenyl)-5-cyano-6-oxo-1,6-dihydropyridin-2-yl)-2-hydroxybenzyl)-N-methyl-1-pentylpiperidine-4-carboxamide
Interactions	PUBCHEM	Benziodarone
Interactions	PUBCHEM	1-(4-Chlorophenyl)-1H-indole
Interactions	PUBCHEM	2-Naphthalenamine, N-(3-chlorophenyl)-
Interactions	PUBCHEM	3-[5-[(E)-[[6-(2,3-Dimethylanilino)-[1,2,5]oxadiazolo[3,4-b]pyrazin-5-yl]hydrazinylidene]methyl]furan-2-yl]benzoic acid
Interactions	UP_KW_LIGAND	Metal-binding
Interactions	UP_KW_LIGAND	Zinc
Literature	GENERIF_SUMMARY	Relative levels of expression of survivin mRNA and survivin protein increased after low- and high-dose CDDP treatment. Survivin mRNA was not detected in normal gastric mucosas.
Literature	GENERIF_SUMMARY	data demonstrate a crucial role of PI3-kinase/Akt/NF-kappaB/survivin signaling in tubular morphogenesis of HCAEC triggered by hypoxic preconditioning.
Literature	GENERIF_SUMMARY	Expression of survivin mRNA and protein in gastric cancer cell line (MKN-45) during cisplatin treatment
Literature	GENERIF_SUMMARY	Survivin was expressed in all thymocyte subsets but its expression level was developmentally regulated.
Literature	GENERIF_SUMMARY	expression associated with reduced survival and reduced apoptotic capacity in gliomas
Literature	GENERIF_SUMMARY	Data demonstrate that the primary survivin pool in mitotic cells is associated with microtubules and participates in the assembly of a bipolar mitotic spindle.
Literature	GENERIF_SUMMARY	expression of alternative survivin splice varients related to gastric tumor progression
Literature	GENERIF_SUMMARY	The ratio of survivin to Fas in primary tumors may be used to predict the risk for recurrent disease in patients with neuroblastoma.
Literature	GENERIF_SUMMARY	survivin may represent a novel neuroendocrine marker for chromaffin cell tumors, survivin does not appear to reliably distinguish benign from malignant pheochromocytomas/paragangliomas and thus does not identify patients at risk of recurrent disease
Literature	GENERIF_SUMMARY	This nuclear shuttling protein is exported actively from the nucleus via the CRM1-dependent pathway.
Literature	GENERIF_SUMMARY	Expression in primary glioblastomas.
Literature	GENERIF_SUMMARY	survivin expression freqently seen in gastric cancer patients and first-degree relatives; co-expression with cyclooxygenase-2 suggests multiple pathways

Main Category	Category	Terms
		contributing to apoptosis inhibition in gastric cancer
Literature	GENERIF_SUMMARY	expression of functionally different survivin splice variants suggests a role in the progression and clinical behavior of human renal cell carcinomas
Literature	GENERIF_SUMMARY	Survivin expression was significantly associated with PCNA-labeling index, clinical stage, histological grade, clinical outcome, and survival rate in ovarian epithelial carcinomas.
Literature	GENERIF_SUMMARY	Higher expression level of surviving mRNA in primary leukemic cells from acute leukemia patients may be one of the reasons that leukemic cells are insensitive to chemotherapy.
Literature	GENERIF_SUMMARY	Survivin was not likely to be related with an inhibition of apoptosis or induction of telomerase but may have some role in gastric cancer extension.
Literature	GENERIF_SUMMARY	comparative study of expression of survivin in primary and metastatic colonic adenocarcinomas
Literature	GENERIF_SUMMARY	expression in laryngeal squamous cell carcinomas and its prognostic implications
Literature	GENERIF_SUMMARY	studies define a novel p53-survivin signaling pathway activated by DNA damage that results in down-regulation of survivin, cell cycle arrest, and apoptosis
Literature	GENERIF_SUMMARY	cellular studies and clinical data suggest a direct link between survivin expression and tumour cell susceptibility to taxol
Literature	GENERIF_SUMMARY	survivin not only suppresses apoptosis but also accelerates cancer cell-proliferative activity and plays an important role in tumor progression in hepatocellular carcinoma
Literature	GENERIF_SUMMARY	Reovirus-induced apoptosis involves reduction of cellular survivin levels
Literature	GENERIF_SUMMARY	findings suggest that activated STAT3 signaling directly contributes to malignant progression of primary effusion lymphoma by preventing apoptosis, acting through the prosurvival protein survivin
Literature	GENERIF_SUMMARY	These results indicate that survivin stimulates Aurora-B kinase activity and helps correctly target Aurora-B to its substrates during the cell cycle, thus providing a mechanism as to how survivin exerts its function in human cells.
Literature	GENERIF_SUMMARY	Survivin may be actively involved in regulating cell viability during HIV-1 infection
Literature	GENERIF_SUMMARY	Suppression of phosphorylation on Thr34 by flavopiridol enhances tumor cell apoptosis.
Literature	GENERIF_SUMMARY	the determination of nuclear survivin expression could be used to individualise therapeutic strategies in oesophageal squamous cell cancer in the future.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Survivin expression in astrocytic tumors varies with histologic malignancy and may play an important role in the oncogenesis and progression of astrocytic tumors
Literature	GENERIF_SUMMARY	The level of survivin was decreased after thapsigargin treatment in hepatoma cell lines indicating that survivin plays an important role in endoplasmic reticulum stress-mediated apoptosis.
Literature	GENERIF_SUMMARY	We conclude that the changes in the level of caspase-3 and survivin play an important role in the transformation from normal gastric mucosa to gastric cancer.
Literature	GENERIF_SUMMARY	Survivin is an important inhibitor of apoptosis in colorectal cancer cell lines.
Literature	GENERIF_SUMMARY	role in pathological conditions-REVIEW
Literature	GENERIF_SUMMARY	Survivin is expressed in embryonic tissues as well as in the majority of human cancers, but is not expressed in most normal adult tissues.
Literature	GENERIF_SUMMARY	Survivin-HBXIP complexes, but neither survivin nor HBXIP individually, bind pro-caspase-9, preventing its recruitment to Apaf1, and thereby selectively suppressing apoptosis.
Literature	GENERIF_SUMMARY	survivin may play an important role in transition from intraductal papillary-mucinous tumor of the pancreas adenoma to carcinoma in situ
Literature	GENERIF_SUMMARY	Data show that mammalian cells lacking survivin are unable to align their chromosomes, fail to recruit Aurora B to kinetochores and become polyploid at a very high frequency.
Literature	GENERIF_SUMMARY	survivin is upregulated by oncogenic c-H-Ras in tumor cells
Literature	GENERIF_SUMMARY	survivin plays an important role in the onset of gastric carcinoma and that high survivin expression is an early event of gastric carcinoma.
Literature	GENERIF_SUMMARY	survivin has a role in the dedifferentiation of thyroid carcinoma
Literature	GENERIF_SUMMARY	Expression of non-destructible beta catenin mutants increased survivin expression and protected against ultraviolet-B-induced apoptosis.
Literature	GENERIF_SUMMARY	All bladder tumor tissues expressed survivin mRNA.
Literature	GENERIF_SUMMARY	Survivin is essential for cell cycle progression in leukemic cells
Literature	GENERIF_SUMMARY	survivin is required to maintain a critical anti-apoptotic threshold in prostate cancer cells
Literature	GENERIF_SUMMARY	adenovirus 2 E1B-55K protein blocks p53 as a transcriptional repressor protein of the survivin and the MAP4 promoters

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	survivin has separable checkpoint functions at multiple phases of mitosis and in the control of mitochondrial-dependent apoptosis.
Literature	GENERIF_SUMMARY	abnormally low levels of expression of the nuclear form of this protein may be a marker of more aggressive disease or higher morphologic grade in ependymal and choroid plexus tumours.
Literature	GENERIF_SUMMARY	Stabilization of survivin may contribute to the apoptosis resistance effect of COX-2 in non-small cell lung cancer.
Literature	GENERIF_SUMMARY	PML-induced apoptosis by down-regulation of Survivin.
Literature	GENERIF_SUMMARY	phosphorylation of survivin at threonine 117 by aurora-B may regulate targeting of survivin
Literature	GENERIF_SUMMARY	Hsp90, a molecular chaperone that is central to the cellular stress response, associates with survivin, an apoptosis inhibitor and essential regulator of mitosis.
Literature	GENERIF_SUMMARY	Data suggests that survivin expression may identify cases of oral squamous cell carcinoma with more aggressive and invasive phenotype.
Literature	GENERIF_SUMMARY	Substitution of single amino acid converts Survivin to proapoptotic agent.
Literature	GENERIF_SUMMARY	Through induction of survivin by IL-11 in a human transplant model, evidence is given that survivin can protect cells from T cell-mediated graft injury.
Literature	GENERIF_SUMMARY	NO inhibited the expression of survivin, which was down-regulated by the p38 MAP kinase pathway
Literature	GENERIF_SUMMARY	Survivin up-regulates Fas ligand (FasL) expression in colon cancer cells by augmenting DNA binding of transcription factor specificity protein 1 (Sp1) to the FasL promoter.
Literature	GENERIF_SUMMARY	STAT3 signaling supports gastric cancer cell survival in association with survivin expression.
Literature	GENERIF_SUMMARY	High expression of survivin is associated with esophageal cancer
Literature	GENERIF_SUMMARY	higher level of expression of survivin is associated with Colonic Neoplasms
Literature	GENERIF_SUMMARY	Testicular survivin was found in infertile men.
Literature	GENERIF_SUMMARY	new insights into the regulation and function of survivin and implications for the pathogenesis, diagnosis, and treatment of inflammatory diseases and cancer
Literature	GENERIF_SUMMARY	survivin, p53, and bcl-2 are elevated in breast carcinoma but not ductal intraepithelial neoplasia
Literature	GENERIF_SUMMARY	An important role is suggested for survivin in progression, recurrence, and treatment of hepatocellular carcinoma

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Changes in the turnover of survivin at centromeres influence the stability of kinetochore-microtubule attachment and signaling of the spindle checkpoint.
Literature	GENERIF_SUMMARY	the high level of survivin in some cancers is, at least in part, due to a genetic defect in the promoter region of the human survivin gene, which causes derepression of survivin transcription apparently due to the mutated repressor binding motifs.
Literature	GENERIF_SUMMARY	Overexpression of survivin plays an important role in the development and progression of pancreatic cancer
Literature	GENERIF_SUMMARY	may enhance tumor cell survival upon radiation exposure such as regulation of double-strand DNA break repair and tumor cell metabolism
Literature	GENERIF_SUMMARY	survivin-2B variant is a proapoptotic factor and its expression is upregulated by p53
Literature	GENERIF_SUMMARY	Survivin is a special tumor marker independent of histopathological characteristics and may play an important role during human colorectal tumorigenesis by inhibiting apoptosis.
Literature	GENERIF_SUMMARY	the high level of survivin in some cancers is, at least in part, due to a genetic defect in the promoter region of the human survivin gene, which causes derepression of survivin transcription
Literature	GENERIF_SUMMARY	survivin can reduce the cell growth inhibition and apoptosis, and p53 elevates the p21 level, which may attenuate the cell death in the quercetin-treated human lung carcinoma cells
Literature	GENERIF_SUMMARY	possible role of survivin in gastric cancer invasiveness and metastasis
Literature	GENERIF_SUMMARY	depletion of survivin caused defects in cell division, followed by an arrest of DNA synthesis
Literature	GENERIF_SUMMARY	results show that survivin exists in a novel mitochondrial pool in tumor cells; mitochondrial survivin orchestrates a novel pathway of apoptosis inhibition, which contributes to tumor progression
Literature	GENERIF_SUMMARY	survivin may have a role in promoting aberrantly increased cell viability and contribute to the altered homeostatic balance between cell growth and cell death in leukemia
Literature	GENERIF_SUMMARY	Inhibition of Cdc2 activity and the subsequent downregulation of survivin and XIAP by subtoxic doses of rottlerin contribute to amplification of caspase cascades.
Literature	GENERIF_SUMMARY	Survivin does not act to inhibit Smac-mediated apoptosis
Literature	GENERIF_SUMMARY	survivin may play an important role in tumor progression of hepatocellular carcinoma.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	G(1) cell cycle arrest sensitizes breast cancer cells to TRAIL at least in part by reducing levels of the anti-apoptotic protein survivin
Literature	GENERIF_SUMMARY	Our results suggest that the different forms of survivin may have different roles in apoptosis in breast cancer.
Literature	GENERIF_SUMMARY	involvement of Survivin (and the associated events) in UV-B response in vitro and in vivo in experimental models regarded to have relevance to human situations
Literature	GENERIF_SUMMARY	The homodimer interface of this protein interacts with Smac/Diablo.
Literature	GENERIF_SUMMARY	Protein binding to the 21-bp DNA element in the survivin promoter contributes at least in part to the inhibitory effect of hedamycin on survivin gene transcription.
Literature	GENERIF_SUMMARY	Patients with non-Hodgkin's lymphoma presented a high level of survivin expression, it was more pronounced in aggressive than in indolent nHL.
Literature	GENERIF_SUMMARY	a role for Survivin as a marker of tumour morphology and clinical outcome in medulloblastoma.
Literature	GENERIF_SUMMARY	Survivin expression provides some prognostic information regarding uterine cervical cancer, even though it was not correlated with survival. It could represent a possible new prognostic marker of uterine cervical squamous cell carcinoma patients.
Literature	GENERIF_SUMMARY	Use of the adeno-associated virus mediated survivan mutant (Cys84Ala) is a promising strategy in colon cancer gene therapy.
Literature	GENERIF_SUMMARY	Targeting specific survivin isoforms, rather than survivin alone, may selectively and effectively destroy tumor cells.
Literature	GENERIF_SUMMARY	Survivin is involved in supporting luteal function, and that HCG contributes to this role.
Literature	GENERIF_SUMMARY	Our findings suggest that activated NF-kappaB signaling contributes directly to malignant progression of ATL by preventing apoptosis, acting through the prosurvival protein survivin.
Literature	GENERIF_SUMMARY	upregulation of Survivin via IGF-1 signaling confers resistance to flutamide in prostate cancer cells
Literature	GENERIF_SUMMARY	a novel survivin splice variant designated as survivin 2alpha
Literature	GENERIF_SUMMARY	A high level of expression of survivin mRNA indicates a high grade of malignancy, high likelihood of recurrence, and poor prognosis.
Literature	GENERIF_SUMMARY	High expression level of survivin and its splice variants have important roles in the progression of human colorectal carcinomas.
Literature	GENERIF_SUMMARY	Survivin up-regulates telomerase activity by augmenting

Main Category	Category	Terms
		gene transcription of hTERT.
Literature	GENERIF_SUMMARY	apoptotic defects in pleural mesotheliomas are linked to increased levels of survivin; gene may be important in mesothelial carcinogenesis, and overexpression may be involved in pleural mesothelioma drug resistance
Literature	GENERIF_SUMMARY	Survivin-2B specifically expressed in lung cancer tissue and found to specifically elicit a cellular immune response in lung cancer patients
Literature	GENERIF_SUMMARY	survivin is expressed in human germ cells and might be involved in apoptosis control during spermatogenesis
Literature	GENERIF_SUMMARY	Overexpression of survivin is associated with metastatic melanomas
Literature	GENERIF_SUMMARY	Exposure of human neuroblastoma cells to exogenous VEGF results in an increased expression of survivin protein.
Literature	GENERIF_SUMMARY	Survivin expression correlates with the differentiation state of testicular germ cell tumors and may be relevant for their tumorigenesis and malignant potential
Literature	GENERIF_SUMMARY	Results indicate that the differential p53-regulated expression of survivin at different stages of the cell cycle results in different cellular outputs under the same apoptosis-inducer.
Literature	GENERIF_SUMMARY	Up-regulated survivin expression was an independent predictor of high-risk human papillomavirus in cervical lesions
Literature	GENERIF_SUMMARY	Survivin is related to differentiation of K562 cells, and it can be a target of gene therapy for leukemia.
Literature	GENERIF_SUMMARY	This is the first study reporting a statistical significant prognostic relevance of nuclear and cytoplasmic Survivin overexpression in pancreatic cancer.
Literature	GENERIF_SUMMARY	survivin-2B and survivin-DeltaEx3 play opposite roles in disease relapse and NSCLC cell survival, which is likely through the differential modulation of mitochondrial potential
Literature	GENERIF_SUMMARY	Aurora-c is directly associated with survivin and is required for cytokinesis.
Literature	GENERIF_SUMMARY	Survivin might be a good predictor of diffuse large B-cell lymphoma and its subtypes.
Literature	GENERIF_SUMMARY	studies demonstrate that lower levels of survivin in recurrent metastatic melanomas are associated with significantly improved survival in patients receiving postoperative adjuvant immunotherapy
Literature	GENERIF_SUMMARY	Overexpression of survivin in hepatocellular carcinoma(HCC) closely correlated to expression of VEGF. Stronger expression of survivin and VEGF also found in metastasis. Could become new target of gene therapy of HCC.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Rebamipide significantly downregulates in AGS cell survivin expression, its association with Aurora-B and cell proliferation. Rebamipide-induced downregulation of survivin is at the transcription level and does not involve ubiquitin-proteasome pathway.
Literature	GENERIF_SUMMARY	Like survivin, livin may act as a major cancer antigen in breast cancer patients
Literature	GENERIF_SUMMARY	Testicular survivin levels in infertile patients were related inversely to the severity of spermatogenic failure (p
Literature	GENERIF_SUMMARY	Whereas survivin expression is essential for megakaryocyte and erythroid progenitors, its down-regulation is required for terminal differentiation of megakaryocytes.
Literature	GENERIF_SUMMARY	survivin is downregulated by p53, and that siRNA targeting of survivin increases cell sensitivity to adriamycin and promotes apoptosis
Literature	GENERIF_SUMMARY	Survivin is a critical regulator of multiple processes, including proliferation, apoptosis, and angiogenesis, that determine the remodeling response of vein grafts following arterialization
Literature	GENERIF_SUMMARY	Survivin variants are differentially expressed in breast cancer according to tumour progression
Literature	GENERIF_SUMMARY	IAP-2, XIAP, and survivin may make an important contribution to the resistance to the apoptotic effect of cisplatin in prostate cancer
Literature	GENERIF_SUMMARY	Survivin should accelerate tumorigenesis by inhibiting the anti-proliferative effect of p53 tumor suppressor that is activated by aurora B in normal and glioblastoma cells containing intact p53.
Literature	GENERIF_SUMMARY	upregulated by Hepatitis B virus X (HBx) protein in in hepatoma tissues, suggesting that HBx and survivin may both be involved in carcinogenesis of hepatocellular carcinoma
Literature	GENERIF_SUMMARY	Survivin may have a role in venous or perineural invasion of metastastic pancreatic cancer
Literature	GENERIF_SUMMARY	survivin gene expression is negatively regulated by p53 in NSCLC, and survivin expression could inhibit apoptosis and accelerate tumor cell proliferation to produce more aggressive carcinomas
Literature	GENERIF_SUMMARY	downregulation of the XIAP or survivin enhances cell death by TRAIL and increases sensitivity against some chemotherapeutic agents in hepatocellular carcinoma cells
Literature	GENERIF_SUMMARY	Data show that INCENP has an important role in stabilizing the chromosomal passenger complex, and that Borealin acts to promote binding of Survivin to INCENP.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Inhibition of Bcr-Abl by imatinib significantly decreased survivin expression and cell viability in omatinib-resistantchronic myelogenous leukemia cells.
Literature	GENERIF_SUMMARY	alternatively spliced variants of survivin do not play a role in mitosis
Literature	GENERIF_SUMMARY	SI gene (GenBank accession No.AY830084) might be a novel alternatively spliced isoforms of the survivin gene involved in other functional significances related to tumorigenesis
Literature	GENERIF_SUMMARY	thymosin beta-4 increases colon carcinoma apoptosis resistance triggered by FasL and apoptosis inhibitors by downregulating Fas and upregulating Survivin expression
Literature	GENERIF_SUMMARY	Espression of survivin dropped in testes with postmeiotic spermatogenic arrest. These findings substantiate results from studies in the rodent testis suggesting a predominant expression of survivin in meiotically dividing germ cells.
Literature	GENERIF_SUMMARY	Overexpression of survivin was strongly associated with tumor grade, stage and lervical lymph node metastasis in laryngeal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	There was a significant correlation between Survivin/ VEGF expression and stage, lymph node metastasis and prognosis in larygeal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	increased accumulation of Survivin in the cytoplasm of more malignant glioma cells may prove to be a selective advantage, thus accelerating progression to a more aggressive phenotype
Literature	GENERIF_SUMMARY	Depletion of survivin by small interfering RNAs (siRNAs) increased both the number of microtubules nucleated by centrosomes and the incidence of microtubule catastrophe, the transition from microtubule growth to shrinking.
Literature	GENERIF_SUMMARY	Induction of survivin is at least in part responsible for DEC1 antiapoptosis.
Literature	GENERIF_SUMMARY	transgenic mice expressing survivin in pancreatic islet beta-cells show no changes in cell proliferation, as determined by islet size or islet number
Literature	GENERIF_SUMMARY	Antisense survivin RNA can induce apoptosis and inhibit the proliferation of Hep-2 cells by down-regulating the expression of endogenous survivin in vitro.
Literature	GENERIF_SUMMARY	combination of gene dosages of MYCN and Survivin and the expression level of BIN1 using the quantitative polymerase chain reaction method was significantly correlated with the clinical stage and the patients' outcome in neuroblastoma
Literature	GENERIF_SUMMARY	Survivin expression was associated with cell proliferation in renal cell carcinoma cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Combined detection of survivin and PTEN protein smight be more valuable in the evaluation of malignancy and prognosis in epithelial ovarian tumors.
Literature	GENERIF_SUMMARY	Survivin overexpression, by disrupting the balance between cell proliferation/differentiation and apoptosis, may play an important role in chronic myelomonocytic leukemia's pathophysiology.
Literature	GENERIF_SUMMARY	Survivin is positively correlated with PTEN expression in gastric cancer and is a molecular marker of lymph node metastasis
Literature	GENERIF_SUMMARY	Expression of HIF-1alpha mRNA was closely correlated with the expression of survivin, PCNA, and CD34 proteins in pancreatic cancer.
Literature	GENERIF_SUMMARY	A functional module within the chromosomal passenger complex involving the inner centromere protein INCENP, Survivin, and Borealin.
Literature	GENERIF_SUMMARY	analysis of a novel regulatory pathway for the expression of survivin under the control of KLF5 and p53
Literature	GENERIF_SUMMARY	Results suggest that anti-proliferative and pro-apoptotic properties of caveolin-1 may be attributed to reduced survivin expression via a mechanism involving diminished beta-catenin-Tcf/Lef-dependent transcription.
Literature	GENERIF_SUMMARY	Results suggest that survivin and splice variant survivin-EX3 may be closely linked to escape from apoptosis and the development of endometriosis.
Literature	GENERIF_SUMMARY	In esophageal adenocarcinoma, survivin expression and pattern of distribution had no prognostic relevance
Literature	GENERIF_SUMMARY	survivin promoter polymorphism at position 31 may not represent an increased risk for the development of cervical cancer
Literature	GENERIF_SUMMARY	Apoptosis of HL-60 cells induced by bufalin is related to downregulation of expressions of bcl-2 and survivin.
Literature	GENERIF_SUMMARY	Eosinophils of hypereosinophilic syndrome (HES) patients (but not normal eosinophils)express high levels of survivin and inhibit the caspase cascade in HES eosinophils.
Literature	GENERIF_SUMMARY	the direct interaction of Survivin and Aurora B was critical for the correct location of Survivin and the function of the Survivin complex in cell division
Literature	GENERIF_SUMMARY	RTQ-LDA and RTQ identified survivin as significantly over-expressed in dead-of-disease compared to long-term survival glioblastoma multiforme patients.
Literature	GENERIF_SUMMARY	survivin-aryl hydrocarbon receptor-interacting protein (AIP) complex may influence the cellular xenobiotic response to environmental toxin(s) and contribute to subcellular chaperone trafficking during cell death

Main Category	Category	Terms
		regulation
Literature	GENERIF_SUMMARY	Survivin, an inhibitor of apoptosis, is key players in cellular apoptotic mechanisms. High survivin mRNA levels were associated with worse overall survival in children's neuroblastoma.
Literature	GENERIF_SUMMARY	High levels of survivin mRNA and protein expression is associated with endometrial adenocarcinomas
Literature	GENERIF_SUMMARY	Data show that activation of survivin gene expression is mediated by oxygen-independent hypoxia-inducible factor-1alpha up-regulation, mediated by epidermal growth factor (EGF) receptor signaling in EGF-treated cancer cells.
Literature	GENERIF_SUMMARY	84% of the examined primary central nervous system lymphoma specimens showed expression of survivin. Tumours expressing survivin co-expressed PDGFRalpha frequently and PDGF-A occasionally.
Literature	GENERIF_SUMMARY	TRAIL induces apoptotic cell death in neuroblastoma and survivin inhibits TRAIL induced apoptosis. Neuroblastoma cell sensitivity to TRAIL is governed by the ratio of TRAIL-R to survivin expression.
Literature	GENERIF_SUMMARY	All variants of survivin exhibited particular associations with clinicopathologic characteristics of breast cancer patients. Survival analyses suggest a modulating role of survivin 2alpha and survivin-3B on the biological function of total survivin.
Literature	GENERIF_SUMMARY	anti-apoptotic and mitotic roles of survivin can be separated through mutation of its nuclear exportation signal
Literature	GENERIF_SUMMARY	Nuclear survivin could reflect the different behavior between keratinocyte stem cells in vitro and in vivo, in terms of proliferation.
Literature	GENERIF_SUMMARY	Survivin gene promoter-driven stathmin siRNA expression vector may have potential use in tumor gene therapy with targeted tumor gene silencing effect.
Literature	GENERIF_SUMMARY	CD3/CD46-induced T regulatory cell 1-like cells die through a process of abortive proliferation
Literature	GENERIF_SUMMARY	serum survivin could have a role in metastases in lymph nodes from breast cancer
Literature	GENERIF_SUMMARY	a novel pathway regulating angiogenesis through Survivin DeltaEx3 and a novel mechanism for Rac1 activation during angiogenesis
Literature	GENERIF_SUMMARY	This suggests that the balance between cellular Smac/DIABLO and Survivin levels could be critical for cellular destiny in taxane-treated cancer cells.
Literature	GENERIF_SUMMARY	A dominant negative mutant of survivin (surDN) was expressed in colorectal tumor cells. Its expression induced by RU486 was dose- and time-dependent. surDN suppressed tumor cell growth and had a

Main Category	Category	Terms
		synergistic effect with antineoplastic agents.
Literature	GENERIF_SUMMARY	findings suggest that a predominant nuclear localization of Survivin increases the sensitivity for pro-apoptotic stimuli, whereas nuclear export enables Survivin to fulfill its inhibitor of apoptosis function
Literature	GENERIF_SUMMARY	Molecular data based on gene transcriptional profiles of a 3-gene set (GPC3, LYVE1, and survivin) allow a reliable diagnosis of early hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	functional relationship between TEL/AML1, HSP90, and survivin provides the rational for targeted therapy
Literature	GENERIF_SUMMARY	Survivin gene RNAi not only could inhibit the proliferation of human cervical carcinoma cells (HeLa), but also could enhance the radiosensitivity of those cells. RNAi-targeted survivin gene strategy is an approach to radiosensitization therapy
Literature	GENERIF_SUMMARY	Our report shows the functional significance of the Survivin-Crm1 interface and provides a novel link between the mitotic effector Crm1 and the CPC.
Literature	GENERIF_SUMMARY	results indicate that Ad-PTEN exerts its tumor suppressive effect on bladder cancer cells through inhibiting survivin and upregulating caspase-related proteins
Literature	GENERIF_SUMMARY	Results suggest an active role of Survivin in normal and in myelodysplastic hematopoiesis.
Literature	GENERIF_SUMMARY	A review of the roles of survivin including its role in the mechanisms of tumor invasion and metastasis.
Literature	GENERIF_SUMMARY	COX-2 and beta-Catenin may have roles in regulating intracellular Survivin levels in mouse and human colon cancer
Literature	GENERIF_SUMMARY	Patients with meningiomas and malignant gliomas mount a high-titer IgG immune response against the 'universal' tumor-associated antigen survivin.
Literature	GENERIF_SUMMARY	Nuclear Survivin expression was found to be an independent predictor of progression-free survival.
Literature	GENERIF_SUMMARY	indicate a role for c-myc in the transcriptional regulation of survivin in breast cancer
Literature	GENERIF_SUMMARY	The degree of survivin staining was significantly higher in the high grade refluxing ureters compared to low grade reflux and to the bladder wall in the same patients.
Literature	GENERIF_SUMMARY	The up-regulation of Survivin expression and its close relationship with angiogenesis may play an important role in carcinogenesis and development of differentiated thyroid carcinomas.
Literature	GENERIF_SUMMARY	The expression of STAT3 was positively related to that of Survivin and Ki67 in in nasal NK/T cell lymphoma.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	overexpression of survivin mRNA is a common feature in medulloblastoma, may contribute to their anti-apoptosis properties and clinical behaviours
Literature	GENERIF_SUMMARY	Survivin is a critical modulator of atherosclerotic macrophage apoptosis under dual control by growth factors and oxidized lipids accumulating in atheroma.
Literature	GENERIF_SUMMARY	An inverse correlation between survivin expression and the level of spontaneous apoptosis in nasal and paranasal sinus carcinoma suggests that survivin may play an important role in tumorgenesis.
Literature	GENERIF_SUMMARY	The presence of the inhibitor of apoptosis protein survivin is a highly significant independent predictor of shorter duration of survival of patients with poor prognostic features, and merits investigation as a marker in other prognostic groups.
Literature	GENERIF_SUMMARY	CD8+ T cells react against survivin antigen in patients with multiple myeloma
Literature	GENERIF_SUMMARY	Sensitization towards TRAIL was due to the transcriptional downregulation of survivin
Literature	GENERIF_SUMMARY	Survivin expression is a bad prognostic indicator in cases with acute leukemia especially in AML. Survivin negativity and EPR-1 positivity show good clinical outcome in acute leukemias
Literature	GENERIF_SUMMARY	Cell hypoxia promotes an antiapoptotic phenotype in brain endothelial cells by activating survivin via the PI3-kinase/Akt pathway.
Literature	GENERIF_SUMMARY	Review highlights how better knowledge of the role of survivin in tumor versus normal cells will be instrumental for the design of optimal strategies to selectively disrupt survivin in cancer.
Literature	GENERIF_SUMMARY	The abnormal expression of Survivin and PTEN gene may be related to carcinogenesis, tumor progress and prognosis in laryngeal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Review highlights the role of survivin as a subunit of the chromosomal passenger complex and its export as a novel strategy to antagonize survivin's activity.
Literature	GENERIF_SUMMARY	Intracellular delivery of survivin or Survivin DeltaEx3 by fusion with protein transduction domain would enhance skin flap survival through accelerating revascularization.
Literature	GENERIF_SUMMARY	Changes of survivin mRNA and protein expression during paclitaxel treatment in breast cancer cells are reported.
Literature	GENERIF_SUMMARY	relationship between the expression of survivin and the development of ovarian germ cell tumors
Literature	GENERIF_SUMMARY	Downregulation of survivin results in significant inhibition of tumor growth in vitro.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Phosphorylation by aurora-B negatively regulates survivin function during mitosis.
Literature	GENERIF_SUMMARY	survivin expression was up-regulated in cervical carcinoma associated with humanpapillomavirus (HPV) infection
Literature	GENERIF_SUMMARY	Survivin-DeltaEx3 plays a key role in the inhibition of caspase-3 activity, and in the control of the mitochondrial checkpoint of apoptosis.
Literature	GENERIF_SUMMARY	Survivin may be used as an indicator of the success of in vitro fertilization-embryo transfer.
Literature	GENERIF_SUMMARY	GM-CSF-stimulated CD34(+) cell proliferation is regulated by the JAK/STAT3/survivin signaling pathway.
Literature	GENERIF_SUMMARY	XIAP and Survivin are not involved in the cytoprotective effect of HGF against antineoplastic agents.
Literature	GENERIF_SUMMARY	findings suggest that survivin is involved in tumor cell apoptosis and proliferation and may play a role in critical steps of cancer progression in NSCLC
Literature	GENERIF_SUMMARY	Nuclear presence of survivin may be an independent biomarker for disease recurrence and overall survival in patients with Stage I and II melanoma.
Literature	GENERIF_SUMMARY	Ehe expression of survivin was potently inhibited by quercetinin non-small cell lung cancer cells.
Literature	GENERIF_SUMMARY	mRNA levels of survivin and its splice variants in liver tumour tissue were significantly higher than in paired non-tumour tissue.
Literature	GENERIF_SUMMARY	Survivin is a potentially significant protein in the diagnosis, prognosis and treatment of gastric tumors.
Literature	GENERIF_SUMMARY	Survivin isoforms are unlikely to modulate survivin in trans in cancer patients.
Literature	GENERIF_SUMMARY	Chromogranin A can increase prostatic cancer cell survival through sutvivin up-regulatoin.
Literature	GENERIF_SUMMARY	subcellular localisation of survivin was significantly correlated with the lymph node involvement indicating its value in predicting the aggressiveness of tumour cells, since it increases the resistance to apoptosis and promotes cell proliferation
Literature	GENERIF_SUMMARY	XAF1 mediates Survivin down-regulation through a complex containing XIAP, supporting dual roles for XAF1 in apoptosis
Literature	GENERIF_SUMMARY	the direct interaction of survivin with the nuclear export receptor Crm1 is critically involved in its intracellular localization and cancer-relevant functions [review]
Literature	GENERIF_SUMMARY	PIN1 down-regulates BIRC5.
Literature	GENERIF_SUMMARY	SVV is abnormally expressed in the early stage of colorectal carcinogenesis, which may be correlated with

Main Category	Category	Terms
		the carcinogenesis of colorectal ademoma.
Literature	GENERIF_SUMMARY	The positive rate of Survivin protein expression in laryngeal carcinoma was significantly higher than that in the adjacent nontumorous laryngeal epithelium, laryngeal papilloma and normal laryngeal epithelium.
Literature	GENERIF_SUMMARY	This review discusses the possible connections between viral infection, survivin expression, and subsequent joint inflammation in rheumatoid arthritis.
Literature	GENERIF_SUMMARY	Histone deacetylase inhibitors enhance lexatumumab-induced apoptosis via a p21Cip1-dependent decrease in survivin levels
Literature	GENERIF_SUMMARY	Elevated expression of cell-cycle regulators p16(INK4A), p21(CIP1), and cytoplasmic/nuclear beta-catenin correlated with increased colorectal cancers risk, as did elevated expression of survivin and human telomerase reverse transcriptase.
Literature	GENERIF_SUMMARY	Forced expression of survivin-2B abrogates mitotic cells and induces mitochondria-dependent apoptosis by blockade of tubulin polymerization and modulation of Bcl-2, Bax, and survivin.
Literature	GENERIF_SUMMARY	Survivin may affect invasion and metastasis of human colorectal carcinoma cells via regulating the production of MMPs.
Literature	GENERIF_SUMMARY	Both Survivin and Livin were correlated to tumor recurrence and/or metastasis, and shorter survival.
Literature	GENERIF_SUMMARY	Survivin expression in bladder cancer cell line can be markedly inhibited by the application of siRNA-survivin, that could also induce apoptosis and inhibit the growth of the tumor.
Literature	GENERIF_SUMMARY	Heavy ion beams induce survivin expression in human hepatoma cells more effectively than X-rays.
Literature	GENERIF_SUMMARY	prognosis of stage I-IIIA non small cell lung cancer may be linked to the cellular pattern of distribution of survivin.
Literature	GENERIF_SUMMARY	Survivin expression is associated with progression of ovarian carcinoma.
Literature	GENERIF_SUMMARY	survivin expression was detected in 53 of the cases with NHL, and is associated with aggressive histology and shorter overall survival.
Literature	GENERIF_SUMMARY	Survivin contains a monomer-specific NES and that dimerization modulates cytoplasmic access of the protein
Literature	GENERIF_SUMMARY	The present review is a summary of survivin's functions, focused on its role in cell division in normal and cancer cells, as well as introduction to discussion about anticancer therapies based on survivin depletion.
Literature	GENERIF_SUMMARY	Re-expression of E-cadherin in HT29(US) cells restored

Main Category	Category	Terms
		the ability of caveolin-1 to down-regulate beta-catenin-Tcf/Lef-dependent transcription and survivin expression, as seen in HT29(ATCC) cells.
Literature	GENERIF_SUMMARY	Survivin levels were not helpful in identifying patients who would benefit from neoadjuvant treatment of rectal cancer.
Literature	GENERIF_SUMMARY	Survivin-2a may act as an antagonist of survivin-DeltaEX3 in gastric cancer.
Literature	GENERIF_SUMMARY	Inhibition of survivin expression by siRNA could sensitize hepatocellular carcinoma cells to chemotherapeutic drugs.
Literature	GENERIF_SUMMARY	Expression of survivin mRNA in bladder cancer tissue was much higher than in adjacent normal tissues, and expression was correlated with pathological grades and clinical stages.
Literature	GENERIF_SUMMARY	We concluded that As(2)O(3) induced the apoptosis of Hep-2 cells via down-regulating the expression of survivin mRNA.
Literature	GENERIF_SUMMARY	Survivin and its splice variants might be involved in the pathogenesis and development of cervical carcinomas.
Literature	GENERIF_SUMMARY	The survivin protein concentration in hepatocellular carcinoma cells was increased by the combination of hypoxia and anti-cancer drugs.
Literature	GENERIF_SUMMARY	mitotic regulator Survivin binds as a monomer to its functional interactor Borealin
Literature	GENERIF_SUMMARY	survivin was not expressed in normal bladder urothelium; survivin was overexpressed in 53% of non-muscle-invasive bladder tumors; overexpression of survivin was associated with greater tumor grade
Literature	GENERIF_SUMMARY	The expression of Survivin and PTEN proteins are closely related to the prognosis of adrenal tumors.
Literature	GENERIF_SUMMARY	Compared with survivin, PLK1 seems to be a better independent prognostic factor for diffuse large B-cell lymphoma.
Literature	GENERIF_SUMMARY	Borealin and INCENP associate with the helical domain of Survivin to form a tight three-helical bundle.
Literature	GENERIF_SUMMARY	Expression of survivin and Smac regulated cell cycle and apoptosis in ovarian cancer cells.
Literature	GENERIF_SUMMARY	survivin -31C>G polymorphism influences survivin expression, thus contributing to the genetic susceptibility to lung cancer.
Literature	GENERIF_SUMMARY	Inhibition of Survivin expression and/or function has been shown to impede tumor cell proliferation, and markedly induce spontaneous or chemotherapy induced apoptosis.
Literature	GENERIF_SUMMARY	The antiapoptotic isoform of survivin which was

Main Category	Category	Terms
		decreased during differentiation, was re-induced by GM-CSF in neutrophil-like, differentiated HL60.
Literature	GENERIF_SUMMARY	Radiation-induced apoptosis of tumor cells is increased by inhibition of the interaction between Survivin and Smac/DIABLO.
Literature	GENERIF_SUMMARY	survivin may be of importance for renal pathophysiology and pathology
Literature	GENERIF_SUMMARY	Data show that in both HeLa cells and chondrocytes, Cartilage oligomeric matrix protein induced survivin mRNA by 5-fold. Thus survivin is the first gene identified to be up-regulated transcriptionally by COMP.
Literature	GENERIF_SUMMARY	expression is negatively regulated by Egr-1
Literature	GENERIF_SUMMARY	Survivin expression was observed in 29 (91%) surgical malignant peritoneal mesothelioma (MPM) specimens; transfection of MPM cells with the survivin siRNA inhibited survivin protein expression and cell growth, and enhanced drug-induced apoptosis.
Literature	GENERIF_SUMMARY	tumor vaccine consisted of a cell-penetrating peptide of Tat(49-57), a CTL epitope peptide survivin(85-93) and a plasmid encoding murine interleukin-15 (IL-15)could effectively mediate the expression of the gene and presentation of the tumor antigen
Literature	GENERIF_SUMMARY	These data shed more light on the molecular biology of meningioma cells and suggest that survivin and proteins of the RB pathway could play a determinant role in the development and the treatment of meningiomas.
Literature	GENERIF_SUMMARY	Survivin mRNA positive cases are associated with bladder tumour recurrence
Literature	GENERIF_SUMMARY	survivin in the cell nucleus has reduced stability and is not cytoprotective
Literature	GENERIF_SUMMARY	Downregulation of survivin causes changes in mitosis and apoptosis, which may be related to changes in TNF receptors and NF-kappaB signaling.
Literature	GENERIF_SUMMARY	Survivin may play an important role in progression of hepatocellular carcinoma by promoting cell proliferation, and may be positively correlated with high risk of disease recurrence and poor prognosis.
Literature	GENERIF_SUMMARY	IGF1 promotes resistance to apoptosis in melanoma cells through an increased expression of BCL2, BCL-X(L), and survivin
Literature	GENERIF_SUMMARY	Survivin expression is induced as an early event in lung carcinogenesis.
Literature	GENERIF_SUMMARY	Our findings suggest an important role of survivin to promote trophoblast cell survival and proliferation during placentation, thus maintaining pregnancy.
Literature	GENERIF_SUMMARY	Survivin and VEGF co-expression is an independent

Main Category	Category	Terms
		prognostic survival factor for patients with advanced nasopharyngeal carcinoma.
Literature	GENERIF_SUMMARY	Increased expression of COX-2 and survivin in ovarian cancer is associated with adverse clinicopathologic parameters, including reduced survival.
Literature	GENERIF_SUMMARY	Up-regulation of of BIRC5 is associated with breast carcinomas
Literature	GENERIF_SUMMARY	-survivin expression might be a prognostic factor for non-small cell lung cancer
Literature	GENERIF_SUMMARY	methylation status of the survivin promoter and occurrence of these SNPs within the promoter region of the survivin gene appear to be of minor importance in leukemogenesis
Literature	GENERIF_SUMMARY	Overexpressions of survivin mRNA and p27 protein and reduced expression of PTEN protein might be valuable markers to predict the presence of hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Survivin and caspase-3 expression correlate with poor prognostic parameters (higher histologic grade and high proliferation), but not with outcome, in breast carcinoma patients.
Literature	GENERIF_SUMMARY	identify leptin/STAT3 signaling as a novel pathway for survivin expression in breast cancer cells
Literature	GENERIF_SUMMARY	Survivin expression in Hepa1-6 cells increases in response to mitomycin treatment at low doses.
Literature	GENERIF_SUMMARY	Wild-type survivin mRNA expression level is significantly associated with docetaxel-resistance in gastric cancer.
Literature	GENERIF_SUMMARY	survivin levels were correlated with activity scores in chronic liver disease.Increased Survivin levels were correlated with high tumor grade and vascular invasion. Tumor recurrences were more frequent in patients with high survivin levels.
Literature	GENERIF_SUMMARY	Survivin immunohistochemical staining can be a useful adjunct tool for the grading of challenging cases of papillary urothelial carcinoma.
Literature	GENERIF_SUMMARY	oxidative stress could lead to a significant activation of survivin expression, suggesting that this might be an important event in the development of pterygium, inducing and supporting a hyperproliferative condition
Literature	GENERIF_SUMMARY	survivin has a role in proliferation of leukemia cells
Literature	GENERIF_SUMMARY	Expression of BIRC5 in non-Hodgkin lymphoma treated with CHOP is reported.
Literature	GENERIF_SUMMARY	The majority (81%) of breast carcinomas show survivin expression which correlates with bcl-x, another antiapoptotic marker, and both markers tend to correlate with prognosis

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	EBV uses its latent protein, LMP2A, to activate the NF-kappaB-survivin pathway to rescue EBV-infected epithelial cells from serum deprivation
Literature	GENERIF_SUMMARY	Hyperexpression of Survivin is associated with gliomas
Literature	GENERIF_SUMMARY	In human cardiomyopathy, survivin was potently induced in the failing heart and downregulated again after hemodynamic support by a left ventricular assist device. Its expression positively correlated with the mean cardiomyocyte DNA content
Literature	GENERIF_SUMMARY	The -31C genotype of survivin promoter is associated with gastric cancer, and may be a risk factor of gastric carcinoma.
Literature	GENERIF_SUMMARY	we believe the elevated expression of survivin might play an important role of development in nasal polyps
Literature	GENERIF_SUMMARY	survivin and its spliced isoforms have associations with PCa cell proliferation and aggressive phenotypes
Literature	GENERIF_SUMMARY	Survivin may play a critical role in mediating radiation resistance in SMMC-7721 through its up-regulation and in a caspase-3 dependent manner.
Literature	GENERIF_SUMMARY	Survivin shRNA effectively inhibits survivin gene expression in Hep-2 cells leading to growth suppression and apoptotic induction in Hep-2 cells.
Literature	GENERIF_SUMMARY	The expression of survivin gene may play some roles in the pathogenesis of nasopharyngeal carcinoma.
Literature	GENERIF_SUMMARY	Survivin, Smac, and PKC alpha might play important roles in the inhibition of apoptosis by FGF-2 in human small cell lung cancer cells.
Literature	GENERIF_SUMMARY	Immunohistochemical assessment of both Survivin and CD44v6 status in negative surgical margin may be a valuable approach for predicting recurrence and survival after curative surgery for laryngeal cancer
Literature	GENERIF_SUMMARY	survivin is expressed in Merkel cell carcinoma
Literature	GENERIF_SUMMARY	Report effect of antisense-mediated inhibition of survivin, hTERT and VEGF in bladder cancers.
Literature	GENERIF_SUMMARY	The expression of survivin gene significantly increased in in myelodysplastic syndrome patients, as compared with controls.
Literature	GENERIF_SUMMARY	Survivin knockdown combined with apoptin overexpression inhibits cell growth significantly
Literature	GENERIF_SUMMARY	Smac/DIABLO interacts in a similar way with both XIAP and survivin through its amino terminal residues.
Literature	GENERIF_SUMMARY	Inhibition of survivin function by dominant-negative approaches showed that this inhibitor of apoptosis family member is critical to cell survival.
Literature	GENERIF_SUMMARY	in breast cancer, increased nuclear, but not cytoplasmic,

Main Category	Category	Terms
		survivin was associated with reduced overall survival (OS) & disease-specific survival; high cytoplasmic-to-nuclear ratio of survivin was associated with improved OS & disease-specific survival
Literature	GENERIF_SUMMARY	Expression of Survivin, Caspase-3 and p53 in surgical margin of LSCC may play an important role in the tumorigenesis and development of laryngeal carcinoma.
Literature	GENERIF_SUMMARY	Genetically programmed cell death is related to the p27, cathepsin and survivin pathways in Fuchs' dystrophy and to the p21 and p27 pathways in pseudophakic bullous keratopathy.
Literature	GENERIF_SUMMARY	Suppression of survivin expression in HCCLM6 cells can reduce their invasive and adhesive capabilities.
Literature	GENERIF_SUMMARY	These data suggest that the apoptotic response of infected intestinal epithelial cells is actively suppressed by C. parvum via upregulation of survivin, favoring parasite infection.
Literature	GENERIF_SUMMARY	Chemically synthesized human survivn does not inhibit caspase 3.
Literature	GENERIF_SUMMARY	Silencing survivin gene expression promotes cell apoptosis mediated by translocation of the mitochondrial apoptosis inducing factor, while no changes were observed in caspase-3 activation and Bid cleavage.
Literature	GENERIF_SUMMARY	p21 could be a potential mediator of survivin suppression at the transcriptional level
Literature	GENERIF_SUMMARY	tumor-shared survivin does not appear to be the target of immune tolerance in healthy donors and cancer patients and is a relevant candidate for cancer vaccine
Literature	GENERIF_SUMMARY	These results indicate that FSH increases the expression of VEGF by upregulating the expression of survivin, which is activated by the PI3K/AKT signaling pathway.[survivin]
Literature	GENERIF_SUMMARY	Survivin is a novel effector of Ran signaling, and this pathway may be preferentially exploited for spindle assembly in tumor cells.
Literature	GENERIF_SUMMARY	A novel molecular mechanism of how NO signaling may contribute to therapy resistance in ovarian cancer by modulating survivin expression.
Literature	GENERIF_SUMMARY	ER- breast cancer cells become dependent on Notch-survivin signaling for their maintenance, in vivo.
Literature	GENERIF_SUMMARY	There was significant statistical correlation between survivin and p53 expression but no correlation of survivin and Bcl2 expression in laryngeal and hypopharyngeal cancer.
Literature	GENERIF_SUMMARY	overexpression of survivin in endothelial cells serves as a protective mechanism that defends the vasculature

Main Category	Category	Terms
		from drug cytotoxicity.
Literature	GENERIF_SUMMARY	CCL2 protects prostate cancer PC3 cells from autophagic death via the phosphatidylinositol 3-kinase/ Akt/survivin pathway
Literature	GENERIF_SUMMARY	survivin expression in non-small cell lung cancer was significantly higher than in normal tissues; survivin mRNA was up-regulated in tumors of 42 patients (66%); survivin overexpression did not correlate with survival of lung cancer
Literature	GENERIF_SUMMARY	survivin greatly enhances the activity of an oncolytic adenovirus in the presence of low-dose radiotherapy
Literature	GENERIF_SUMMARY	Up-regulation of survivin is associated with oral squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Results suggest that survivin appears to participate in the multidrug resistance mechanism of H460/cDDP lung cancer cells.
Literature	GENERIF_SUMMARY	BIRC5 expressing circulating cancer cells are associated with metastasis and recurrence in gastric and colorectal cancer.
Literature	GENERIF_SUMMARY	T-cell survivin in prostate or colorectal neoplasms has a role in response to radiation therapy
Literature	GENERIF_SUMMARY	survivin gene acts as a critical mediator of AML1/ETO-induced late oncogeneic events.
Literature	GENERIF_SUMMARY	survivin is an important predictive factor for effectiveness of neoadjuvant chemotherapy with TE regimen in locally advanced breast cancer
Literature	GENERIF_SUMMARY	Survivin expression was markedly downregulated after the siRNA expression vector against HIF-1 alpha was transfected into the LS174T cells.
Literature	GENERIF_SUMMARY	BIRC5 is a novel target of transcriptional regulation by histone deacetylase inhibitors in non-small cell lung cancer.
Literature	GENERIF_SUMMARY	chromosomal binding of the mitotic regulators such as survivin is essential for mitotic progression.the localization of survivin was altered in cells.
Literature	GENERIF_SUMMARY	Results demonstrate that the survivin C-31G polymorphism may be involved in distal gastric carcinogenesis and tumor differentiation in a Chinese population.
Literature	GENERIF_SUMMARY	The function of survivin in the control of autophagy and the interaction between CCL2, survivin and autophagy in the complex program of tumor progression is discussed.
Literature	GENERIF_SUMMARY	Survivin was down-regulated in the growth inhibition of hepatoma cells induced by a selective Cyclooxygenase 2 Inhibitors , N-(2-cyclohexyloxy-4-nitrophenyl)methanesulfonamide , in a concentration-

Main Category	Category	Terms
		and time-dependent manner.
Literature	GENERIF_SUMMARY	survivin may function as a factor inducing breast cancer resistance against tamoxifen-induced apoptosis.
Literature	GENERIF_SUMMARY	Survivin expression in myeloid precursors was increased in bone marrow biopsies of all severe congenital neutropenia patients compared to reactive bone marrow, irrespective of the underlying genetic defect & it was correlated with the dose of rG-CSF.
Literature	GENERIF_SUMMARY	survivin may have a role in progression of lung cancer
Literature	GENERIF_SUMMARY	The greatest upregulation of IL-23p19, Foxp3 and survivin mRNA was seen in colorectal carcinomas than normal mucosa.
Literature	GENERIF_SUMMARY	Adenoviral activity is facilitated by abrogation of multiple cell cycle checkpoints and can be enhanced by expression of survivin within the nucleus.
Literature	GENERIF_SUMMARY	The main purpose of this study was to compare (in the same cells) survivin expression driven by chromosomal survivin promoters (endogenous survivin) with survivin expression from plasmids driven by recombinant survivin promoter.
Literature	GENERIF_SUMMARY	Akt2 down-regulation sensitizes ovarian cancer cells to paclitaxel-induced apoptosis, and inhibits survivin expression
Literature	GENERIF_SUMMARY	the increase of level of survivin in tumor tissues is not the result of decrease in content of its inhibitors SMAC and PML, as their content in tumor and normal cells is the same.
Literature	GENERIF_SUMMARY	Survivin may serve as a universal target antigen for anti-cancer immunotherapy.
Literature	GENERIF_SUMMARY	Hedyotis diffusa wild injection suppresses Bcl2 and survivin expression which inhibits the growth of HL-60 cells.
Literature	GENERIF_SUMMARY	survivin and NF-kappaB are widely expressed in lymphoma cells and they play an important role in enhancing proliferation and inhibiting apoptosis of tumor cells.
Literature	GENERIF_SUMMARY	The -31CC genotype of survivin promoter is associated with colorectal cancer (CRC) and may be a risk factor for CRC.
Literature	GENERIF_SUMMARY	the phosphorylation of survivin on Thr34 critically regulate survivin and plays an important role during the malignant transformation of oral submucous fibrosis
Literature	GENERIF_SUMMARY	Survivin has a dual role in mitosis & in apoptosis resistance. It plays an important role in development, but is absent in differentiated adult tissues. Aberrant survivin expression is found in most human cancers (Review)

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Synergy between methylselenininc acid and paclitaxel against ovarian cancer is associated with silencing survivin expression.
Literature	GENERIF_SUMMARY	silibinin induced apoptosis of laryngeal squamous carcinoma cells by a mechanism involving decreased survivin expression, which suggests the possibility that silibinin may be an effective treatment of laryngeal cancers.
Literature	GENERIF_SUMMARY	Our results suggested that the survivin expression depended on the p53 status and the C allele in the survivin promoter polymorphism -625G/C might increase the possibility of the survivin overexpression in esophageal cancer patients
Literature	GENERIF_SUMMARY	Survivin is upregulated in human and rodent liver regeneration and correlates with proliferation, suggesting an association of survivin and cell division.
Literature	GENERIF_SUMMARY	High expression of BIRC5 is associated with breast cancer.
Literature	GENERIF_SUMMARY	survivin gene expression correlated well with disease activity and radiographic damage in patients with rheumatoid arthritis
Literature	GENERIF_SUMMARY	There is a potential diagnostic utility of survivin to assisting in the microscopic differentiation of benign and malignant sebaceous neoplasms [survivin].
Literature	GENERIF_SUMMARY	Patients with low survivin expression had statistically significant better survival rates than patients with high survivin expression (p
Literature	GENERIF_SUMMARY	polymorphism of the survivin gene promoter is associated with both the clinical tumor stage and the pathologic tumor grade and might be involved in the development of urothelial carcinoma
Literature	GENERIF_SUMMARY	After treatment of HepG2 cells with DOX, the expression of p53 and p21 was increased, whereas that of survivin was reduced during 24 hours of the treatment.
Literature	GENERIF_SUMMARY	Overexpression of survivin is characteristic for pituitary tumors.
Literature	GENERIF_SUMMARY	Mutation within a putative Sp1 binding site in BIRC5 promoter led to a sharp decrease of promoter activity.
Literature	GENERIF_SUMMARY	Overexpression of survivin is associated with head and neck cancer.
Literature	GENERIF_SUMMARY	TCF4 regulates BIRC5 gene expression
Literature	GENERIF_SUMMARY	Cyclin B1, p34(cdc2) and p-survivin may play a key role during the mitosis and proliferation of oral submucosal fibrosis.
Literature	GENERIF_SUMMARY	STAT pathways and survivin could be potential targets for reducing resistance developed in patients receiving

Main Category	Category	Terms
		FLT3 inhibitors
Literature	GENERIF_SUMMARY	inhibition of apoptosis in rheumatoid arthritis synovium was observed downstream of caspase-3 and may involve the caspase-3 inhibitors, survivin and xIAP.
Literature	GENERIF_SUMMARY	Insulin induces survivin expression via the PI3K signalling pathway.
Literature	GENERIF_SUMMARY	Survivin specific siRNA could inhibit SO-Rb50 cell proliferation and induced apoptosis by knockdown of Survivin gene.
Literature	GENERIF_SUMMARY	Higher survivin scores positively related to tumor differentiation of esophageal SqCC. Significantly worse prognosis in patients with high scores of survivin, poor differentiation, T4 stage, positive for lymph node metastasis and distant metastasis.
Literature	GENERIF_SUMMARY	Repression of Survivin by FKHRL1 facilitates FKHRL1-induced apoptosis and sensitizes to cell death induced by DNA-damaging agents.
Literature	GENERIF_SUMMARY	Survivin and COX-2 expressions may participate in the pathogenesis and development of giant cell tumor of bone, but is not indicative of the prognosis.
Literature	GENERIF_SUMMARY	Higher expression of cortactin and survivin significantly correlated with advanced clinicopathological stage in renal cell carcinomas.
Literature	GENERIF_SUMMARY	Higher expression of survivin and cortactin correlates significantly with tumor stages and shorter survival time. Survivin and cortactin may be good biomarkers of aggressiveness of colorectal adenocarcinomas.
Literature	GENERIF_SUMMARY	By reducing COX-2 expression, caveolin-1 interrupts a feedback amplification loop involving PGE(2)-induced signaling events linked to beta-catenin/Tcf/Lef-dependent transcription of tumor survival genes including cox-2 itself and survivin.
Literature	GENERIF_SUMMARY	cinobufocini induces apoptosis of A 549 cells, which is associated with the decreasing expression of survivin mRNA and protein, increasing caspase-3 activity of A 549 cells.
Literature	GENERIF_SUMMARY	Significant correlation between COX-2 and survivin in primary pterygium. Pterygium may originate through an anti-apoptotic mechanism.
Literature	GENERIF_SUMMARY	We suggest that continued expression of survivin upon differentiation in vivo may contribute to teratoma formation by hES cells.
Literature	GENERIF_SUMMARY	Tolfenamic acid enhanced the radiation-induced apoptosis associated with decreased survivin expression in tumors and this correlates with the enhanced response of these tumors to the radiation.
Literature	GENERIF_SUMMARY	Data indicate an almost universal expression of survivin in benign and malignant salivary gland tumors.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Knockdown of BIRC5 expression by siRNAs enhances chemosensitivity of prostate cancer cells and attenuates its tumorgenicity.
Literature	GENERIF_SUMMARY	Dendritic cells transduced with the survivin gene induced potent survivin-specific CTL responses in vitro. This suggests that cancer immunotherapy targets for survivin might offer a novel approach to treating various urologic cancers
Literature	GENERIF_SUMMARY	dysregulation of survivin expression in hyperplastic endometrium may be part of the molecular mechanisms for progestin resistance.
Literature	GENERIF_SUMMARY	the interaction between survivin and aurora-B kinase may be essential for survivin to increase hTERT expression
Literature	GENERIF_SUMMARY	a novel microenvironmental function for survivin in the control of cancer aggressiveness and spread.
Literature	GENERIF_SUMMARY	survivin is an essential mediator of arthritogenic properties of uPA regulating its synthesis in synovial fibroblasts and uPAR expression in leucocytes
Literature	GENERIF_SUMMARY	survivin plays a critical role in epidermal homeostasis in normal conditions and during UVB exposure, with possible implication in skin carcinogenesis
Literature	GENERIF_SUMMARY	Survivin protein is an important predictive and prognostic parameter of poor outcome in gastric carcinoma.
Literature	GENERIF_SUMMARY	A significant association was detected between pre-therapeutic survivin levels and prognosis in gastric cancer patients
Literature	GENERIF_SUMMARY	Polymorphisms of survivin promoter are associated with esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Luciferase reporter gene assays revealed that survivin promoter activity is repressed upon overexpression of KLF4 in EC9706 cells.
Literature	GENERIF_SUMMARY	Data indicates that demethylation of the survivin promoter by decitabine results in p53-dependent survivin repression and that p53 binding can be inhibited by DNA methylation.
Literature	GENERIF_SUMMARY	RET interacts in vivo with aryl hydrocarbon receptor-interacting protein to alter survivin availability.
Literature	GENERIF_SUMMARY	2-methoxyestradiol treatment decreased expression of survivin in K562 cells.
Literature	GENERIF_SUMMARY	Study demonstrates that alterations in the expression of IAP family members, including Survivin and Livin/ML-IAP, are frequent in hepatocellular carcinoma (HCC).
Literature	GENERIF_SUMMARY	GPC-3 is a reliable and more specific immunohistochemical marker than survivin for the diagnosis of hepatocellular carcinoma in fine-needle

Main Category	Category	Terms
		aspiration biopsies
Literature	GENERIF_SUMMARY	Elevated expressions of MMP7, TROP2, and survivin are associated with survival, disease recurrence, and liver metastasis of colon cancer.
Literature	GENERIF_SUMMARY	On the basis of the results obtained the use of exhaled breath condensate COX-2 and survivin levels could be suggested as two potential markers within an early non-invasive screening of populations of smokers at risk of lung cancer.
Literature	GENERIF_SUMMARY	Emodin potentiates gemcitabine antitumor activity through suppression of survivin gene in pancreatic cancer in a tumor cell line.
Literature	GENERIF_SUMMARY	Upregulation of survivin expression in KSHV-associated human cells contributes to cell proliferation.
Literature	GENERIF_SUMMARY	Quercetin inhibits human SW480 colon cancer growth in association with inhibition of cyclin D1 and survivin expression through Wnt/beta-catenin signaling pathway.
Literature	GENERIF_SUMMARY	Results show that the folate-induced DNA methylation limits proliferation and increases the sensitivity to temozolomide-induced apoptosis in glioma cells through methylation of PDGF-B, MGMT, survivin, and bcl-w genes.
Literature	GENERIF_SUMMARY	Decreasing trend for BIRC5 expression is associated with early to advanced stages of myelodysplastic syndrome.
Literature	GENERIF_SUMMARY	results showed that overexpression of survivin and survivin-deltaEx3 in bladder tumors correlates with poor prognosis of bladder cancer
Literature	GENERIF_SUMMARY	These results support the notion that the upregulation of survivin in cancer cells is reflective of an oncogenic event.
Literature	GENERIF_SUMMARY	Conclude that BIRC5 and BIRC5-2B are induced via a functional cooperation between MYCN and E2F1 in neuroblastomas.
Literature	GENERIF_SUMMARY	Data show that antagonism with Smac could downregulate the protein expression of survivin and sensitize the response of pancreatic cancer to chemotherapeutic agents, which emerged as a potential and effective therapeutic strategy for pancreatic cancer.
Literature	GENERIF_SUMMARY	Data show that combination therapy resulted in robust apoptotic induction with a concomitant survivin and XIAP reduction in the MIA PaCa-2 cells with little effect in the PANC-1 cells.
Literature	GENERIF_SUMMARY	High BIRC5 is associated with malignant cancers than in the benign group.
Literature	GENERIF_SUMMARY	Nuclear degradation of Wilms tumor 1-associating protein and survivin splice variant switching underlie

Main Category	Category	Terms
		IGF-1-mediated survival
Literature	GENERIF_SUMMARY	Survivin RNAi can partly suppress the expression of survivin in SKOV3/DDP cells, inhibit the cell proliferation and promote cell apoptosis.
Literature	GENERIF_SUMMARY	Ectopic expression of wild-type CaSR in colon carcinoma cells inhibited the expression of survivin. Vitamin D suppressed the expression of survivin and survivin gene transcriptional activities.
Literature	GENERIF_SUMMARY	MS-275 could suppress the proliferation, induce apoptosis and reduce the expression of survivin in human myeloma cell line U266.
Literature	GENERIF_SUMMARY	siRNA targeting survivin gene downregulates survivin gene expression in renal clear cell carcinoma cells, inhibiting growth, and inducing apoptosis.
Literature	GENERIF_SUMMARY	Survivin was expressed in about two-thirds of the male breast neoplasms in cohort, and was not corrected with prognosis.
Literature	GENERIF_SUMMARY	Data show that with SF1126 treatment, caspase 3 and PARP proteins were cleaved, survivin was down-regulated, and inhibition of PI3-kinase was confirmed by reduced phosphorylation of Akt.
Literature	GENERIF_SUMMARY	Intracellular localization of survivin determines biological behavior in colorectal cancer.
Literature	GENERIF_SUMMARY	Upregulation of survivin in non-small cell lung cancer is associated with its neoplasm staging and correlated with p53 and Bcl-2 level.
Literature	GENERIF_SUMMARY	Data show that p53-independent curcumin-induced apoptosis in ovarian carcinoma cells involves p38 MAPK activation, ablation of Akt signaling, and reduced expression of the antiapoptotic proteins Bcl-2 and survivin.
Literature	GENERIF_SUMMARY	High Survivin and its spliced isoform gene expression is associated with proliferation of renal cancer cells and clinical stage of renal cancer.
Literature	GENERIF_SUMMARY	Yes-associated protein and survivin have roles in gastric carcinoma and precancerous lesions
Literature	GENERIF_SUMMARY	Hyperglycemia critically alters survivin expression in vitro and in vivo, which leads to attenuation of angiogenic activity and impacts endothelial metabolism.
Literature	GENERIF_SUMMARY	survivin was not related to lymph node metastases
Literature	GENERIF_SUMMARY	MMP-7 and survivin are related to the generation of colon cancer, and are independent factors for prognosis of colon cancer.
Literature	GENERIF_SUMMARY	The positive expression of survivin in nucleus might be associated with invasion, progression and poor prognosis of non-small cell lung cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data indicate survivin plays a critical role in reducing susceptibility of endometriotic stromal cells to staurosporine-induced apoptosis.
Literature	GENERIF_SUMMARY	expression in oral potentially malignant lesions might indicate a progressive deregulation of expression paralleling oncogenesis
Literature	GENERIF_SUMMARY	Survivin-c-erbB2-cox2 co-expression score has independent prognostic value for endometrial cancer.
Literature	GENERIF_SUMMARY	Arsenic trioxide suppresses transcription of survivin mRNA in NB-4 cells.
Literature	GENERIF_SUMMARY	Deletion of cav-1 protects hyperoxia-induced apoptosis via modulation of survivin expression.
Literature	GENERIF_SUMMARY	Ginsenoside Rg3 significantly downregulated the expression of survivin protein in the lung squamous cell carcinoma SK-MES-1 cell line.
Literature	GENERIF_SUMMARY	presents a potent anti-proliferative effect in part via the down-regulation of survivin expression and STAT3 phosphorylation in HTLV-1-transformed cells.
Literature	GENERIF_SUMMARY	treatment with 1,1-Bis(3'-indolyl)-1-(p-bromophenyl)methane and the 2,2'-dimethyl analog decreased survivin protein and mRNA expression in cancer cell lines
Literature	GENERIF_SUMMARY	Inhibiting Livin and survivin expression with siRNA reduces colon cancer cell proliferation.
Literature	GENERIF_SUMMARY	Upregulated p27kip1 can downregulate survivin expression and inhibit telomerase activity in gastric carcinoma cells.
Literature	GENERIF_SUMMARY	C-reactive protein can stimulate the proliferation of multiple myeloma U266 cells directly by up-regulating the expression of survivin and HSP90alpha.
Literature	GENERIF_SUMMARY	the overexpression of survivin in the majority of NSCLCs together with the abundant or upregulated expression of HBXIP and XIAP suggest that tumours are endowed with resistance against a variety of apoptosis-inducing conditions.
Literature	GENERIF_SUMMARY	t(17;19)+ acute lymphoblastic leukemias cells express Survivin at high levels and that a dominant negative mutant of E2A-HLF suppresses Survivin expression.
Literature	GENERIF_SUMMARY	Fe3O4-magnetic nanoparticles down-regulates bcl-2 and survivin in ovarian carcinoma cells.
Literature	GENERIF_SUMMARY	The expressions of survivin and survivin-DeltaEx3 mRNA were positively correlated with the pathologic grades and clinical stages of bladder cancer, but survivin-2B mRNA was negatively correlated.
Literature	GENERIF_SUMMARY	Survivin expression may have a role in urothelial carcinoma recurrence

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Survivin levels were higher in aqueous humor and serum of retinoblastoma patients than the controls.
Literature	GENERIF_SUMMARY	Survivin might be an independent prognostic factor and a potential target for the chemoradiotherapy of colorectal cancer patients.
Literature	GENERIF_SUMMARY	Over-expression of survivin mRNA is associated with malignant pleural effusion.
Literature	GENERIF_SUMMARY	Treatment of various gliomas with capsaicin downregulated the caspase inhibitor survivin. The reduction in survivin protein level was associated with downregulation of cyclin B and Cdc2 expression.
Literature	GENERIF_SUMMARY	Runx2 (runt-related transcription factor 2) regulates survivin expression in prostate cancer cells
Literature	GENERIF_SUMMARY	Down regulation of survivin gene by RNAi can increase chemosensitivity of HCC both in vitro and in vivo. The reversal of drug resistance may be reduced through the inhibitions of LRP.
Literature	GENERIF_SUMMARY	survivin is downregulated by resveratrol, which induces apoptosis in human multidrug-resistant SPC-A-1/CDDP cells
Literature	GENERIF_SUMMARY	The expression of survivin in both normal and cancer pituitary cells suggests that it may play an important role in regulation of the gland's proliferation
Literature	GENERIF_SUMMARY	TNF signaling activates Notch2 that sensitizes endothelial cells to apoptosis via modulation of the key apoptosis regulator survivin.
Literature	GENERIF_SUMMARY	findings suggest that CCL2 hyperactivates mTORC1 through simultaneous regulation of both AMPK and Akt pathways and reveals a new network that promotes prostate cancer
Literature	GENERIF_SUMMARY	Enhanced expression of survivin is associated with ulcerative colitis-associated neoplasia.
Literature	GENERIF_SUMMARY	Gene expression of Survivin, CK19, CEA and VEGF was higher than in normal controls in 98.6%, 97.1%, 42.9% and 38.6% of cases, respectively, suggesting a potential diagnostic value of both Survivin and CK19.
Literature	GENERIF_SUMMARY	Survivin was found in the majority of colorectal carcinomas, suggesting that its expression is an early event in colorectal carcinogenesis; it is statistically significantly associated with tumor differentiation but not with patient survival
Literature	GENERIF_SUMMARY	The expression of survivin increased after each surgical procedure. However it was lower after CO(2) pneumoperitoneum than after laparotomy.
Literature	GENERIF_SUMMARY	significant relationship between expression and carcinogenesis and aggressiveness of oral squamous cell carcinoma

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Our data suggest that assessment of survivin and second mitochondria-derived activator of caspase mRNA expression may be useful for predicting survival in non-small cell lung cancer patients receiving platinum-based chemotherapy after surgical resection
Literature	GENERIF_SUMMARY	Data show that LLL12 inhibited STAT3 phosphorylation, induced apoptosis as indicated by the increases of cleaved caspase-3 and poly (ADP-ribose) polymerase, and also downregulated cyclin D1, Bcl-2, and survivin at both protein and messenger RNA levels.
Literature	GENERIF_SUMMARY	Data show that the methylated sense oligonucleotide reduce specifically survivin mRNA expression and induce cell apoptosis.
Literature	GENERIF_SUMMARY	significant correlation between the levels of anti-survivin antibody in the serum and survivin mRNA in head and neck cancer tissues
Literature	GENERIF_SUMMARY	polymorphism in the survivin promoter creates an 'alternative' binding site for the transcription factor GATA-1, which can enhance the expression of the protein in breast carcinomas.
Literature	GENERIF_SUMMARY	The abnormal expression of survivin and caspase-3 may play important role in the pathogenesis of juvenile laryngeal papilloma.
Literature	GENERIF_SUMMARY	nuclear factor-kappaB activation and survivin expression are inhibited by PSK in combination with docetaxel in gastric neoplasms
Literature	GENERIF_SUMMARY	Expression of Survivin was significantly higher in sinonasal inverted papilloma and sinonasal squamous cell carcinoma than in normal tissues.
Literature	GENERIF_SUMMARY	This study demonstrated survivin expression in normal canine epidermis; nuclear survivin expression in pre-neoplastic and neoplastic lesions demonstrated a possible association of survivin with development of squamous cell carcinomas in humans and dogs.
Literature	GENERIF_SUMMARY	Survivin protein expression in the cytoplasm and in the nucleus detected by immunohistochemistry is significantly associated with prognosis of leiomyosarcoma and synovial sarcoma patients
Literature	GENERIF_SUMMARY	The expression of survivin in glioma cells was significantly down-regulated as the concentration of celecoxib increased.
Literature	GENERIF_SUMMARY	The expression of Survivin protein in extrahepatic cholangiocarcinoma has a negative correlation with histological differentiation, but has a positive correlation with lymphatic vessel infiltration and serum CA19-9 concentrations.
Literature	GENERIF_SUMMARY	study revealed survivin expression in synovial tissues and fibroblast-like synoviocytes (FLS) of rheumatoid arthritis patients; suggests extracellular survivin may be produced from rheumatoid FLS and may play an

Main Category	Category	Terms
		important role in destructive RA process
Literature	GENERIF_SUMMARY	Preoperative survivin expression levels appear to be associated with overall survival in patients with esophageal cancers. Increasing levels could potentially indicate a higher risk for shorter overall survival.
Literature	GENERIF_SUMMARY	Loss of p53, rather than beta-catenin overexpression, induces survivin-mediated resistance to apoptosis in an esophageal cancer cell line.
Literature	GENERIF_SUMMARY	Data show that survivin is expressed in most cases of CML patients and its over expression is associated with bad prognosis.
Literature	GENERIF_SUMMARY	The expression of Elf-1 and survivin in non-small cell lung cancer is related to differentiation, lymphatic metastasis, clinical stage, and prognosis, and both are positively correlated with microvessel density.
Literature	GENERIF_SUMMARY	caspase-9 G allele confers increased susceptibility to pancreatic cancer development, and the survivin C carriage status may be related to aggressive features of this malignancy.
Literature	GENERIF_SUMMARY	these results demonstrated that PI3K/Akt/p70S6K1 pathway is essential for regulating survivin mRNA expression.
Literature	GENERIF_SUMMARY	new information on the BIRC5 gene expression regulation
Literature	GENERIF_SUMMARY	The expression of survivin in cervical carcinoma with lymphatic metastasis was higher than that without lymphatic metastasis. There was no relationship between survivin or bcl-2 expression and the pathological type or tumor type of cervical carcinoma.
Literature	GENERIF_SUMMARY	Survivin transcription was associated with P-glycoprotein/MDR1 overexpression, PI3k/Akt pathway was involved in P-glycoprotein/MDR1 associated survivin transcription activity in the multidrug resistant MCF-7 breast cancer cells.
Literature	GENERIF_SUMMARY	We showed for the first time that down-regulation of survivin is not a definite therapeutic function of Hsp90 inhibitors
Literature	GENERIF_SUMMARY	This study investigated the expression of Cox-2, Survivin and Bcl-2 in hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	33% of patients with hepatocellular carcinoma had promoter methylation of the survivin gene. Immunohistochemical staining of survivin was observed in 32%. Nuclear survivin was associated with high proliferative index and poor prognosis.
Literature	GENERIF_SUMMARY	survivin cleavage by Granzyme M triggers degradation of the survivin-X-linked inhibitor of apoptosis protein (XIAP) complex to free caspase activity leading to cytolysis of target tumor cells

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Phosphorylation of survivin at Ser(20) by Plk1 kinase is essential for accurate chromosome alignment and cell proliferation but is dispensable for its anti-apoptotic activity in cancer cells.
Literature	GENERIF_SUMMARY	Data suggest that determination of HMGB1, survivin and VEGF concentrations has no clinical significance in the prognosis of the survival time in lung cancer.
Literature	GENERIF_SUMMARY	Survivin nuclear translocation in tumor cells did not differ between HCCs (10%) and FLCs (9%).
Literature	GENERIF_SUMMARY	Survivin downregulation by siRNA/cationic liposome inhibited proliferation, induced apoptosis and enhanced radiosensitivity in human hepatoma cells in vitro and in vivo.
Literature	GENERIF_SUMMARY	The positive expression rates of OPN mRNA and protein in prostate cancer tissue were significantly correlated to survivin expression.
Literature	GENERIF_SUMMARY	survivin is an important determinant of senescence reversibility in tumor cells
Literature	GENERIF_SUMMARY	These results demonstrate that IL-4, through activation of the STAT-6 signaling pathway, is involved in survivin expression levels as well as its localization.
Literature	GENERIF_SUMMARY	Rac1 knockdown by RNAi interference confirmed the specificity of NSC23766 and requirement for Rac1 in the regulation of cyclin D1, survivin, and XIAP in breast cancer cells.
Literature	GENERIF_SUMMARY	apoptosis inhibition by survivin is a significant prognostic parameter of poor response to chemotherapy in Burkitt's lymphoma.
Literature	GENERIF_SUMMARY	Results suggest that autoantibody against survivin is generated in patients with systemic sclerosis, especially those with long disease duration.
Literature	GENERIF_SUMMARY	Periplocin from Cortex periplocae inhibits cell growth and down-regulates survivin and c-myc expression in colon cancer.
Literature	GENERIF_SUMMARY	Results support the idea that isolating survivin and MRP5+ CTCs may help in the selection of metastatic colorectal cancer patients resistant to standard 5-FU and L-OHP based chemotherapy, for which alternative regimens may be appropriate.
Literature	GENERIF_SUMMARY	Data suggest that modulation of Birc5 expression may represent a novel physiological mechanism to protect the inner ear against stress-induced cell damage.
Literature	GENERIF_SUMMARY	mesenchymal stem cells (MSCs) had anti-apoptotic effect on U266 and H929 cells that was mediated by survivin.
Literature	GENERIF_SUMMARY	knock-down of Beclin 1 down-regulated survivin protein, and the turnover rate of survivin was increased when Beclin 1 expression was silenced

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	expression higher in carcinomatous components than in sarcomatous components ofuterine cervical carcinosarcoma
Literature	GENERIF_SUMMARY	Results indicate that Survivin autoantibody as a latent value of tumor marker could be used for clinical diagnosis in NSCLC patients.
Literature	GENERIF_SUMMARY	SF2/ASF and survivin expression is involved in non-small cell lung cancer progression
Literature	GENERIF_SUMMARY	High BIRC5 expression is associated with prostate cancer.
Literature	GENERIF_SUMMARY	Data revealed the significance of HER2 in the radio-resistance of HER2(+) breast cancer cells through induction of transcription factors NF-kappaB and c-myc, leading to activation of survivin, a downstream target oncogene preventing apoptosis.
Literature	GENERIF_SUMMARY	Survivin is released from exosomes to the extracellular space both basally and after proton irradiation-induced stress. Survivin was able to enhance cellular proliferation, survival and tumor cell invasion.
Literature	GENERIF_SUMMARY	Nuclear survivin expression is a strong, independent prognostic marker for poor clinical outcomes in epithelial ovarian carcinoma.
Literature	GENERIF_SUMMARY	survivin is a direct target of miR-542.
Literature	GENERIF_SUMMARY	These findings implicate survivin as an important survival factor in the gastric mucosa of humans.
Literature	GENERIF_SUMMARY	Increased expression of survivin is associated with T4 breast carcinoma.
Literature	GENERIF_SUMMARY	These data suggest that survivin transcript is regulated by estrogens and progestins in the disease-free human endometrium.
Literature	GENERIF_SUMMARY	TEL-AML1 might exert its antiapoptotic action at least in part by suppressing miRNA-494 and miRNA-320a, lowering their expression causing enhanced survivin expression.
Literature	GENERIF_SUMMARY	Data indicate that the expression of Bax mRNA was up-regulated, while the expression of Bcl-2 and survivin were decreased in both cell lines following zinc treatment.
Literature	GENERIF_SUMMARY	Data identified the oncogenic transcription factor STAT3 as a binding partner of nuclear survivin.
Literature	GENERIF_SUMMARY	1,25(OH)2D3 inhibits survivin and TP73 variants in colon and breast carcinomas; findings point to positive regulation of TP73 isoforms by survivin, reinforcing the possibility that downregulation of TP73 forms by 1,25(OH)2D3 could be survivin-dependent
Literature	GENERIF_SUMMARY	After ventricular unloading, decreased survivin protein expression might contribute to cardiac hypertrophy

Main Category	Category	Terms
		decrease by lowering the number of cyclin D1/cdk4 complexes.
Literature	GENERIF_SUMMARY	Increased survivin expression was correlated with decreased apoptosis and increased proliferation in laryngeal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	levels of survivin mRNA expression were significantly higher in oral squamous cell carcinoma cells or tissues than those in normal human oral keratinocyte or corresponding noncancerous tissues
Literature	GENERIF_SUMMARY	Single nucleotide polymorphisms in survivin gene is associated with non-small cell lung cancer.
Literature	GENERIF_SUMMARY	Human survivin expressed in mice may promote tumorigenesis by imparting a survival advantage to cells that acquire additional genetic lesions.
Literature	GENERIF_SUMMARY	After UVB treatment, Bcl-2 levels increased and casp3 and survivin levels decreased in patients with psoriasis vulgaris, restoring expression to near-normal.
Literature	GENERIF_SUMMARY	survivin-3B expression in colon cancer is an important factor involved in the invasive capacity of cancer cells in the presence of anticancer drug.
Literature	GENERIF_SUMMARY	Survivin promoter polymorphism (-31G>C) is associated with esophageal cancer
Literature	GENERIF_SUMMARY	our results support a role of survivin in colorectal carcinogenesis while the -31G/C single nucleotide polymorphism may constitute a marker of survival.
Literature	GENERIF_SUMMARY	High survivin level was associated with high grade, late stages and lymph node involvement, whereas low Bak1 mRNA was associated with late stages of breast cancer.
Literature	GENERIF_SUMMARY	Cardiopulmonary bypass could inhibit neutrophils apoptosis and increase the expression of survivin.
Literature	GENERIF_SUMMARY	Nuclear survivin expression may be a useful biomarker for predicting prognosis in patients with head and neck adenoid cystic carcinoma who were treated with surgical resection.
Literature	GENERIF_SUMMARY	Tumor survivin is downregulated by the antisense oligonucleotide LY2181308.
Literature	GENERIF_SUMMARY	With the malignant development of ovarian mucinous tumors, the expressions of Survivin are up-regulated, and the expressions of Smac are down-regulated.
Literature	GENERIF_SUMMARY	APC mutation-induced up-regulation of the survivin/ABK cascade can explain delayed crypt cell maturation, expansion of proliferative cell populations (including mitotic figures), and promotion of colon tumorigenesis.
Literature	GENERIF_SUMMARY	direct evidence that the N-terminal sequence of Survivin is not critical for dimer formation but may contribute to correct folding and function of baculovirus IAP repeat

Main Category	Category	Terms
		domain.
Literature	GENERIF_SUMMARY	Survivin expression is up-regulated by LH via ERK1/2 pathway in ovarian tumor cell line and transformed cell line (but not normal cell line); when survivin is upregulated by LH, tumor cell line is less responsive to cisplatin (apoptosis is reduced).
Literature	GENERIF_SUMMARY	results strongly suggest that IGF-1 inhibits 5-fluorouracil induced apoptosis through increasing survivin levels, which prevents Smac/DIABLO release and blocks the activation of caspases
Literature	GENERIF_SUMMARY	Cytoplasmic expression of survivin is an independent predictor of poor prognosis in patients with salivary gland cancer
Literature	GENERIF_SUMMARY	Findings provide robust evidence of the importance of survivin cytoplasmic-to-nuclear ratio as a breast cancer biomarker, and its potential to predict outcome in tamoxifen-treated patients.
Literature	GENERIF_SUMMARY	These results suggest that Epstein-Barr virus nuclear antigen 1 can contribute to the oncogenic process by up-regulating the apoptosis suppressor protein survivin.
Literature	GENERIF_SUMMARY	impact of survivin on myocardial homeostasis in hypertrophic cardiomyopathy
Literature	GENERIF_SUMMARY	Decreased immunoexpression of survivin could be a potential marker in human non-alcoholic fatty liver disease progression.
Literature	GENERIF_SUMMARY	This study suggests that histone deacetylation guides SurKex-induced DNA methylation in survivin silencing possibly through increasing the expression of DNMT1 mRNA.
Literature	GENERIF_SUMMARY	survivin physically binds to polo-like kinase 1 domain.
Literature	GENERIF_SUMMARY	These results indicate that the two SNPs have a significant and cooperative influence on bladder cancer susceptibility.
Literature	GENERIF_SUMMARY	High survivin expression is associated with adenocarcinomas than in squamous cell subtype of lung cancer.
Literature	GENERIF_SUMMARY	regulation of transcription of survivin
Literature	GENERIF_SUMMARY	Data indicate that survivin is a direct target of miR-203.
Literature	GENERIF_SUMMARY	The present result suggests that the presence of the C allele of -31C/G Survivin promoter polymorphism in combination with D17S250 instability may be used as a risk factor for gastric cancer in our population.
Literature	GENERIF_SUMMARY	These results revealed a cooperative role of TLS and PRMT1 in transcriptional regulation of survivin.
Literature	GENERIF_SUMMARY	HER-2 expression significantly correlates with Survivin at the protein level in primary invasive ductal breast

Main Category	Category	Terms
		cancer tissues
Literature	GENERIF_SUMMARY	The changes of cell cycle distribution caused by survivin(T34A) or survivin(N-8AA) enhanced the G(2)/M cell cycle-dependent chemosensitivity of paclitaxel on epithelial ovarian cancer HO-8910 cells.
Literature	GENERIF_SUMMARY	Silencing survivin with siRNA inhibited the proliferation of leukemia cell line K562.
Literature	GENERIF_SUMMARY	Downregulation of survivin by siRNA transfection suppressed proliferation and induced apoptosis in SPCA1 and SH77 cells.
Literature	GENERIF_SUMMARY	Data show that ectopic expression of survivin significantly attenuated YM155-induced apoptosis and autophagy, whereas survivin siRNA induced autophagy.
Literature	GENERIF_SUMMARY	Tamoxifen down-regulates survivin expression in HepG2 cells and it is mediated by transcriptional and post-transcriptional level via PI3K/Akt/mTOR pathway to induce apoptosis.
Literature	GENERIF_SUMMARY	The protein kinase CK2 phosphorylates survivin specifically on threonine 48 within its BIR domain. T48 is critical to its anti-apoptotic and mitotic roles.
Literature	GENERIF_SUMMARY	Adeno-associated virus-mediated survivin mutant Thr34Ala cooperates with oxaliplatin to inhibit tumor growth and angiogenesis in colon cancer.
Literature	GENERIF_SUMMARY	role of beta-catenin/Tcf4/survivin signaling in determining the fate of human corneal epithelial stem cells in different media
Literature	GENERIF_SUMMARY	the survivin -31G/C polymorphism have not been any major role in genetic susceptibilty to hepatocellular carcinogenesis
Literature	GENERIF_SUMMARY	The functional polymorphism -31C/G in the promoter of BIRC5 gene may play a role in mediating the susceptibility to nasopharyngeal carcinoma among Chinese.
Literature	GENERIF_SUMMARY	Increased survivin is associated with urothelial carcinoma of the bladder.
Literature	GENERIF_SUMMARY	BIRC5 cell cycle network for the understanding of molecular mechanism between no-tumor hepatitis/ cirrhosis and HCC transformation was identified.
Literature	GENERIF_SUMMARY	Polo-like kinase 1 contributes to the tumorigenicity of hepatoma cells via regulation of Survivin expression
Literature	GENERIF_SUMMARY	Chronic myeloid leukemia patients at advanced-stage displayed an increased expression of the standard-survivin form along with a significant decrease of survivin-2B and -DeltaEx3 levels. Anti-survivin CD8(+) T cells were undetectable at diagnosis.
Literature	GENERIF_SUMMARY	The expressions of survivin and GRIM-19 may be closely correlated with the pathogenesis of prostate

Main Category	Category	Terms
		cancer.
Literature	GENERIF_SUMMARY	The difference in expression of survivin between platinum-resistant and platinum-sensitive epithelial ovarian carcinoma was not significant.
Literature	GENERIF_SUMMARY	Overexpression of wild type p53 protein may suppress the expression of survivin and its antiapoptotic activity in basal cell carcinomas cells.
Literature	GENERIF_SUMMARY	5-Bromotetrandrine and daunorubicin can effectively reverse the multidrug resistance of K562/A02 cells by downregulating survivin.
Literature	GENERIF_SUMMARY	survivin represents an innate cytoprotective resistor against stress conditions in the auditory system.
Literature	GENERIF_SUMMARY	The withdrawal of Survivin is a novel and active physiological regulatory mechanism that tilts the survival balance and promotes the progression of apoptosis.
Literature	GENERIF_SUMMARY	Studies demonstrate that targeting survivin may be an effective approach for radiosensitization of malignant glioblastoma.
Literature	GENERIF_SUMMARY	Data show direct interaction between miR-218 and the 3'UTR of mRNAs encoding ROBO1, survivin (BIRC5), and connexin43 (GJA1).
Literature	GENERIF_SUMMARY	high levels of survivin protein expression are indispensable for survival of HER2 gene-amplified breast cancer cells with de novo resistance to HER1/2-targeted therapies
Literature	GENERIF_SUMMARY	Data show that ALK inhibitor-induced apoptosis is mediated both by BIM upregulation resulting from inhibition of ERK signaling as well as by survivin downregulation resulting from inhibition of STAT3 signaling in EML4-ALK-positive lung cancer cells.
Literature	GENERIF_SUMMARY	High survivin expression in pretreatment tumor biopsy was associated with less tumor downstaging after preoperative chemoradiotherapy for locally advanced rectal cancer.
Literature	GENERIF_SUMMARY	survivin may contribute to an apoptosis-resistant phenotype by inhibiting idarubicin-induced apoptosis
Literature	GENERIF_SUMMARY	the increased survivin expression observed in oesophageal cancer cells that have lost p53.
Literature	GENERIF_SUMMARY	Studies indicate that survivin has been identified by molecular profiling analysis to be associated with higher tumor grade, more advanced disease, abbreviated survival, accelerated rates of recurrence, and chemotherapy and radiation resistance.
Literature	GENERIF_SUMMARY	Survivin isoforms may play a role in cell apoptosis and their quantification could provide information about clinical management of patients suffering from colorectal cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Survivin may mediate metastasis in hepatocellular carcinoma. The knockdown of survivin expression may enhance cancer metastasis through the downregulation of GRP78 and upregulation of vimentin expression
Literature	GENERIF_SUMMARY	Tetrandrine inhibits the expression of Survivin and increase the apoptosis of K562 cells.
Literature	GENERIF_SUMMARY	The findings of study suggest that increased expression of Survivin may indicate a worse prognosis in breast cancer patients.
Literature	GENERIF_SUMMARY	The mechanism of dimer-monomer balance regulation may also work as a ""switcher"" between survivin functions and thereby explain remarkable functional diversities of this protein."
Literature	GENERIF_SUMMARY	These results support a role of survivin in colorectal carcinogenesis while the -31 G/C snp may constitute a marker of survival.
Literature	GENERIF_SUMMARY	We conclude that low p21 and high survivin expression are poor prognosis indicators in FIGO stage I endometrial endometrioid adenocarcinoma
Literature	GENERIF_SUMMARY	MAGE-A regulates Bax and survivin through p53-dependent mechanisms
Literature	GENERIF_SUMMARY	High expression levels of survivin in the primary tumor may be used as a potential predictive biomarker of pCR to neoadjuvant chemotherapy in patients with stage II and stage III breast cancer.
Literature	GENERIF_SUMMARY	Cytoplasmic survivin in adenocarcinomas and squamous cell carcinomas could be useful independent prognostic marker in lung neoplasms.
Literature	GENERIF_SUMMARY	current data of survivin in skin biology and a comprehensive compilation of survivin expression in skin pathologies (review)
Literature	GENERIF_SUMMARY	An important role of survivin -31 G/C polymorphism with cancer risk especially in Asian population.
Literature	GENERIF_SUMMARY	The functions of survivin in anti-apoptosis and regulation of cell division might mediate the role of NS5A in HCV-induced HCC.
Literature	GENERIF_SUMMARY	Centromere protein F and survivin are malignant behaviour markers for colorectal gastrointestinal stromal tumours.
Literature	GENERIF_SUMMARY	Survivin expression may be a prognostic indicator for survival in early-stage neuroendocrine carcinoma of the uterine cervix without lymph node metastasis.
Literature	GENERIF_SUMMARY	BIRC5 protein expression in tumor samples versus corresponding normal tissue was 1.3 times more often elevated in the squamous cell lung cancer group (in 76% tumor samples).
Literature	GENERIF_SUMMARY	Survivin is a critical factor for cell division and cell

Main Category	Category	Terms
		viability in primary human osteoblastic cells.
Literature	GENERIF_SUMMARY	The high expression of survivin was shown to be correlated with the lower-grade differentiation of a tumor and its higher stage.
Literature	GENERIF_SUMMARY	Data suggest that the development of strategies to concurrently induce both CD4(+) and CD8(+) T-cell responses against survivin is warranted for optimal anti-survivin cancer immunotherapy.
Literature	GENERIF_SUMMARY	Survivin is a novel target of CD44-promoted breast tumor invasion
Literature	GENERIF_SUMMARY	Survivin is expressed at protein and mRNA levels in human osteoarthritic cartilage-- the so-called chondroid nests, comprising accumulations of ""stressed"" chondrocytes where survivin gene transcription and translation are reactivated."
Literature	GENERIF_SUMMARY	Glycyrrhetinic acid can inhibit proliferation of U266 cells in a time- and dose-dependent manner and induce apoptosis through arresting G(0)/G(1) phase and down-regulating expression of survivin.
Literature	GENERIF_SUMMARY	mesenchymal stem cells have lower sensitivity to radiation, which may associate with the expression of radioresistant genes survivin and HO-1.
Literature	GENERIF_SUMMARY	Data show that siRNA-mediated survivin knockdown induced spontaneous apoptosis in melanoma cells.
Literature	GENERIF_SUMMARY	Histone deacetylase inhibitor belinostat represses survivin expression through reactivation of transforming growth factor beta (TGFbeta) receptor II leading to cancer cell death
Literature	GENERIF_SUMMARY	Survivin expression correlates with PPARdelta expression in human colorectal cancer
Literature	GENERIF_SUMMARY	We conclude that survivin is functionally relevant in gliomas
Literature	GENERIF_SUMMARY	Paracrine factors from normal endometrial stromal cells can inhibit hormone-stimulated cell proliferation in Ishikawa cells by regulating cell survival and apoptosis through PI3K/AKt/Survivin signaling
Literature	GENERIF_SUMMARY	Data show that survivin was an independent prognostic factor for predicting overall survival of patients with esophageal cancer.
Literature	GENERIF_SUMMARY	Nuclear accumulation of Survivin and interaction with components of the DNA-double-strand break (DSB) repair machinery indicates Survivin to regulate DSB damage repair that leads to a significant improvement of survival of LN229 glioblastoma cells.
Literature	GENERIF_SUMMARY	High BIRC5 expression is associated with colon cancer.
Literature	GENERIF_SUMMARY	RNAi-mediated survivin knockdown in vascular smooth muscle cells leads to profound cell cycle arrest at G(2)/

Main Category	Category	Terms
		M and impaired chemotaxis without cytotoxicity.
Literature	GENERIF_SUMMARY	BIRC5 promoter SNPs do not affect nuclear survivin expression and survival of malignant pleural mesothelioma patients
Literature	GENERIF_SUMMARY	Livin and survivin may be involved in the pathogenesis and progression of adult acute lymphoblastic leukemia.
Literature	GENERIF_SUMMARY	survivin expression correlated with TNM stage and lymph node metastasis in patients with non-small-cell lung cancer
Literature	GENERIF_SUMMARY	These data suggest that splice variant Survivin-3B contributes to cell survival through the anti-apoptotic pathway and that its expression level may be an important factor in determining therapeutic strategies for breast carcinoma
Literature	GENERIF_SUMMARY	survivin may play an important role in the apoptosis pathway in cardiac myxomas.
Literature	GENERIF_SUMMARY	The presence of survivin gene allele C was found to be significantly increased in endometrial cancer tissues compared to the healthy tissues.
Literature	GENERIF_SUMMARY	Additional studies on the expression of survivin isoforms and their subcellular localization in relation to function will confirm the key role of survivin in the skin.
Literature	GENERIF_SUMMARY	Jiedu Huayu Recipe could block the NF-kappaB signal pathway of K562/A02 cells and reverse drug resistance by influencing the expression of NF-kappaB and survivin genes.
Literature	GENERIF_SUMMARY	Survivin but not TNFalpha may have an important role in the development of acute leukemia and it could serve as a significant prognostic mark
Literature	GENERIF_SUMMARY	Expression patterns of survivin and gamma-H2AX were different, suggesting their involvement in different biologic mechanisms as an early event leading to resistance to apoptosis in vulvar carcinogenesis.
Literature	GENERIF_SUMMARY	although survivin was highly expressed in gastrointestinal stromal sarcoma, there was no correlation between survivin expression and disease-free or overall survival. Survivin may be a biomarker for disease progression in gastrointestinal stromal sarcoma.
Literature	GENERIF_SUMMARY	the BIR domain of Survivin presents an accessible and preformed binding site to recognize histone H3
Literature	GENERIF_SUMMARY	Data suggest that expression of survivin mRNA and protein in preeclamptic placenta is significantly reduced compared to that in normal placenta; these levels are significantly reduced in severe preeclampsia compared to mild preeclampsia.
Literature	GENERIF_SUMMARY	High survivin levels predict poor clinical response to infliximab treatment in patients with rheumatoid

Main Category	Category	Terms
		arthritis.
Literature	GENERIF_SUMMARY	findings suggest that positive survivin expression may be used as a predictor of a poor response to chemotherapy and shorter survival in lung cancer patients with malignant pleural effusion
Literature	GENERIF_SUMMARY	The results suggest that the survivin T9809C single nucleotide polymorphism might contribute to the prediction of susceptibility and pathological development to hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Survivin likely has different functions in distinct breast cancer subtypes.
Literature	GENERIF_SUMMARY	Describe the baseline levels of urine levels of survivin after radical cystectomy for primary invasive bladder cancer.
Literature	GENERIF_SUMMARY	High survivin expression was an independent factor for predicting lymph node metastasis of colorectal cancer. The expression levels of survivin mRNA and protein were higher in CRC cells than in normal intestinal epithelial cell line.
Literature	GENERIF_SUMMARY	Persistent activation of signaling by the AKT-survivin pathway induced by PTEN loss underlies a mechanism of resistance to erlotinib-induced apoptosis in EGFR mutation-positive non-small cell lung cancer.
Literature	GENERIF_SUMMARY	The survivin protein acted as mitotic regulator and apoptosis inhibitor.
Literature	GENERIF_SUMMARY	High expression of nuclear survivin is a significant marker of a poor prognosis in patients with a pancreatic endocrine tumor.
Literature	GENERIF_SUMMARY	Survivin was overexpressed in carcinoma cells. It was present in different subcellular compartments. Nuclear positivity of survivin or combined nuclear & cytoplasmic expression was a poor prognostic parameter in ductal breast carcinoma.
Literature	GENERIF_SUMMARY	A novel dendritic cell(DC) progenitor regulatory pathway in which PGE(2) signaling through EP1/EP3 receptors regulates Flt3 expression and downstream STAT3 activation and survivin expression, required for optimal progenitor survival and differentiation.
Literature	GENERIF_SUMMARY	findings showed down-regulation of survivin expression in cancer cells; study allows the conclusion that survivin is essential for proper chromosome segregation and cytokinesis
Literature	GENERIF_SUMMARY	Silencing survivin causes growth suppression of MGC-803 cells and enhances sensitivity to celecoxib in vitro.
Literature	GENERIF_SUMMARY	Downregulation of survivin by shRN inhibited cell proliferation and arrested cell at G2/M phase.
Literature	GENERIF_SUMMARY	Nuclear survivin typifies aggressive high-grade salivary

Main Category	Category	Terms
		gland carcinomas with worse prognosis
Literature	GENERIF_SUMMARY	results demonstrate that Aurora B confers TRAIL resistance to cancer cells via phosphorylation of survivin.
Literature	GENERIF_SUMMARY	Expressions of SMAC/DIABLO and survivin were significantly reciprocal in breast cancer and benign tumor tissues.
Literature	GENERIF_SUMMARY	High expression of survivin is associated with ovarian epithelial tumors.
Literature	GENERIF_SUMMARY	IL-4 triggers a simultaneous activation of the JNK-pathway and the up-regulation of survivin turning on a cancer proliferation mechanism
Literature	GENERIF_SUMMARY	Ang-2 mediated Survivin expression in response to OxLDL-induced endothelial apoptosis.
Literature	GENERIF_SUMMARY	The functional polymorphism -31G>C, -1547A>G and -241C>T in the promoter of Survivin gene may play a significant role in mediating the bladder cancer risk among North Indian cohort.
Literature	GENERIF_SUMMARY	REG4, BIRC5 and NEIL2 genes might have a role in the sensitivity of cancer patients to radiotherapy
Literature	GENERIF_SUMMARY	No association between rs8073069, rs9904341 or rs1042489 in survivin gene and the risk of hepatocellular carcinoma (HCC) is found in Chinese han population, but rs8073069G-rs9904341C- rs1042489T is perhaps a protective haplotype for HCC.
Literature	GENERIF_SUMMARY	Data show that the expression of TSLC1 was lower in hepatocellular carcinoma (HCC) cells than in normal cells, and survivin expression is higher in detected HCC cells than in normal cells.
Literature	GENERIF_SUMMARY	Survivin primarily recognizes N-terminal Ala1 of H3, H3T3ph, and SmacN peptides and exhibits a 25-fold preference for H3T3ph over SamcN peptide.
Literature	GENERIF_SUMMARY	LMP1 upregulated survivin protein expression and its phosphorylation by p53 due to the transactivation of the survivin promoter in nasopharyngeal carcinoma cells.
Literature	GENERIF_SUMMARY	Survivin -31G>C promoter polymorphism might be associated with an increased risk of cancer, especially in the Asian populations.
Literature	GENERIF_SUMMARY	GC heterozygotes had considerably decreased susceptibility for KCOTs compared with GG homozygotes. same was true for GC+CC vs. GG. GG genotype of -31G/C polymorphism might be risk factor for KCOT development.
Literature	GENERIF_SUMMARY	Survivin polymorphism at position -31 (G/C) (rs 9904341) was associated with papillary thyroid carcinoma.
Literature	GENERIF_SUMMARY	functional -31G>C polymorphism in the promoter of

Main Category	Category	Terms
		survivin may influence the susceptibility and progression of renal cell cancer in Chinese population
Literature	GENERIF_SUMMARY	study indicates that miR-203 exerts a tumor-suppressive effect during laryngeal carcinogenesis via suppressing the expression of survivin
Literature	GENERIF_SUMMARY	Patients who showed high expression of cytoplasmic and nuclear survivin and Aurora B had significantly shorter overall survival
Literature	GENERIF_SUMMARY	These results suggest that survivin gene polymorphisms and their interactions with environmental carcinogens may increase susceptibility to oral cancer in Taiwanese men.
Literature	GENERIF_SUMMARY	Survivin expression was not related differentiation, clinical stage, and lymph node metastasis of lung cancer.
Literature	GENERIF_SUMMARY	The over-expression of PTEN gene may inhibit the proliferation of K562 cells and promote cell apoptosis via the regulation of Survivin, Xiap and Smac genes.
Literature	GENERIF_SUMMARY	HDAC6 deacetylates survivin to regulate its nuclear export
Literature	GENERIF_SUMMARY	The BIRC5 might be an important regulator of the asthmatic processes and it provides some evidence that its effect might be affected by single nucleotide polymorphisms(SNPs) located in the gene regulatory regions.
Literature	GENERIF_SUMMARY	The expression of Survivin in combination with BCRP and HER-2 is associated with clinical response to TE neoadjuvant chemotherapy in breast cancer.
Literature	GENERIF_SUMMARY	Our data demonstrate that Survivin paralogues have different characteristics of phosphospecific binding to threonine-3 of histone H3, providing new insight into the biology of the inner centromere.
Literature	GENERIF_SUMMARY	nuclear survivin, including survivin itself and the survivin-Ex3 splice variants, may participate in modulation of altered cell kinetics of uterine cervical carcinomas during radiation therapy.
Literature	GENERIF_SUMMARY	Gene silencing of survivin reduces the proliferation of lymphoma cell line Jeko-1.
Literature	GENERIF_SUMMARY	High survivin expression is associated with malignant peripheral nerve sheath tumors.
Literature	GENERIF_SUMMARY	review of the role of survivin as a target in cancer [review]
Literature	GENERIF_SUMMARY	There was a significant association between a higher nuclear survivin expression and laryngeal squamous cell carcinoma recurrence. Wild-type survivin was the most frequently detected splice variant.
Literature	GENERIF_SUMMARY	Upregulation of survivin is associated with disease

Main Category	Category	Terms
		progression in the myelodysplastic syndromes.
Literature	GENERIF_SUMMARY	Downregulation of survivin expression decreased the growth and viability of tumor cells in culture and reduced growth of the cancer cells upon transplantation into immunodeficient mice
Literature	GENERIF_SUMMARY	survivin and Bcl-2 may act synergistically in the occurrence, development, invasion and metastasis of non-small cell lung carcinoma (NSCLC), both of which are up-regulated in NSCLC tissues
Literature	GENERIF_SUMMARY	These findings suggest that survivin and XIAP may have a role in leukemogenesis and provide prognostic information.
Literature	GENERIF_SUMMARY	This study identified a novel role for survivin in erythroblast enucleation through previously unknown protein partners.
Literature	GENERIF_SUMMARY	High survivin expression is associated with drug resistance to docetaxel in patients with advanced breast cancer.
Literature	GENERIF_SUMMARY	survivin may act as a transcription factor or cofactor to interact with p53 on the p21(WAF1/CIP1) promoter leading to the inhibition of p21(WAF1/CIP1) expression at least in part by neutralizing p53-mediated transcriptional activation of the p21 gene.
Literature	GENERIF_SUMMARY	Survivin knockdown induced morphological features of neuronal differentiation in neuroblastoma cell lines.
Literature	GENERIF_SUMMARY	analysis of a promiscuous survivin-derived T-cell epitope restricted to the HLA-A3 super-type alleles
Literature	GENERIF_SUMMARY	Decreased survivin cytoplasmic expression or nuclear overexpression may be related to favorable prognosis of Wilms tumor.
Literature	GENERIF_SUMMARY	Nuclear survivin could serve as a marker for indicating disease status in esophageal squamous cell carcinoma patients.
Literature	GENERIF_SUMMARY	The results of this study showed that disregulation of survivin expression had an important role in the determination of the course of the disease in patients with nodal diffuse large B cell lymphoma treated with R-CHOP
Literature	GENERIF_SUMMARY	survivin was not influenced by confounders like inflammation and exhibited a relatively low number of false-positives. Therefore, despite the low sensitivity, survivin may still be considered as a component of a multimarker panel.
Literature	GENERIF_SUMMARY	concluded that the survive gene expression was lower in APL than those in any other types of leukemia, thus closely associated with clinical manifestation
Literature	GENERIF_SUMMARY	CXCL8 and CXCR1 are involved in the pathogenesis of endometriosis by up-regulating proliferation and growth

Main Category	Category	Terms
		and restricting apoptosis in endometrial stromal cells by activating the PTEN/Akt pathway and mediating the expression of survivin and Bcl-2
Literature	GENERIF_SUMMARY	The survivin promoter exhibited a higher transcriptional activity than PSMA promoter and enhancer in prostate tumor cell lines.
Literature	GENERIF_SUMMARY	The mechanism of estrogen-promoted ovarian cancer metastasis may due to the up-regulation of survivin conducted through the ERalpha signalling pathway.
Literature	GENERIF_SUMMARY	Birc5 and Hspa5 could be important survival factors for hepatoma carcinoma cells
Literature	GENERIF_SUMMARY	Survivin regulates VEGF-C expression in lymphatic metastasis in breast cancer.
Literature	GENERIF_SUMMARY	Survivin expression correlated with the expressions of multiple proteins involved with cell proliferation and survival.
Literature	GENERIF_SUMMARY	CUG-BP1 associates with the 3'-untranslated region of survivin mRNA, thereby stabilizing the transcript and elevating its expression in oesophageal cancer cells.
Literature	GENERIF_SUMMARY	A review of the prognostic and predictive relevance of survivin detection in cancer patients.
Literature	GENERIF_SUMMARY	survivin mRNA assay is a useful tumor marker for discriminating malignant from benign pleural effusion.
Literature	GENERIF_SUMMARY	survivin expression alters the sensitivity of hepatocellular carcinoma cells to curcumin-induced cytotoxicity.
Literature	GENERIF_SUMMARY	Up-regulated of BIRC5 and LASP1 was able to abrogate the effects induced by transfection with the miR-203 precursor in triple negative breast cancer.
Literature	GENERIF_SUMMARY	Survivin expression was also significantly upregulated in all head and neck cancer cell lines and levels were further elevated in cisplatin-resistant cells.
Literature	GENERIF_SUMMARY	Overexpression of survivin is associated with drug resistance to docetaxel-based chemotherapy in advanced gastric cancer.
Literature	GENERIF_SUMMARY	survivin levels are increased in lamina propria T cells(LPTs) from patients with Crohn's disease (compared with ulcerative colitis and controls) because survivin interacts with HSP90 and prevents proteasome degradation; this allows LPTs to avoid apoptosis
Literature	GENERIF_SUMMARY	Overexpression of survivin is associated with multidrug-resistance in multiple myeloma.
Literature	GENERIF_SUMMARY	novel insight into the molecular circuitry of Noscapine-induced apoptosis to indicate suppression of survivin expression as a critical mediator of this process.
Literature	GENERIF_SUMMARY	Allele & genotype frequencies were significantly

Main Category	Category	Terms
		different between Wilms tumor patients & controls for the -31 G/C polymorphism. Those with CC & CG genotypes had decreased risk compared with GG individuals.
Literature	GENERIF_SUMMARY	miR-203 overexpression significantly inhibited survivin expression in HepG2 cells, while underexpression of miR-203 significantly promoted it. Low miR-203 expression contributes to the progression of hepatocellular carcinoma via targeting survivin.
Literature	GENERIF_SUMMARY	Data suggest that survivin expression in up-regulated following induction of HIF1A (hypoxia inducible factor 1 alpha subunit) by cellular hypoxia cervical cancer cells possibly leading to tumor formation and tumor progression.
Literature	GENERIF_SUMMARY	Survivin and livin mRNA levels in patients with lung cancer were significantly higher than in those with benign lung disease.
Literature	GENERIF_SUMMARY	It seems that vascular endothelial growth factor and survivin play role in local invasion and spread of gastric adenocarcinoma and negatively influences survival.
Literature	GENERIF_SUMMARY	Survivin expression in biopsy could be an important predictive factor of preoperative CRT response.
Literature	GENERIF_SUMMARY	Data suggest that BIBR1532 caused rapid cell death in Nalm-6 cells probably through transcriptional suppression of survivin-mediated c-Myc and human telomerase reverse transcriptase (hTERT) expression in a concentration-dependent manner.
Literature	GENERIF_SUMMARY	High levels of survivin mRNA expression in luteinized granulosa cells in cases with tubal infertility seem to protect ovaries from follicular apoptosis.
Literature	GENERIF_SUMMARY	BIRC5 is over-expressed in Ewing sarcoma (ES) cell lines and tumor samples. Further, it plays an important role in cell growth and viability in vitro.
Literature	GENERIF_SUMMARY	High survivin mRNA expression is associated with breast cancer.
Literature	GENERIF_SUMMARY	The Nuclear survivin expression is associated with poor prognosis for Caucasian non-small cell lung cancer(NSCLC) patients.
Literature	GENERIF_SUMMARY	These data suggest that TFF3 and survivin expressions play a vital role in gastric cancer development, and these two proteins are important markers for prognosis in gastric cancer.
Literature	GENERIF_SUMMARY	The study showed that survivin expression detected by immunohistochemistry seems to be associated with a worse prognosis of ESCC patients.
Literature	GENERIF_SUMMARY	study demonstrated the expression of inhibitor of apoptosis proteins, survivin and Livin and the pro-apoptotic protein TSP-1 in endemic Burkitt lymphoma (BL); no significant difference was found between BL

Main Category	Category	Terms
		and reactive lymphoid hyperplasia
Literature	GENERIF_SUMMARY	Liver cancer initiation is controlled by AP-1 through SIRT6-dependent inhibition of survivin.
Literature	GENERIF_SUMMARY	With the cutoff point at 5%, 14% of all cases were classified nuclear survivin positive and 86% nuclear survivin negative
Literature	GENERIF_SUMMARY	Survivin knockdown in combination with radiation inhibits cell proliferation, repair, and colony formation significantly and increases apoptosis more than each single treatment alone.
Literature	GENERIF_SUMMARY	Survivin overexpression had a poor prognosis value in patients with non-small cell lung cancer. [Meta-analysis]
Literature	GENERIF_SUMMARY	data have demonstrated for the first time that human oocytes/embryos not only express but also secret survivin, suggesting that survivin may play an important role in human oogenesis and embryogenesis
Literature	GENERIF_SUMMARY	Survivin protein gene expression is a key predictor of malignant progression in patients diagnosed with laryngeal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Plasma-derived exosomal survivin, a plausible biomarker for early detection of prostate cancer.
Literature	GENERIF_SUMMARY	Expression of survivin modulated by sodium valproate is under the regulation of AKT.
Literature	GENERIF_SUMMARY	TERT immunoexpression was closely related to survivin reactivity, increased from normal vulvar squamous epithelia to lichen sclerosus and to vulvar intraepithelial neoplasia, and invasive squamous cell carcinoma
Literature	GENERIF_SUMMARY	High expression of BIRC5 is associated with shorter overall survival and progression-free survival in patients with resected pancreatic adenocarcinoma after adjusting for other histopathological factors.
Literature	GENERIF_SUMMARY	Nuclear survivin expression and Bcl-2 expression warrant investigation as potential predictive biomarkers to enable individualized treatments for patients whose LSCCs strongly express nuclear survivin or/and Bcl-2.
Literature	GENERIF_SUMMARY	promoter polymorphisms are not associated with susceptibility to endometriosis in the Estonian population
Literature	GENERIF_SUMMARY	Data suggest that survivin mRNA is significantly overexpressed in adrenocortical carcinoma as compared to adrenocortical adenoma or normal adrenal gland; moderate/high intensity of staining for survivin seems to indicate poor prognosis.
Literature	GENERIF_SUMMARY	suggests that glycolysis inhibitors target an 'archilles heel' of BIRC5/Survivin-overexpressing neuroblastoma and may be highly useful as chemosensitizers in the treatment of high-stage NB

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Literature	GENERIF_SUMMARY	Transfection of the NSCLC cell lines with specific siRNAs against SRF, E2F1 and survivin resulted in a very considerable reduction of the intracellular mRNA concentration.
Literature	GENERIF_SUMMARY	Survivin mutations confer improved response to chemotherapy in vivo.
Literature	GENERIF_SUMMARY	The positive expression rates of survivin and PTEN in vocal cord polyps were obviously lower than in papilloma of larynx and laryngeal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	SLC5A8 suppresses tumour progression through depletion of survivin without involving its transport function.
Literature	GENERIF_SUMMARY	these data support the conceptof a dysregulated survivin expression, especially in individualswith AERD, which might be part of the pathogenesis of nasal polyps.
Literature	GENERIF_SUMMARY	Presence of the C allele was significantly associated with tumor stage and size; therefore, survivin might be an important marker in the prognosis of tongue SCC that requires further investigation.
Literature	GENERIF_SUMMARY	OCT4 positively regulates Survivin expression to promote cancer cell proliferation and leads to poor prognosis in esophageal squamous cell carcinoma
Literature	GENERIF_SUMMARY	Our results suggest that survivin may be a valuable diagnostic marker, as well as a new independent prognostic parameter, in lobular breast carcinoma.
Literature	GENERIF_SUMMARY	this study validated the functions of the rs2239680 SNP in the mir-355-related regulation of BIRC5 in Chinese Han with lung cancer.
Literature	GENERIF_SUMMARY	High expression of survivin is associated with oral squamous cell carcinoma and peritumoral tissue.
Literature	GENERIF_SUMMARY	Suppression of survivin expression by RNAi attenuates the malignant phenotype of hepatocellular carcinoma cells.
Literature	GENERIF_SUMMARY	these results indicated that stimulation of ovarian cancer cell VEGF, Cox2 and survivin expression by FSH involves the microRNA27a
Literature	GENERIF_SUMMARY	The survivin promoter -31G/C polymorphism may confer an increased susceptibility to gastric cancer.
Literature	GENERIF_SUMMARY	expression of Survivin was negatively correlated with that of Fas (r = -0.517, p
Literature	GENERIF_SUMMARY	Data using recombinant proteins in cell-free lysates suggest that interaction of survivin with phosphorylated histone 3 (H3T3-Phos; a mitotic biological marker appearing during cell division) is abolished by trimethylation at neighboring lysine.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Analysis of survivin-specific T cells in breast cancer patients using human DCs engineered with survivin mRNA.
Literature	GENERIF_SUMMARY	Results showed that PTEN expression progressively decreased with the continuum from normal epithelium to squamous cell carcinoma (P
Literature	GENERIF_SUMMARY	High expression of survivin is associated with T-cell lymphoma.
Literature	GENERIF_SUMMARY	HER2 stabilizes survivin while concomitantly down-regulating survivin gene transcription by suppressing Notch cleavage.
Literature	GENERIF_SUMMARY	mutant K-Ras contributes to the maintenance of the aberrantly high levels of survivin in tumors by regulating its stability, and the ability of mutant K-Ras to induce malignant transformation is, in part, dependent on these high levels of survivin.
Literature	GENERIF_SUMMARY	Data indicate that siRNA down-regulation of survivin retarded tumor growth in cervical cancer mouse xenograft.
Literature	GENERIF_SUMMARY	Functional interaction between PIN1 and survivin may therefore play an important role in hepatocarcinogenesis and chemoresistance.
Literature	GENERIF_SUMMARY	autophagy induced by harmol represented a pro-apoptotic mechanism, and harmol suppressed the expression of survivin and subsequently induced apoptosis.
Literature	GENERIF_SUMMARY	show that these events can be at least in part attributed to the STAT3/Survivin pathway
Literature	GENERIF_SUMMARY	Survivin protein expression is an independent molecular marker for predicting epithelial ovarian cancer prognosis. shRNA targeting survivin can effectively suppress survivin expression, enhance apoptosis, and increase the sensitivity of ovarian cancer cells to paclitaxel.
Literature	GENERIF_SUMMARY	Survivin was highly expressed in tumour cell nuclei in all samples of malignant pleural mesothelioma and this correlated positively with both apoptosis and proliferation, but did not have a significant prognostic value
Literature	GENERIF_SUMMARY	We report for the first time that survivin-3alpha is expressed in benign and malignant breat tumors with no expression in normal adjacent tissues.
Literature	GENERIF_SUMMARY	The survivin and eRF3 complex may function in spindle formation and segregation of chromosomes and cytokinesis.
Literature	GENERIF_SUMMARY	Our findings suggest that miR-16 represses colorectal cancer cell growth in vitro by regulating the p53/survivin signaling pathway.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	results from the current meta-analysis suggests that survivin -31G>C polymorphism might increase the risk of gastrointestinal tract cancer, especially among gastric and colorectal cancers
Literature	GENERIF_SUMMARY	High Survivin is associated with lymph node and distant metastases of colorectal cancer.
Literature	GENERIF_SUMMARY	the survivin (A9194G) polymorphism was found to play a protective role in the susceptibility to papillary thyroid carcinoma
Literature	GENERIF_SUMMARY	Results indicate that most primary hepatoblastoma tissue specimens express survivin. Expression increased after chemotherapy, suggesting that survivin may be important for survival of tumor cells.
Literature	GENERIF_SUMMARY	Up-regulation of survivin during immortalization of human myofibroblasts is linked to repression of tumor suppressor p16(INK4a) protein and confers resistance to oxidative stress.
Literature	GENERIF_SUMMARY	The simultaneous RNA interferencemediated downregulation of survivin and VEGF expression inhibited proliferation and induced the apoptosis of Panc1 cells and human umbilical vein endothelial cells.
Literature	GENERIF_SUMMARY	Treatment of transplanted laryngeal squamous cell carcinoma with Gold Throat Tablets reduced expression of survivin and PTEN.
Literature	GENERIF_SUMMARY	Our data showed that individuals carrying the survivin 31 GC genotype and those carrying the survivin 644 CC genotype had a significantly decreased risk of having non-small cell lung cancer.
Literature	GENERIF_SUMMARY	Levels of survivin and mRNA expression rates and its four splice variants are elevated in colorectal cancer tissues.
Literature	GENERIF_SUMMARY	Survivin expression could correlate as an independent predictor of recurrence, while it could also be a potential bio-target gene of angiogenesis in sacral chordoma.
Literature	GENERIF_SUMMARY	results suggest that survivin plays an important role in the response to radiotherapy and may be a useful marker for predicting radiotherapy response in patients with head and neck squamous cell carcinoma
Literature	GENERIF_SUMMARY	beta-catenin is delocalized and survivin is over-expressed in recurrent sporadic and nevoid basal cell carcinoma syndrome-associated keratocystic odontogenic tumor
Literature	GENERIF_SUMMARY	Silibinin induced the apoptosis of Hep-2 cells via oxidative stress and down-regulating survivin expression.
Literature	GENERIF_SUMMARY	Survivin 31G/C may adjust the Survivin expression, and it might contribute to a risk of developing colorectal cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Cases of adenoid cystic carcinoma of the lacrimal gland with high nuclear BIRC5 levels had a shorter progression-free survival.
Literature	GENERIF_SUMMARY	overexpression of survivin and VEGF accompany a higher risk of BTCC recurrence, making survivin and VEGF biomarkers for predicting the relapse of bladder cancer.
Literature	GENERIF_SUMMARY	Data show that YM155 induces apoptosis in leukemia cells through downregulation of survivin and Mcl-1, and that caspase-8 may play a critical role during apoptosis.
Literature	GENERIF_SUMMARY	survivin expression is regulated by ERK and AKT signaling for metastatic progression of colorectal cancer
Literature	GENERIF_SUMMARY	These findings support a direct role for Survivin in melanoma metastasis, which requires alpha5 integrin and suggest that inhibitors of alpha5 integrin may be useful in combating this process
Literature	GENERIF_SUMMARY	Angiopoietin 2-mediated signaling via survivin/ref-1/MSK-1 pathway promotes doxorubicin resistance in HepG2 cells.
Literature	GENERIF_SUMMARY	BIRC5-31CC and CASP9+83CT/TT genotypes were associated with an increased risk for renal cell carcinoma development in the female group of our southern European study population.
Literature	GENERIF_SUMMARY	These findings indicate that bFGF could rapidly activate the protein kinase B activities, enhance the expression of survivin and the proliferation of hepatocellular carcinoma cells via the PI3K pathway.
Literature	GENERIF_SUMMARY	High survivin expression is associated with high HPV infections in precancerous cervical lesions.
Literature	GENERIF_SUMMARY	results suggested that high survivin expression had an unfavorable prognostic role for patients with breast cancer [meta-analysis]
Literature	GENERIF_SUMMARY	Data indicate that MAGE3, Survivin and B-cell maturation antigen (BCMA) mRNA-pulsed dendritic cells (DCs) are capable of stimulating tumor-associated antigens (TAA)-specific T-cell responses in multiple myeloma (MM) patients.
Literature	GENERIF_SUMMARY	Data support biological model suggesting over-expression of survivin/BIRC5 in endometriosis serves as protective mechanism that defends pancreatic beta-cells from apoptosis during Th1-cell (helper-inducer T-lymphocytes) autoimmune assault. [LETTER]
Literature	GENERIF_SUMMARY	MiR203 regulates the proliferation, apoptosis and cell cycle progression of pancreatic cancer cells by targeting Survivin.
Literature	GENERIF_SUMMARY	Report a new pathway of glioma radioresistance via Report the regulation of radiation-induced centrosome duplication and therefore mitotic cell death by ILK,

Main Category	Category	Terms
		HIF-1alpha and survivin.
Literature	GENERIF_SUMMARY	the combination of simultaneous siRNA-mediated knockdown of antiapoptotic Bcl-xL and survivin-a multitarget molecular-based therapy-and conventional chemotherapy shows great potential for improving bladder cancer treatment.
Literature	GENERIF_SUMMARY	Expression of survivin in colorectal cancer patients indicates poor prognosis and a pro-metastatic tumor phenotype. (Meta-analysis)
Literature	GENERIF_SUMMARY	Cytoplasmic survivin is a critical downstream effector of the PI3K/Akt pathway and a more favorable outcome in acute myeloid leukemia.
Literature	GENERIF_SUMMARY	Combinatorial expression of WT1, survivin, and TERT determine the survival outcome in adult acute myeloid leukemia.
Literature	GENERIF_SUMMARY	TGF-beta1 regulates the cell cycle via survivin.
Literature	GENERIF_SUMMARY	The results of this study indicated that the loss of survivin due to Helicobacter pylori infection involves proteasome-mediated degradation of the protein induced by secreted bacterial gamma-glutamyl transpeptidase.
Literature	GENERIF_SUMMARY	Overexpression of alternative splice isoform survivin-3B is associated with tumorigenesis by facilitating immune escape in a manner associated with resistance to immune cell toxicity.
Literature	GENERIF_SUMMARY	In Serbian population, -31G/C polymorphism in the promoter of the survivin gene cannot be considered as a risk factor for oral squamous cell carcinoma and skin basal cell carcinoma.
Literature	GENERIF_SUMMARY	Functional promoter -31G/C variant of Survivin gene predict prostate cancer susceptibility among Chinese.
Literature	GENERIF_SUMMARY	Survivin is overexpressed in blood and tissue of patients with non-small cell lung cancer and is associated with histological type and tumor grade.
Literature	GENERIF_SUMMARY	Survivin BIR-domain mutant Cys84Ala was transduced by adeno-associated viral vectors into mice with a xenograft human gastric cancer. This increased apoptosis & 5-FU sensitivity & inhibited cancer growth in vivo.
Literature	GENERIF_SUMMARY	High survivin expression is associated with gastric cancer.
Literature	GENERIF_SUMMARY	The present review explores the role of post-translational modifications on protein-protein interactions focused on BIRC5 to provide new insights into the functions and cell localization of this BIRC5 in pathophysiological conditions. [review]
Literature	GENERIF_SUMMARY	A correlation between the expression of survivin and overall survival for patients with gastric cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data suggest that survivin is a key mediator of cytoprotection in acute lung injury (as seen in adult respiratory distress syndrome); survivin appears to act at epithelial/mucosal cell level; action of survivin depends partly on apoptosis inhibition.
Literature	GENERIF_SUMMARY	The survivin -31G>C polymorphism is associated with elevated cancer risk.
Literature	GENERIF_SUMMARY	In this updated meta-analysis of 26 studies, we conclude that the BIRC5 rs9904341 polymorphism might contribute to risk of various cancers, especially in Asian populations.
Literature	GENERIF_SUMMARY	The survivin rs9904341 most likely contributed to increased susceptibility to tumor in Asians as well as to gastric, colorectal and bladder cancers.
Literature	GENERIF_SUMMARY	Upregulation of survivin expression may be a molecular marker of dedifferentiation in thyroid epithelial carcinomas, likely being responsible for survival responses of tumor cells and, thus, favoring progression toward a poorly differentiated phenotype.
Literature	GENERIF_SUMMARY	BIRC5 splice variant expression correlates with refractory disease in pediatric acute myeloid leukemia.
Literature	GENERIF_SUMMARY	CR108 is a novel vitamin K3 derivative that induces apoptosis and tumor inhibition by ROS production and mitochondrial dysfunction and associates with the phosphorylation of p38 MAP kinase and the inhibition of survivin in the human breast cancer.
Literature	GENERIF_SUMMARY	silencing of survivin and over-expression of GRIM-19 can significantly inhibit the growth and induce the apoptosis of Hep-2 laryngeal cancer cells in vitro and in vivo.
Literature	GENERIF_SUMMARY	Up-regulation of survivin by AKT and hypoxia-inducible factor 1alpha contributes to cisplatin resistance in gastric cancer.
Literature	GENERIF_SUMMARY	The regulation of survivin mRNA expression. [Review]
Literature	GENERIF_SUMMARY	survivin is an unfavorable molecule for papillary thyroid carcinoma
Literature	GENERIF_SUMMARY	Studies indicate that survivin expression indicates worse prognosis in patients with bladder cancer but the results should be interpreted with caution.
Literature	GENERIF_SUMMARY	Our results indicate that indomethacin can potentiate TRAIL-induced apoptosis through upregulation of death receptors and downregulation of survivin.
Literature	GENERIF_SUMMARY	Cigarette smoking is associated with the higher nuclear survivin expression in lung adenocarcinomas at the early stage.
Literature	GENERIF_SUMMARY	Expression of NgBR has a strong correlation with the expression of survivin, which is a well-known apoptosis inhibitor, in breast cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Higher expression of survivin in tumor tissue was associated with increased risk of death; while surviving expression in adjacent tissue was a superior predictor of recurrence.
Literature	GENERIF_SUMMARY	survivin and EGFR expression are useful prognostic markers of triple-negative breast cancer
Literature	GENERIF_SUMMARY	The present studies aim to examine whether and how cilia function (Pkd1 or Pkd2) and structure (Tg737) play a role in cystic kidney and aneurysm through survivin downregulation.
Literature	GENERIF_SUMMARY	Data indicate that verapamil suppresses MDR1 and survivin expressions and increases Bim expression via suppression of Src activation.
Literature	GENERIF_SUMMARY	Suggest that levels of nuclear and cytoplasmic survivin expression differ between gastric carcinoma and colorectal carcinoma.
Literature	GENERIF_SUMMARY	these data support the concept that survivin can be regarded as a useful prognostic marker for pediatric MPNST and a promising target for therapeutic interventions.
Literature	GENERIF_SUMMARY	The present study indicated the significance of Smac and survivin in determining the breast cancer response to anthracyclinebased chemotherapy, and may permit further stratifying of prechemotherapy patients to undertake more tailored treatments.
Literature	GENERIF_SUMMARY	Survivin variants are associated with aggressive breast cancer.
Literature	GENERIF_SUMMARY	Survivin may be a regulator of VEGFC expression in gastric cancer cells, and is essential in invasion and lymphatic metastasis.
Literature	GENERIF_SUMMARY	survivin and VEGF are over-expressed in small-cell lung cancer and may have a role in tumorigenesis
Literature	GENERIF_SUMMARY	Cell free BIRC5 mRNA levels were significantly increased in serum of colorectal cancer. A significantly higher diagnosis capability was showed when combined BIRC5 mRNA and CEA
Literature	GENERIF_SUMMARY	survivin expression was correlated with poor prognosis in patients with hepatocellular carcinoma, regardless whether they were assessed by overall survival or disease-free survival.
Literature	GENERIF_SUMMARY	Survivin expression may serve as a biomarker to predict treatment efficacy in hepatocellular carcinoma
Literature	GENERIF_SUMMARY	has a dual cellular function
Literature	GENERIF_SUMMARY	The ectopic expression of survivin significantly promoted the invasive activity of colon carcinoma cells and resulted in increased levels of MMP7 activation.
Literature	GENERIF_SUMMARY	Here, we critically review the current data on YM155

Main Category	Category	Terms
		and other new experimental agents supposed to antagonize survivin
Literature	GENERIF_SUMMARY	Report no correlation between survivin expression and prognosis in epithelial ovarian carcinoma.
Literature	GENERIF_SUMMARY	Survivin expression did not have prognostic significance in synovial sarcoma.
Literature	GENERIF_SUMMARY	siRNA-mediated silencing of survivin inhibits proliferation and enhances etoposide chemosensitivity in acute myeloid leukemia.
Literature	GENERIF_SUMMARY	LANA of Human herpesvirus 8interacts with Aurora kinase B and induces phosphorylation of survivin at residue T34.
Literature	GENERIF_SUMMARY	Proto-oncogene survivin was increased in individuals prior to onset of symptoms of rheumatoid arthritis and was correlated to cytokines suggesting its role at pre-clinical stages of the disease
Literature	GENERIF_SUMMARY	Data show that knockdown of Yin Yang 1 (YY1) releases the survivin promoter from the observed repression and leads to a 3- to 5-fold increase in promoter activity above basal levels.
Literature	GENERIF_SUMMARY	downregulation of STAT3 or Survivin can suppress the proliferation of T24 bladder cancer cells.
Literature	GENERIF_SUMMARY	Survivin-2B promoted autophagy and further regulated cell death by accumulating and stabilizing IKK alpha in the nucleus of selenite treated leukemia cells.
Literature	GENERIF_SUMMARY	survivin is expressed in blood and may have a role in response to EGFR-tyrosine kinase inhibitors and prognosis in patients with non-small cell lung cancer
Literature	GENERIF_SUMMARY	High nuclear survivin expression in stromal cells is associated with phyllodes tumors and fibroadenomas of the breast.
Literature	GENERIF_SUMMARY	Cyclin D1 and survivin may be used as molecular markers to predict the sensitivity of radiotherapy.
Literature	GENERIF_SUMMARY	survivin protein has a role in the proliferative and chemoresistant compartment in chronic lymphocytic leukemia
Literature	GENERIF_SUMMARY	Study shows that like Survivin, the Survivin splice variants are also exosomally packaged in the breast cancer patients' sera, mimicking the survivin splice variant pattern reported in breast cancer tissues.
Literature	GENERIF_SUMMARY	RUNX3-binding site in the promoter of survivin gene is identified. This leads to significantly inhibition of survivin expression. Negative correlation of RUNX3 and survivin expression in clinical specimens of gastric cancer exists.
Literature	GENERIF_SUMMARY	HIF-1alpha/PTEN/CD44v6/Survivin, as measured by immunohistochemistry, can be used to predict the

Main Category	Category	Terms
		prognosis and potential for invasion and metastasis in Asian patients with gastric cancer.
Literature	GENERIF_SUMMARY	Survivin expression is associated with tumor stage and grade and may present a predictive marker of overall survival in urothelial carcinoma of the bladder.
Literature	GENERIF_SUMMARY	Low expression of survivin is associated with low response to chemotherapy in epithelial ovarian cancer or primary peritoneal cancer.
Literature	GENERIF_SUMMARY	Survivin expression in low-grade gliomas is associated with poor survival and is preferentially expressed in proneural gliomas.
Literature	GENERIF_SUMMARY	The sensitivity of gastric cancer patients to paclitaxel treatment was inversely correlated with the mRNA and protein expression of BIRC5.
Literature	GENERIF_SUMMARY	Protein expression of PI3K, Akt and survivin were significantly associated with the development, progression and metastasis of gastric cancer and may have value as diagnostic and prognostic markers in gastric cancer
Literature	GENERIF_SUMMARY	Data indicate that Go6976 displays a significant antiproliferative activity against leukemia cells through the profound down-regulation of survivin and MCL-1.
Literature	GENERIF_SUMMARY	these data identified combined expression levels of Aldh1, Survivin, and EpCAM as strong independent prognostic factors, with high hazard ratios, for survival and tumour recurrence in colorectal cancer patients, and therefore reflect tumour aggressiveness.
Literature	GENERIF_SUMMARY	The expression of PIK3CA and survivin had a significant correlation with pathological grade and lymph node metastasis in the primary adenosquamous lung carcinoma patients.
Literature	GENERIF_SUMMARY	Overexpression of survivin but reduced HLA-I expression is useful in the prediction of tumor-free survival of clear cell renal cell carcinoma patients
Literature	GENERIF_SUMMARY	Mismatch repair proteins may suppress the antiapoptotic function of survivin through p53 inactivation of its promoter in grade 1 and grade 2 colon carcinomas.
Literature	GENERIF_SUMMARY	Survivin plays an essential role in proper amphitelic kinetochore-spindle assembly; constraining Survivin's mitotic function results in polyploidy and aneuploidy which cannot be controlled by p53.
Literature	GENERIF_SUMMARY	Hepatitis B protein HBx accelerates hepatocarcinogenesis with partner survivin through modulating tumor suppressor miR-520b and oncoprotein HBXIP.
Literature	GENERIF_SUMMARY	Survivin family proteins are novel effectors of doxorubicin resistance in chemotherapy-naive breast cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	survivin contributes to the maintenance of pluripotency and post-transcriptional control of survivin isoform expression is selectively regulated by microRNAs.
Literature	GENERIF_SUMMARY	Results indicate that PKCzeta regulates survivin expression levels and inhibits apoptosis in colon cancer cells.
Literature	GENERIF_SUMMARY	Expression of AR significantly impacts prognosis and is thus promising prognostic marker in periocular SGC.
Literature	GENERIF_SUMMARY	Data indicate that expression of survivin in peripheral blood could be useful both in the diagnosis and prognosis of gallbladder cancer (GBC).
Literature	GENERIF_SUMMARY	The results suggest that survivin, especially survivin delta Ex3 splice variant being overexpress, is a characteristic feature of thyroid malignancy.
Literature	GENERIF_SUMMARY	we show that survivin is essential for melanoma survival in vitro and in vivo
Literature	GENERIF_SUMMARY	Tumor N-survivin expression is a predictive biomarker in the chemotherapy setting as a surrogate marker of high proliferation status in lung cancer.
Literature	GENERIF_SUMMARY	transfecting mTOR siRNA into A549 cells significantly decreased the expression of hypoxia-inducible factor 1alpha (HIF-1alpha) and survivin
Literature	GENERIF_SUMMARY	The overexpressions of both Survivin and NF-kappaB are associated with worse survival outcome in patients with hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	Our study suggests that an autophagy signal may be activated after the anti-apoptotic molecule survivin is suppressed in hepatocellular carcinoma cells
Literature	GENERIF_SUMMARY	variations in Sp1 binding sites may be associated with survivin expression in breast cancer
Literature	GENERIF_SUMMARY	The BIRC5 gene alternate splice variant survivin 2-alpha is expressed in thyroid nodules.
Literature	GENERIF_SUMMARY	vimentin, survivin and p53 may have a role in clear cell renal cell carcinoma
Literature	GENERIF_SUMMARY	MEKK3 expression was positively correlated with survivin.
Literature	GENERIF_SUMMARY	Downregulating Survivin using siRNA copied the phenotype of miR-542-3p expression in neuroblastoma cell lines, while cDNA-mediated ectopic expression of Survivin partially rescued the phenotype induced by miR-542-3p expression
Literature	GENERIF_SUMMARY	Survivin expression may be part of a mechanism that inhibits the apoptosis of autoreactive B cells in myasthenia gravis and other autoimmune disorders.
Literature	GENERIF_SUMMARY	Analysis indicates that the genes BIRC5, HOXA1 and RARB are critical targets that play an important

Main Category	Category	Terms
		regulatory role in cervical cancer pathogenesis.
Literature	GENERIF_SUMMARY	Survival time was associated with survivin expression in pancreas ductal adenocarcinoma.
Literature	GENERIF_SUMMARY	The expressions of HER-2/neu (c-erbB2), survivin and cycline D1 are positively correlated with the malignant potential of serous ovarian neoplasms.
Literature	GENERIF_SUMMARY	No significant differences between survivin and its splice variants Ex3 and 2beta expression in pituitary tumors and in normal pituitary glands as well as in invasive and in non-invasive tumors were found.
Literature	GENERIF_SUMMARY	Studied survivin protein expression in serum of patients with gallbladder diseases (cholelithiasis, n = 30; GBC, n = 39) and compared with healthy controls.
Literature	GENERIF_SUMMARY	Data suggest that expression of survivin is up-regulated in adenomyosis as compared to normal endometrium; expression of PP2A (protein phosphatase 2 catalytic subunit alpha) and PTEN (phosphatase and tensin homolog) appear to be down-regulated.
Literature	GENERIF_SUMMARY	there was no statistically significant relationship between the rates of cytoplasmic and nuclear survivin expression with identified prognostic factors and total or non-disease survival.
Literature	GENERIF_SUMMARY	Data indicate that siRNA-survivin promotes gastric cancer cell apoptosis and inhibits cell proliferation by downregulating survivin mRNA and protein expression.
Literature	GENERIF_SUMMARY	Decreased plasma survivin levels in pregnancy (compared to nonpregnant healthy controls) but not in asthmatic pregnancy (compared to nonpregnant asthmatic controls) suggest that normal apoptotic mechanisms are compromised in asthmatic gestation.
Literature	GENERIF_SUMMARY	Co-silencing of Livin and Survivin can effectively inhibit the cell proliferation and apoptosis of lung cancer cells.
Literature	GENERIF_SUMMARY	data suggest that elevated expression of AURKA and SURVIVIN, together with AURKA gene amplification, are associated with increased chromosomal instability in breast cancer
Literature	GENERIF_SUMMARY	In refractory aggressive non-Hodgkin's lymphoma, surviving is associated with low overall response and shorter overall survival.
Literature	GENERIF_SUMMARY	Survivin silencing enhances sensitivity of oral squamous cell carcinoma cells to radiotherapy.
Literature	GENERIF_SUMMARY	Silencing survivin expression inhibits the tumor growth of non-small-cell lung cancer cells in vitro and in vivo.
Literature	GENERIF_SUMMARY	Pleural survivin plays a role in the diagnosis of malignant pleural effusion.
Literature	GENERIF_SUMMARY	The -241 C/T (rs17878467) polymorphism in the promoter region is associated with a lower risk of

Main Category	Category	Terms
		hemorrhagic transformation in patients with ischemic stroke.
Literature	GENERIF_SUMMARY	Survivin -31 G/C and -241 C/T polymorphisms are associated with colon cancer development.
Literature	GENERIF_SUMMARY	Pancreatic ductal adenocarcinoma patients with elevated serum survivin levels demonstrated poor overall survival.
Literature	GENERIF_SUMMARY	these results demonstrated that irradiation could regulate glioblastomas cell dedifferentiation via a survivin-dependent pathway.
Literature	GENERIF_SUMMARY	this study documented down-regulation of galectin-3 and Bcl-2 (antiapoptotic molecules) and stepwise increase of survivin (inhibitor of apoptosis), during thyroid tumor progression from PTC to ATC.
Literature	GENERIF_SUMMARY	Data indicate that survivin is an important target of microRNA-218 (miR-218) in cervical cancer cells.
Literature	GENERIF_SUMMARY	Serum survivin and VEGF expression can predict survival in a nasal type extranodal NK/T-cell lymphoma.
Literature	GENERIF_SUMMARY	Patients that presented at the diagnosis a survinin gene expression >/= 1.05 are 17 times more likely to develop metastatic disease.
Literature	GENERIF_SUMMARY	Preoperative measurement of serum VEGF, survivin, and Smac/DIABLO may be of help in early detection of serous ovarian cancer and may provide important information about the patient's outcome and prognosis.
Literature	GENERIF_SUMMARY	Unlike in transitional cell carcinoma, alteration of survivin expression in squamous cell carcinoma occurs less frequently and is not associated with features of tumor aggression or patient outcome
Literature	GENERIF_SUMMARY	Selective inhibition of HDAC2 in lung tumor cell causes survivin downregulation through activation of p53, which is mediated by downregulation of Mdm2.
Literature	GENERIF_SUMMARY	The last 22 amino acids of survivin are required to bind to borealin and support mitosis and cytokinesis, whereas the first 10 amino acids are needed to protect cells from irradiation.
Literature	GENERIF_SUMMARY	Survivin affected the stability of cIAP1 through binding, contributing to cell sensitivity to antineoplastic drug YM155.
Literature	GENERIF_SUMMARY	cell cycle-progressive role in chordomas
Literature	GENERIF_SUMMARY	Multisite mutation mutants of survivin remained as monomers under ambient conditions, and induced cells (breast cancer Bcap-37 cells) apoptosis even more efficiently.
Literature	GENERIF_SUMMARY	Survivin (-31CC) genotype was associated with significantly increased survivin gene expression and ultimately may contribute in the poor clinical outcome

Main Category	Category	Terms
		of non-small cell lung cancer patients.
Literature	GENERIF_SUMMARY	LapR cells possessed increased levels of 2 of the inhibitors of apoptosis (IAPs), survivin and c-IAP-2, which are reported to block caspase activation downstream of cytosolic cytochrome C release.
Literature	GENERIF_SUMMARY	survivin may have a role in radioresistance of laryngeal squamous cell carcinoma
Literature	GENERIF_SUMMARY	these results suggested that simultaneous inhibition of XIAP and survivin may be a promising strategy for the treatment of pancreatic cancer
Literature	GENERIF_SUMMARY	Overexpression of survivin via activation of ERK1/2, Akt, and NF-kappaB is associated with vincristine resistance in multiple myeloma.
Literature	GENERIF_SUMMARY	Survivin modulation influences radiation response differently in estrogen receptor-positive and estrogen receptor-negative breast cancer subtypes, warranting further investigation.
Literature	GENERIF_SUMMARY	Protein expression of BIRC5, TK1, and TOP2A in malignant peripheral nerve sheath tumours--A prognostic test after surgical resection.
Literature	GENERIF_SUMMARY	Patients whose MPM tissues expressed elevated mRNA levels of BIRC5, DSP, NME2, and THBS2 showed a statistically significant shorter overall survival.
Literature	GENERIF_SUMMARY	the Skp1.Cul1.F-box protein complex subunit Fbxl7 modulates mitochondrial function by controlling the cellular abundance of survivin
Literature	GENERIF_SUMMARY	Findings indicate survivin inhibition as an approach to improve renal cell carcinoma (RCC) therapy.
Literature	GENERIF_SUMMARY	Deptor enhances triple-negative breast cancer metastasis and chemoresistance through coupling to survivin expression
Literature	GENERIF_SUMMARY	elevated survivin expression in PBMCs, particularly survivin and CD8, VEGF, and ICAM-1 double-positive PBMCs, may be crucial for good collateral formation in patients with coronary CTO, as confirmed by assessment of a rat model.
Literature	GENERIF_SUMMARY	Overexpression of the survivin gene plays an important role in onset and development of nasopharyngeal carcinoma.
Literature	GENERIF_SUMMARY	siRNA-mediated survivin gene silencing induces nasopharyngeal carcinoma cell apoptosis.
Literature	GENERIF_SUMMARY	MGMT inhibition is associated with decrease in survivin expression and increase in sensitivity to gemcitabine in pancreatic cancer.
Literature	GENERIF_SUMMARY	Survivin -31 G/C polymorphism is associated with the risk of colorectal cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	ERK/MSK/Sp1/c-Myc axis functioned as a critical regulator of survivin expression.
Literature	GENERIF_SUMMARY	Decrease of survivin expression and induction of PUMA expression.
Literature	GENERIF_SUMMARY	In this meta-analysis, survivin overexpression is associated with poor prognosis in esophageal cancer patients.
Literature	GENERIF_SUMMARY	miR-218 targets survivin and regulates resistance to chemotherapeutics in breast cancer
Literature	GENERIF_SUMMARY	p16 and survivin expression might be correlated with the clinico-pathological and prognosis of cervical cancer
Literature	GENERIF_SUMMARY	MiR-205/miR-218 expression is associated with carboplatin chemoresistance and regulation of apoptosis via Mcl-1 and survivin signaling in lung cancer cells.
Literature	GENERIF_SUMMARY	Mitotic activity of survivin is regulated by acetylation at K129.
Literature	GENERIF_SUMMARY	Overall, we have demonstrated a marked efficacy of SINE in DMPM preclinical models that may relay on the interference with survivin intracellular distribution and function.
Literature	GENERIF_SUMMARY	Conversely, we found a positive relationship between clinical stage and nuclear survivin LI, which is considered to be useful in deciding treatment strategies.
Literature	GENERIF_SUMMARY	Data show that silencing of alpha-fetoprotein (AFP) expression in HepG2 cells can effectively inhibit the growth of hepatoma cells and promote apoptosis, which may be useful for reducing intracellular Survivin mRNA levels.
Literature	GENERIF_SUMMARY	our results suggest that survivin play an important role in gastric cancer
Literature	GENERIF_SUMMARY	Survivin expression and chemoresistance were increased in CD133+ colon cancer cells.
Literature	GENERIF_SUMMARY	High Survivin expression and serum levels are associated with pancreatic cancer.
Literature	GENERIF_SUMMARY	In prostate cancer the immunoexpression of survivin is augmented and positively correlated with parameters of tumor aggressiveness.
Literature	GENERIF_SUMMARY	targeting survivin by YM155 can benefit head neck squamous cell carcinoma therapy by increasing apoptotic and autophagic cell death, and suppressing prosurvival pathways.
Literature	GENERIF_SUMMARY	Ectopic expression of survivin in U937 cells attenuated tunicamycin-induced apoptosis by suppression of caspase-3 cleavage, mitochondrial membrane potential,

Main Category	Category	Terms
		and cytochrome c release in U937 cells.
Literature	GENERIF_SUMMARY	Our results indicated that aberrant high cytoplasmic survivin expression in gastric cancer cells is associated with increased proliferation index and tumor growth
Literature	GENERIF_SUMMARY	Simultaneous silencing of NET-1 and survivin using one-chain-double-target siRNA thus provides an advantageous alternative in the development of therapeutics for skin squamous cell carcinoma .
Literature	GENERIF_SUMMARY	Molecular docking computations also show that the studied compounds can bind to BIR domain of Survivin in the same binding site as that of Smac/DIABLO with a suitable binding energy.
Literature	GENERIF_SUMMARY	This review discusses the complex circuitry of survivin in human cancers and gene variants of survivin, and highlights novel therapy that targets this important protein.
Literature	GENERIF_SUMMARY	may play an important role in the development of colorectal cancer
Literature	GENERIF_SUMMARY	Survivin expression was higher in condyloma acuminatum than in normal cells and was correlated with livin and Ki-67.
Literature	GENERIF_SUMMARY	Using multivariate logistic regression analysis, the CC genotype of survivin was found to be associated with a significantly increased risk of pancreatic cancer risk compared with the GG genotype
Literature	GENERIF_SUMMARY	selective activity of 2DG treatment on Survivin-overexpressing tumor cells was also confirmed in a xenograft mouse model, which further supports our hypothesis that glycolysis inhibitors might be useful drugs in the treatment of NB
Literature	GENERIF_SUMMARY	Data show that SIRT6, a member of the sirtuin family, induces apoptosis, and its induction is attributed to the repression of survivin protein, which could be a target for endometrial cancer treatment.
Literature	GENERIF_SUMMARY	miR-204 targets JAK2 and suppressed JAK2 and p-JAK2 expression in breast cancer, which further inhibit the activation of STAT3, BCl-2 and survivin.
Literature	GENERIF_SUMMARY	Findings suggest that overexpression of Class III beta-tubulin, Sox2, and nuclear Survivin might be predictive of taxane resistance and poor progression-free survival in patients with stage III ovarian epithelial cancer.
Literature	GENERIF_SUMMARY	Circulating tumor cell CK20 and survivin expression may effectively predict overall survival in metastic colorectal cancer patients receiving chemotherapy.
Literature	GENERIF_SUMMARY	survivin and its splice variants hold a great biological significance, particularly in the diagnosis of cancer.
Literature	GENERIF_SUMMARY	Antisense microRNA targeting survivin induced apoptosis and prevented cell proliferation and migration

Main Category	Category	Terms
		in rectal tumor cells.
Literature	GENERIF_SUMMARY	Survivin -31G/C and 9194A/G polymorphisms were associated with the risk of lung cancer
Literature	GENERIF_SUMMARY	mitochondrial respiration enhanced by survivin contributes to cancer metabolism, and relocalized mitochondria may provide a ""regional"" energy source to fuel tumor cell invasion and metastasis."
Literature	GENERIF_SUMMARY	IGF-1 signaling activated survivin expression and controlled the expression of EMT biomarkers in the development of GC.
Literature	GENERIF_SUMMARY	MiR-335 and miR-34a can also regulate GC cell growth, apoptosis, and invasion at least partly through survivin.
Literature	GENERIF_SUMMARY	Data show that molecular chaperone GRP78 and survivin cooperatively conferred multi-drug resistance (MDR) by decreasing folate deficiency (FD)-induced reactive oxygen species (ROS) generation
Literature	GENERIF_SUMMARY	Results showed that survivin expression was higher in esophageal cancer and associated with poor prognosis.
Literature	GENERIF_SUMMARY	There is significant correlation between survivin expression with poor prognosis in Diffuse Large B-cell Lymphoma.
Literature	GENERIF_SUMMARY	Reduced levels of survivin can sensitize tumor cells to apoptosis.
Literature	GENERIF_SUMMARY	Overexpression of FOXM1, XIAP, and Survivin contributes to the development of drug-resistance and is associated with poor clinical outcome in breast cancer patients.
Literature	GENERIF_SUMMARY	survivin can function as a new diagnostic biomarker for osteosarcoma and be used as a reference index to determine pathology classification of osteosarcoma
Literature	GENERIF_SUMMARY	Findings suggest that dermal survivin+CD1a+ (CD1a antigen) cell infiltration may be a potential biomarker of systemic sclerosis skin lesions.
Literature	GENERIF_SUMMARY	A decrease of survivin levels during treatment is associated with better clinical responsesin active rheumatoid arthritis.
Literature	GENERIF_SUMMARY	MiR-218 promoted apoptosis, inhibited cell proliferation and caused cell cycle arrest in colorectal cancer cells by suppressing BIRC5 expression.
Literature	GENERIF_SUMMARY	IWR-1 has the potential to suppress tumor metastasis by inhibiting Wnt/beta-catenin pathway as well as survivin expression
Literature	GENERIF_SUMMARY	results suggest that high-grade glioma expressing high level of alpha5beta1 integrin may benefit from associated therapies including integrin antagonists and repressors of survivin expression.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	the Notch1/survivin crosstalk contributes to the maintenance of stemness in human keratinocytes.
Literature	GENERIF_SUMMARY	Suppression of survivin with specific inhibitor YM155 significantly increased the susceptibility of KG1a cells to BUS.
Literature	GENERIF_SUMMARY	Suggest that the p53-p21-DREAM-CDE/CHR pathway regulates p53-dependent repression of Survivin, CDC25C, and PLK1 in HCT116 cells.
Literature	GENERIF_SUMMARY	Survivin and cyclin E2 genes expression may have clinical relevance and can be considered as molecular risk factors for AL. Also they may be useful as predictive markers for treatment outcome in leukemic patients.
Literature	GENERIF_SUMMARY	Survivin, p53, and beta-catenin overexpression, as prognostic markers, might suggest that the Urothelial carcinomas are biologically aggressive with the poor prognosis
Literature	GENERIF_SUMMARY	we discovered a novel mechanism mediated by Smad4 to trigger 5-FU chemosensitivity through cell cycle arrest by inhibiting the PI3K/Akt/CDC2/survivin cascade.
Literature	GENERIF_SUMMARY	Studies on survivin in NP cells may aid in increasing the understanding of the complex processes underlying NP cell degeneration, and could provide fundamental information for gene therapy to inhibit this degeneration in vitro
Literature	GENERIF_SUMMARY	BIRC5 -31C>G polymorphism could be useful pharmacogenetic marker for methotrexate treatment response in rheumatoid arthritis
Literature	GENERIF_SUMMARY	results indicate that survivin functions as a common mediator of gemcitabine- and TRAIL-induced cell death.
Literature	GENERIF_SUMMARY	We defined Survivin as a constitutive actor of HR in breast cancers, and implies that its inhibition would enhance cell vulnerability upon PARP inhibition.
Literature	GENERIF_SUMMARY	Survivin, Bcl-2, and KAI1 are metastasis-related factors in cervical cancer. Overexpression of survivin and Bcl-2, and low expression of KAI1 promotes cervical cancer progress and metastasis.
Literature	GENERIF_SUMMARY	High surviving expression is associated with adenocarcinoma in non-small cell lung cancer.
Literature	GENERIF_SUMMARY	The apoptosisinducing effects of livin and surivin knockdown were investigated using a Hoechst staining kit.
Literature	GENERIF_SUMMARY	Survivin overexpression activates NFkappaB p65, which is important in the acquisition and maintenance of the oncogenic characteristics of esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Overexpression of miR494 and silencing survivin RNA

Main Category	Category	Terms
		through the use of survivin shRNA inhibited the expression of survivin and attenuated PC3 cell growth in vitro and in vivo.
Literature	GENERIF_SUMMARY	MiR-542-3p and its target gene survivin may play crucial roles in the aggressive progression of human bladder cancer.
Literature	GENERIF_SUMMARY	High survivin expression is associated with radioresistance in HER2-positive breast cancers.
Literature	GENERIF_SUMMARY	High survivin expression is associated with gastric cancer.
Literature	GENERIF_SUMMARY	Survivin is a key gene for regulating squamous cell carcinoma (SCC) cancer stem cell formation and cancer SCC development.
Literature	GENERIF_SUMMARY	The expressions of survivin and cycline D1 are positively correlated with the malignant potential of mucinous ovarian neoplasms.
Literature	GENERIF_SUMMARY	Expression of livin, survivin and caspase-3 are closely related to the occurrence and development of prostatic cancer.
Literature	GENERIF_SUMMARY	It also inhibits the mRNA levels of Survivin gene.
Literature	GENERIF_SUMMARY	Survivin is a key regulator of gut tissue integrity by regulating epithelial homeostasis in the stem cell niche.
Literature	GENERIF_SUMMARY	Data show that down regulation of survivin gene in HO8910PM cells with small interference RNA (siRNA) resulted in increased sensitivity to S-allylmercaptocysteine (SAMC) together with a decrease in invasiveness of tumor cells.
Literature	GENERIF_SUMMARY	Nuclear accumulation of YAP could promote the survival of senescent cells by increasing survivin expression.
Literature	GENERIF_SUMMARY	this study shows that c-IAP-2 may prevent HIV-viral protein -mediated mitochondrial depolarization through stabilizing TRAF-1/2 expression and sequential inhibition of caspase-8, Bid, and Bax
Literature	GENERIF_SUMMARY	Expression of Caspase 3, including that of survivin, was significantly more frequently detected in the patients who had received combined cytotoxic and anti-VEGF therapy as compared to both those treated with only cytotoxic agents and the control group
Literature	GENERIF_SUMMARY	The aim of the study was to evaluate the caspase-3 and survivin expression in correlation with MIB-1 expression in gliomas of various grade.
Literature	GENERIF_SUMMARY	Survivin and miRNA-16-1 as potential biomarkers.
Literature	GENERIF_SUMMARY	Present study indicated that survivin expression is a significant prognostic factor and that survivin is a promising therapeutic target for endometrial cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Survivin expression was associated with the clinicopathologic characteristics of non-small cell lung cancer (NSCLC) and may serves as an important biomarker for NSCLC progression.
Literature	GENERIF_SUMMARY	Mi335 might function as a tumor suppressor in osteosarcoma(OS), and downregulation of miR-335 in OS cells contributes to the decreased apoptotic potential of OS cells through derepression of survivin.
Literature	GENERIF_SUMMARY	Significantly higher expression of survivin protein in gallbladder cancer as compared to cholelithiasis group suggests its role in gallbladder carcinogenesis though it may not have prognostic value.
Literature	GENERIF_SUMMARY	increased expression of both DNA methytransferase I (DNMT1) and Survivin was observed and significantly correlated with the reduced miR-203 in NSCLC
Literature	GENERIF_SUMMARY	p53 and Bax might play an important role in microcystin-LR (MC-LR)-induced apoptosis in HepG2 cells in which PUMA and survivin were involved.
Literature	GENERIF_SUMMARY	the N-terminus of survivin is a mitochondrial-targeting sequence that regulates Src.
Literature	GENERIF_SUMMARY	Effective internalization of SurR9-C84A in Y79 cells induced cytotoxic effects.
Literature	GENERIF_SUMMARY	Aurora-C interactions with members of the Chromosome Passenger Complex (CPC), Survivin and Inner Centromere Protein (INCENP) in reference to known Aurora-B interactions to understand the functional significance of Aurora-C overexpression in human cancer cells, is reported.
Literature	GENERIF_SUMMARY	A higher survivin expression seems to represent an unfavorable and robust prognostic factor for clinical outcomes in renal cell carcinoma patients [review; meta-analysis].
Literature	GENERIF_SUMMARY	Data show that knockdown of signal transducer and activator of transcription (STAT3)- significantly inhibited the mRNA and protein levels of survivin.
Literature	GENERIF_SUMMARY	BIRC5 gene has the potential to be a marker for the detection and prognosis of cancer at an early age.
Literature	GENERIF_SUMMARY	These findings uncover the significance of Merlin protein expression and Survivin labeling index as prognosticators for poor clinical outcome in two independent Malignant pleural mesothelioma cohorts.
Literature	GENERIF_SUMMARY	survivin may have a role in recurrence in rectal cancer patients treated with surgery and postoperative concurrent chemo-radiation therapy
Literature	GENERIF_SUMMARY	This study aimed to demonstrate the usefulness of mRNA expression levels of BIRC5 in brush cytology specimens combined with cytology for the diagnosis of malignant biliary stricture

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	High expression of survivin is one prognostic factor correlating with low survival rate in glioblastoma.
Literature	GENERIF_SUMMARY	our data suggest the existence of a novel pre-metastatic phase quiescent-breast cancer stem cells (Q-BCSCs) before epithelial to mesenchymal transition and SURVIVIN acts as a marker for Quiescent-BCSCs
Literature	GENERIF_SUMMARY	Studies indicated that high survivin expression in renal cell carcinoma (RCC) was associated with poor overall survival [Review].
Literature	GENERIF_SUMMARY	Increased survivin expression in choledochal cyst tissue correlated with certain clinical parameters of congenital choledochal cysts.
Literature	GENERIF_SUMMARY	Study suggests that BIRC5 may play a role as a protective factor in the development of intracerebral hemorrhage, and also the brain damage inducing after intracerebral hemorrhage
Literature	GENERIF_SUMMARY	Survivin protein expression levels were downregulated in both cell lines.
Literature	GENERIF_SUMMARY	Survivin expression predicted disease-free survival in high grade papillary urothelial carcinomas.
Literature	GENERIF_SUMMARY	results indicate that survivin is of prognostic signi fi cance in patients with renal cell carcinoma
Literature	GENERIF_SUMMARY	Data suggest that KLF4 could promote cell senescence through a complex network
Literature	GENERIF_SUMMARY	Reciprocal regulation of BMF and BIRC5 is linked to Eomes overexpression in colorectal cancer.
Literature	GENERIF_SUMMARY	High survivin expression is associated with ovarian cancer.
Literature	GENERIF_SUMMARY	High survivin expression is associated with lung cancer and colorectal cancer.
Literature	GENERIF_SUMMARY	The high sensitivity, specificity, and accuracy of serum and aqueous humor survivin and TGF-B1 proteins make them promising markers for early detection and follow-up of RB patients.
Literature	GENERIF_SUMMARY	Survivin was regulated by TGF-beta and was found to be highly expressed during mucosal healing following intestinal inflammation
Literature	GENERIF_SUMMARY	the results of the present study demonstrated that injection of the LV carrying survivin into punctured rabbit intervertebral discs acted to delay changes associated with the degeneration of the discs.
Literature	GENERIF_SUMMARY	Studies showed that survivin mRNA and protein levels are overexpressed in cervical cancer cells. [meta-analysis]
Literature	GENERIF_SUMMARY	The results indicate that CXCL12/CXCR4 protects colorectal cancer cells against radiotherapy via survivin,

Main Category	Category	Terms
		implying an important underlying mechanism of resistance to radiotherapy during colorectal cancer therapy.
Literature	GENERIF_SUMMARY	High survivin expression is Associated with Acne Vulgaris.
Literature	GENERIF_SUMMARY	FAT10 promotes tumor proliferation by directly stabilizing Survivin protein in breast cancer cells.
Literature	GENERIF_SUMMARY	Using recombinant proteins, this study investigated the influence of survivin on the inhibition of caspase-9 by XIAP in vitro. With a fluorescence-based assay for the apoptosome-stimulated activity of caspase-9, the study shows that survivin has no effect on the inhibition of caspase-9 by XIAP, neither in the presence nor in the absence of Smac.
Literature	GENERIF_SUMMARY	morphine promotes renal cell carcinoma growth and increases RCC progression via over-expression of Survivin
Literature	GENERIF_SUMMARY	Studying survivin expression in leukocytes of 144 female rheumatoid arthritis patients this study observed that smoking patients had higher survivin transcription and a remarkable spreading of survivin isoforms.
Literature	GENERIF_SUMMARY	study revealed a critical role of miR-138- 5p as a tumor suppressive miRNA in bladder carcinogenesis, explored the molecular mechanisms by which aberrant miR-138-5p expression contributes to bladder cancer progression and identified Survivin as a direct target of miR- 138-5p. Regulation of Survivin by miR-138-5p might explain why the decrease of miR-138-5p during bladder carcinogenesis can promote cancer progression.
Literature	GENERIF_SUMMARY	results showed that survivin genetic variants were related to EGFR mutation in lung adenocarcinoma patients and might contribute to pathological development to NSCLC.
Literature	GENERIF_SUMMARY	Inhibition of survivin by YM155 overcomes statin resistance in renal cancer cells.
Literature	GENERIF_SUMMARY	In gastroenteropancreatic neuroendocrine neoplasms survivin and XIAP regulate cell proliferation and apoptosis functioning as possible prognostic biomarkers and therapeutic targets.
Literature	GENERIF_SUMMARY	the overexpression of miR-141 augments anoikis resistance in ovarian cancer cells by targeting and repressing the expression of KLF12, which, in turn, competes for binding sites in the survivin promoter with Sp1. The subsequent increase in survivin then protects ovarian cancer cells against anoikis by blocking the intrinsic apoptotic activity.
Literature	GENERIF_SUMMARY	Studies indicate that survivin expression was associated with more aggressive clinical features and predicted poor cancer-specific survival (CSS) and

Main Category	Category	Terms
		overall survival (OS) in patients with renal cell carcinoma (RCC).
Literature	GENERIF_SUMMARY	After cancer cell fusion, some fused cells avoid the apoptotic crisis partly owing to survivin, and continue to proliferate, a process that contributes to human cancer progression.
Literature	GENERIF_SUMMARY	This study shows that both survivin and TIIalpha are independent prognostic predictors in human grade II/III astrocytomas stratified for IDH1-mutation status
Literature	GENERIF_SUMMARY	Results showed that both PARP6 and survivin exhibited higher expression in colorectal adenocarcinoma tissues and cell lines. knockdown of either PARP6 or survivin promotes cell apoptosis and inhibits the cell invasion of colorectal adenocarcinoma cells suggesting a significant correlation between theses 2 proteins.
Literature	GENERIF_SUMMARY	Survivin expression in lung squamous cell carcinoma was increased compared to normal tissue, and was unaffected by methylation status of the surviving gene.
Literature	GENERIF_SUMMARY	Survivin gene promoter methylation was more frequent in gastric cancer patients than in controls.
Literature	GENERIF_SUMMARY	No significant differences of p-Akt and survivin expression were found between PTC patients with type 2 diabetes mellitus and papillary thyroid carcinoma patients without type 2 diabetes mellitus
Literature	GENERIF_SUMMARY	Study showed for the first time that the suppression of rheumatiod arthritis fibroblast-like synoviocyte was mediated by phosphatase and tensin homolog involving survivin silencing.
Literature	GENERIF_SUMMARY	In leukoplakia, the expression of survivin associated with that of ki-67 reinforces the assumption that all these lesions are potentially malignant.
Literature	GENERIF_SUMMARY	G/G genotype of the survivin -31C/G polymorphism might be a risk factor for UC development.
Literature	GENERIF_SUMMARY	nuclear survivin is a prognostic marker for the progression of oral squamous cell carcinomas.
Literature	GENERIF_SUMMARY	Nuclear expression of survivin is a factor for a poor prognosis in glioblastoma patients. Subcellular localization of survivin can help to predict overall survival in glioblastoma patients treated with the standard protocol.
Literature	GENERIF_SUMMARY	Nuclear survivin expression correlates with endoglin-assessed microvascularisation in laryngeal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	HIF-2alpha dictates the resistance of human pancreatic cancer cells to TRAIL under normoxic and hypoxic conditions and transcriptionally regulates survivin expression.
Literature	GENERIF_SUMMARY	Findings suggest that lysosome-associated

Main Category	Category	Terms
		transmembrane protein 4B (LAPTM4B), vascular endothelial growth factor (VEGF), and nuclear survivin expression are significantly correlated in breast cancer, which may be predictive of prognosis as well as effective therapeutic targets for anticancer therapies.
Literature	GENERIF_SUMMARY	In III-rd trimester of pregnancy parameters of Timp-1 and Survivin - anti-apoptotic substances concentration were similar in maternal and cord blood in both artery and vein. We found no increased activity of selected antiapoptotic factors.
Literature	GENERIF_SUMMARY	inhibition of apoptosis targeting survivin might represent an effective strategy for both obesity and cancer therapy.
Literature	GENERIF_SUMMARY	we uncovered that folate deficiency could profoundly downregulate two prominent anti-apoptotic effectors including survivin and glucose-regulated protein-78. Silencing of either survivin or glucose-regulated protein-78 via small interfering RNA interfering technique established that both effectors could serve as reactive oxygen species sinker proteins
Literature	GENERIF_SUMMARY	Data suggest that the first 22 bp of exon 3 in BIRC5 contain cis-acting elements that enhance the exclusion of exon 3 to generate the survivin DEx3 mRNA isoform via exclusion/deletion of exon 3; Sam68 is a possible trans-acting factor that binds to this cis-acting element and regulates exon 3 splicing. (BIRC5/survivin = baculoviral IAP repeat containing 5; Sam68 = GAP-associated tyrosine phosphoprotein p62)
Literature	GENERIF_SUMMARY	By downregulation of Sp1 and survivin at the late phase of treatment.
Literature	GENERIF_SUMMARY	Shorter overall survival was associated with BIRC5 and DNMT3B overexpression highlighting its potential as novel prognostic marker for penile cancer.
Literature	GENERIF_SUMMARY	ablation of Glut1 attenuated apoptosis and increased drug resistance via upregulation of p-Akt/p-GSK-3beta (Ser9)/beta-catenin/survivin.
Literature	GENERIF_SUMMARY	survivin expression modes and YAP localization (Hippo pathway activation modes) were not correlated with survival. In addition, we confirmed that survivin small interference RNA (siRNA) transfection and YM155, an anti-survivin drug, elicited decreased nuclear survivin expression and cell proliferation in ISO-HAS-B cells which expressed survivin consistently
Literature	GENERIF_SUMMARY	treatment of DLD1 cells with tamoxifen , betaestradiol, or a combination of these two drugs, inhibited cell viability and migration, promoted cell apoptosis, and reduced the mRNA and protein expression levels of survivin in a dose and timedependent manner. These results provide novel experimental basis for hormonal adjuvant therapy for the treatment of colorectal cancers
Literature	GENERIF_SUMMARY	Expressions of SDF-1, survivin and smac were significantly higher in epithelial ovarian cancer tissue

Main Category	Category	Terms
		than those in normal tissue.
Literature	GENERIF_SUMMARY	These data showed that miR-338-5p, one of the target miRNA biomarkers, was significantly downregulated in TE-4R. Ectopic overexpression of miR-338-5p induced apoptosis and sensitivity to radiation treatment by interfering with survivin, which is a known inhibitor of apoptosis.
Literature	GENERIF_SUMMARY	Results identified BIRC5 to be significantly upregulated in the lung squamous cell carcinoma tissues of smoking patients and may play an important role in diagnosis and prognosis.
Literature	GENERIF_SUMMARY	Results form study in non-small-cell lung cancer cells showed that SphK2 plays a critical role in doxorubicin-induced resistance by regulating key anti-apoptotic gene, survivin.
Literature	GENERIF_SUMMARY	Overexpression of survivin in gallbladder cancer suggests its possible role and association with poor prognosis. But XIAP has not been found to be associated with gallbladder carcinogenesis.
Literature	GENERIF_SUMMARY	The aim of this study was to compare the prognostic value of the histopathological grading and the conventional biomarker MIB-1 with expression of the anti-apoptotic protein survivin to see if this biomarker could complement recurrence prediction.
Literature	GENERIF_SUMMARY	circulating Survivin in prostate cancer patients was higher compared to controls, with the highest levels in African-American (AA) patients. AA patients with prostate cancer contained higher amounts of extracellular vesicles (EVs) in their plasma and sera than European-American (EA) patients. EVs from AA patients with prostate cancer contained higher amounts of Survivin, XIAP and cIAP-2 than EVs from EA patients.
Literature	GENERIF_SUMMARY	The expressions of survivin and STAT2 are up-regulated in skin lesions of PV patients, and their mRNA expressions are positively correlated.
Literature	GENERIF_SUMMARY	Mcl-1 and Survivin can be considered as promising molecular targets for the treatment of AML. The combination treatment with etoposide, and siRNA-mediated silencing of corresponding genes may be a novel strategy in chemoresistance AML treatment.
Literature	GENERIF_SUMMARY	Data sugest that targeting survivin using honokiol in combination with radiation therapy (RT) may provide novel therapeutic opportunities.
Literature	GENERIF_SUMMARY	Survivin expression in gastric cancer cells is regulated by DEC1.
Literature	GENERIF_SUMMARY	Single nucleotide polymorphism genotyping for -31G/C (rs9904341) polymorphism in the survivin promoter region is found in 54.1% of the breast cancer patients compared with 46.5% of controls from a north Indian population. The presence of homozygous CC genotype is associated with increased risk for development of

Main Category	Category	Terms
		breast cancer.
Literature	GENERIF_SUMMARY	Survivin may play an important role in the occurrence and development of laryngeal carcinoma, and its high expression is related to the poor prognosis of patients with laryngeal cancer. (Meta-analysis)
Literature	GENERIF_SUMMARY	Findings indicated that miR-494 synergistically suppressed GISTs by concomitantly targeting survivin and KIT.
Literature	GENERIF_SUMMARY	Overexpression of Survivin in glioma cells induces chromosomal instability
Literature	GENERIF_SUMMARY	Our findings identify survivin as a target of HO-1 and a mediator of adipocyte-induced survival in the metastatic niche.
Literature	GENERIF_SUMMARY	Survivin expression in tumor tissue and its serum concetration significantly correlate with clinical and histopathological parameters in melanoma patients.
Literature	GENERIF_SUMMARY	We identified 16 potentially druggable candidates. Among them, NEK2, BIRC5, and TOP2A were also found to be amplified in breast cancer, suggesting that they could act as strategic players in the obese-deregulated transcriptome.
Literature	GENERIF_SUMMARY	The upregulation of survivin was detected accompanied by the key component elevations of autophagy pathway including LC3, Beclin1, and p62. YM155, a specific inhibitor of survivin, could abrogate hyperglycemia-induced endothelial hyper-activation.
Literature	GENERIF_SUMMARY	miR-195 targets cyclin D3 to cause cell cycle arrest at the G1 phase and that miR-195 targets survivin to induce apoptosis and senescence in NSCLC cells. Overexpression of cyclin D3 or survivin reverses the effects of miR-195 in NSCLC cells.
Literature	GENERIF_SUMMARY	NF-kappaB activation may infer resistance to apoptosis through the expression of anti-apoptotic genes such as Survivin, which showed progressive increase in expression throughout the oesophageal metaplasia-dysplasia-adenocarcinoma sequence.
Literature	GENERIF_SUMMARY	Data suggest that Ki-67 index and survivin may be useful biomarkers for rectal cancer with preoperative chemoradiotherapy.
Literature	GENERIF_SUMMARY	Nuclear accumulation of survivin is associated with proliferative phenotype and was shown to be a worse prognostic marker in breast ductal carcinoma.
Literature	GENERIF_SUMMARY	Knockdown of BIRC5, a member of the inhibitor of apoptosis protein family, using either lentiviral vector based CRISPR/Cas9 nickase gene editing or inhibition of survivin using the small-molecule inhibitor YM155, results in the suppression of epithelial to mesenchymal transition in retinal pigment epithelial cells.
Literature	GENERIF_SUMMARY	Concomitant high expression of survivin and VEGF-C is

Main Category	Category	Terms
		closely associated with LNM status of PTC patients, which suggests their cooperation in the metastatic process.
Literature	GENERIF_SUMMARY	CSN5 directly bound survivin and decreased its ubiquitination to enhance the protein stability of survivin.
Literature	GENERIF_SUMMARY	Our work shows for the first time that YY1 represses survivin transcription by physically interacting with the survivin promoter. Furthermore, YY1 appears to contribute to basal survivin transcriptional activity, indicating that disruption of its binding may in part contribute to survivin overexpression after cellular stress events including chemotherapy and radiotherapy.
Literature	GENERIF_SUMMARY	LNC473 could recruit deubiquitinase USP9X to inhibit the ubiquitination level of survivin and then increase survivin expression.
Literature	GENERIF_SUMMARY	suggests SIRT1 may serve as a predictor of poor prognosis in esophageal squamous cell carcinoma, and its mediated tumor-promoting role might be associated with the overexpression of EGFR protein in esophageal squamous cell carcinoma
Literature	GENERIF_SUMMARY	The survivin gene 3' UTR polymorphisms (rs17878624) show that GG genotype provides substantial protection from non-small-cell lung carcinoma.
Literature	GENERIF_SUMMARY	These findings collectively suggest that the triple combination of survivin knockdown with ABT-263 and trametinib treatment, may be a potential strategy for the treatment of KRAS-mutant lung adenocarcinoma. Furthermore, our findings indicate that the welldifferentiated type of KRAS-mutant lung tumors depends, at least in part, on TTF1 for growth.
Literature	GENERIF_SUMMARY	Overexpression of miR-485-5p suppresses breast cancer progression and enhances chemosensitivity. Further study demonstrated that miR-485-5p directly targeted the 3'-untranslated region of survivin and overexpression of survivin overcomes the miR-485-5p induced effects on breast cancer.
Literature	GENERIF_SUMMARY	Plasma survivin levels are reduced in obstructive sleep apnea, are relate to disease severity, and are associated with high CRP levels.
Literature	GENERIF_SUMMARY	In the present study, liposomeplasmid DNA encoding mutant survivinT34A could inhibit tumor growth of cervical cancer. This inhibition may be associated with an increase in the apoptosis rate of tumor cells and a reduction in angiogenesis.
Literature	GENERIF_SUMMARY	High BIRC5 expression is associated with ovarian cancer.
Literature	GENERIF_SUMMARY	these findings suggest that BIRC5 supports long-term survival of HIV-1-infected cells
Literature	GENERIF_SUMMARY	Targeting the Cripto-1/TAK-1/NF-kappaB/Survivin

Main Category	Category	Terms
		pathway may be an effective approach to combat apoptosis resistance in cancer.
Literature	GENERIF_SUMMARY	enigma siRNA-transfected cells, cell viability, and the protein levels of AKT and survivin decreased.
Literature	GENERIF_SUMMARY	Our study results may suggest that high serum survivin levels can show 4 times increased risk of cancer in a subject with a high suspicion of cancer. Furthermore, survivin level was not influenced with demographic characteristics of breast, gastric, colorectal, prostate, ovarian cancer, and glioblastome multiforme.
Literature	GENERIF_SUMMARY	Oct4 plays a vital role in the malignant progression of HCC cells through the survivin/STAT3 signaling pathway.
Literature	GENERIF_SUMMARY	let-7b targets PLK1 to inhibit hepatocellular carcinoma cell growth and induce their apoptosis by attenuating the PLK1-mediated Survivin phosphorylation
Literature	GENERIF_SUMMARY	The expressions of HIF-1alpha, surviving, and VEGF played important roles in the occurrence, invasion, and metastasis of hepatocarcinoma.
Literature	GENERIF_SUMMARY	Study demonstrate that ZIC1 plays a tumor suppressive role in breast cancer, by targeting surviving, significantly downregulating its expression.
Literature	GENERIF_SUMMARY	simvastatin significantly inhibited the proliferation and invasion of SACC83 cells, induced apoptosis, and reduced the expression of survivin, which suggests that simvastatin may be a novel target for salivary gland adenoid cystic carcinoma therapy.
Literature	GENERIF_SUMMARY	results demonstrated that PLAUR induces geftinib-resistance through EGFR/p-AKT/survivin signaling pathway in gefitinib-resistant human lung adenocarcinoma cells. PLAUR could be a novel therapeutic target for gefitinib-resistant NSCLC patients.
Literature	GENERIF_SUMMARY	Study in hepatocellular carcinoma cell line elicited a new mechanism in which IGF-1 induced epithelial-mesenchymal transition through regulation of survivin and a downstream pathway.
Literature	GENERIF_SUMMARY	The role of PTEN in the process is potentially significant providing a suppressor balance to the p53/ survivin complex.
Literature	GENERIF_SUMMARY	Survivin is significantly up-regulated in hepatocellular carcinoma (HCC)tissues and associated with tumor growth and lymph node metastasis. Clinical detection of survivin level combined with MRI examination might be beneficial for clinical diagnosis and treatment of HCC
Literature	GENERIF_SUMMARY	Survivin overexpression plays a key role in the chemoresistance of ovarian CSCs.
Literature	GENERIF_SUMMARY	ERCC1 expression may also inhibit esophageal

Main Category	Category	Terms
		squamous cell carcinoma cell apoptosis via regulating survivin expression, and ERCC1 and survivin overexpression are independent predictors of prognosis for ESCC patients who receive chemotherapy and/or radiotherapy
Literature	GENERIF_SUMMARY	Survivin in the signal downstream of VEGFR2 played an important role in esophageal cancer cell survival.
Literature	GENERIF_SUMMARY	High BIRC5 expression is associated with gefitinib resistance in lung cancer.
Literature	GENERIF_SUMMARY	Results show nuclear survivin protein expression in colorectal cancer (CRC) patients with the CC genotype higher than in patient with the GG and GC genotypes suggesting that C allele of - 31G/C polymorphism in survivin might be cooperative in CRC development.
Literature	GENERIF_SUMMARY	Results of this study revealed that Survivin is rarely mutated in oral squamous cells carcinoma samples and upregulated when compared to non-cancerous tissue. A negative correlation between the methylation of the island cg25986496 and BIRC5 mRNA expression was detected from the Cancer Genome Atlas data.
Literature	GENERIF_SUMMARY	Our study identified the STAT3 rs1053004 C/C as a high-risk genotype in MA with lower survivin and VEGF transcription levels in the peripheral blood.
Literature	GENERIF_SUMMARY	miR-203 expression also inhibited primary tumor growth in ovaries and metastatic tumors in multiple peritoneal organs including liver and spleen. miR-203 inhibits ovarian tumor metastasis by targeting BIRC5/ survivin and attenuating the TGFbeta pathway.
Literature	GENERIF_SUMMARY	Survivin may be implicated in the bcl-2 and p53 pathways and therefore in the biology of PDAC. Its potential use as a survival predictor and therapeutic target represent a promising field.
Literature	GENERIF_SUMMARY	High serum survivin levels with GG genotype are associated with Brain Tumors.
Literature	GENERIF_SUMMARY	A positive feedback loop was identified between mitochondrial fission and Notch signaling pathway in triple negative breast cancer (TNBC) cells, where elevated Notch signaling enhanced Drp1-mediated mitochondrial fission, while Drp1-mediated mitochondrial fission in turn promoted the activation of Notch signaling, which ultimately promoted the cell survival of TNBC via increasing survivin expression level.
Literature	GENERIF_SUMMARY	Survivin expression was not associated with breast cancer recurrence in this study.
Literature	GENERIF_SUMMARY	BIRC5 can be used as a cardiac risk score if it is used with other highly accurate means of stratification.
Literature	GENERIF_SUMMARY	Overall meta-analysis revealed significant associations between c.-31G>C transversion and risk of urinary tract cancers in dominant (OR

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The results suggest that RAD51 and survivin are potent markers that determine the therapeutic efficacy of proton beam therapy in patients with pancreatic cancer
Literature	GENERIF_SUMMARY	The expression of survivin-wt and its four splice variants was investigated by quantitative real-time polymerase chain reaction in archival RNA samples of 35 pediatric B-cell ALL patients, divided into high- and standard-risk groups. The ratio of survivin-DeltaEx3/ wild type (WT) expression was higher in the low-risk group than in the high-risk group.
Literature	GENERIF_SUMMARY	Enforced expression of miR-494 decreased the expression of survivin, and thus promoted the TRAIL-induced mitochondria collapse and apoptosis pathway in gastric cancer cells.
Literature	GENERIF_SUMMARY	Of 71 patients, 50 had primary chordomas with a mean survivin expression level of 1.09; the other 21 patients had recurrent chordomas with a mean survivin expression level of 2.57, which was 2.36 times higher than the level in the primary chordoma patients. Analysis of 18 paired samples from 9 patients showed that the expression level of survivin was 2.62 times higher in the recurrent ones.
Literature	GENERIF_SUMMARY	Data suggest targeting cyclin dependent kinase 1 (CDK1) and survivin with Sp1 inhibitors such as mithramycin may be an effective approach to treat Nasal natural killer/T-cell lymphoma (NNKTL).
Literature	GENERIF_SUMMARY	differential expression of survivin and its splice variants may perhaps act as a diagnostic marker in patients suffering from thyroid cancer (review).
Literature	GENERIF_SUMMARY	Peripheral CD4(+) T cells of patients with allergic asthma (AS) express high levels of survivin. The expression of survivin is positively correlated with the expression of IL-4 in CD4+ T cells. CD4+ T cells with overexpression of survivin are prone to differentiate into Th2 cells. The activation-induced cell death machinery is defective in CD4+ T cells of AS patients, which is associated with the expression of survivin.
Literature	GENERIF_SUMMARY	High surviving expression is associated with melanoma.
Literature	GENERIF_SUMMARY	survivin isoforms are regulated both by different GLI proteins and BIRC5 polymorphisms in ovarian cancer.
Literature	GENERIF_SUMMARY	The survivin rs9904341 polymorphism may be associated with the risk of cancer either overall or in the Asian population.[meta-analysis]
Literature	GENERIF_SUMMARY	The minor allele C of -31G/C survivin single nucleotide polymorphism is more frequent in Egyptian breast cancer patients compared to the control group. Patients with the GC + CC genotype have higher serum levels of survivin and are at increased risk of breast cancer compared to the GG genotype of the control group and the benign group. There is significant elevation of serum survivin in different breast cancer stages.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	BIRC5 may be responsible for the condition of stem cell pluripotency.
Literature	GENERIF_SUMMARY	the novel functionalized SWCNT loaded with DOX and survivin siRNA was successfully synthesized, and the nanocomplex exhibited effective antitumor effects both in vitro and in vivo, thereby providing an alternative strategy for the codelivery of antitumor drugs and genes.
Literature	GENERIF_SUMMARY	microRNAs are not involved in the regulation of survivin in peripheral blood mononuclear cells from rheumatoid arthritis patients.
Literature	GENERIF_SUMMARY	Survivin was overexpressed in the nucleus of tumor cells, and also in a greater proportion of tumor cells, in comparison with the non-tumoral infiltrating cells.
Literature	GENERIF_SUMMARY	Survivin -31G>C polymorphism is associated with sporadic colorectal cancer (CRC) risk in the Southern Chinese population. The -31G wild-type genotypes and GC, GG genotypes are the independent protective factors against sporadic CRC excluding those with a BMI >/=28.0 kg/m2, family cancer history, and premenopausal.
Literature	GENERIF_SUMMARY	description of the structure, expression and function of survivin, especially in the context of cancer [review]
Literature	GENERIF_SUMMARY	in acute myeloid leukemia, WT1 is associated with increased bone marrow blasts and higher Flt3-ITD mutations, while survivin is associated with higher white blood cell count; both show less remission and survival
Literature	GENERIF_SUMMARY	The proximal BIRC5 promoter possesses specific binding sites for key transcription factors such as nuclear factor kappaB and signal transducer and activator of transcription 3. Upregulation of survivin exacerbates the autoimmune diseases (AIDs) including multiple sclerosis and myasthenia gravis by reducing the activity threshold of survivin-specific cytotoxic T cells.
Literature	GENERIF_SUMMARY	Survivin Promoter Polymorphism (-31 C/G) is associated with Oral Cancer.
Literature	GENERIF_SUMMARY	Validation of survivin and HMGA2 as biomarkers for cisplatin resistance in bladder cancer.
Literature	GENERIF_SUMMARY	Overexpressed in tumors, BIRC5 is associated with unfavorable overall survival in triple negative lung adenocarcinomas.
Literature	GENERIF_SUMMARY	BIRC5 disruption has favorable effects on the Acute Myelocytic Leukemia clinical outcome.
Literature	GENERIF_SUMMARY	The mRNA and protein expression level of SOX2 and Survivin was significantly higher in salivary adenoid cystic carcinoma tissues than in paracancerous normal salivary gland tissues, indicating that both of the two are tissue-specific and may become salivary adenoid

Main Category	Category	Terms
		cystic carcinoma oncogenes.
Literature	GENERIF_SUMMARY	MicroRNA-203 is highly expressed in non-small cell lung cancer (NSCLC) , and is closely related to tumor stage, lymph node metastasis, distant metastasis and poor prognosis of NSCLC patients. It is concluded that microRNA-203 promotes the progression of NSCLC via regulating survivin expression.
Literature	GENERIF_SUMMARY	this study suggests that -31C>G SNP in the promoter of the BIRC5 gene may be associated with risk of developing ESCC. And -241 C>T and -625 G>C polymorphisms are significantly related to the survival of ESCC patients.
Literature	GENERIF_SUMMARY	The survivin molecule as a double-edged sword in cellular physiologic and pathologic conditions and its role as a potential biomarker and therapeutic target in cancer.
Literature	GENERIF_SUMMARY	survivin binds to non-muscle myosin II (NMII), regulating its filament assembly.
Literature	GENERIF_SUMMARY	The protooncogene survivin was identified as a functional target of miR3305p, and this was validated using a luciferase reporter assay.
Literature	GENERIF_SUMMARY	Survivin Impairs the Apoptotic Machinery in CD4+ T Cells of Patients with Ulcerative Colitis.
Literature	GENERIF_SUMMARY	BCR-ABL induces tyrosine phosphorylation of YAP presumably through Src family kinases, which results in expression of Survivin and Cyclin D leading to leukemogenesis in CML cells.
Literature	GENERIF_SUMMARY	BIRC5 (survivin) C allele CC genotype in the rs9904341 polymorphism increased risk of relapsing-remitting multiple sclerosis (RRMS). Survivin mRNA expression in PBMCs and serum survivin level were increased in RRMS patients in comparison to the controls.
Literature	GENERIF_SUMMARY	microRNAs are potentially regulating the survivin gene in PBMCs from systemic sclerosis patients.
Literature	GENERIF_SUMMARY	Effect miR-214-3p on proliferation and apoptosis of breast cancer cells by targeting survivin protein.
Literature	GENERIF_SUMMARY	d-a-Tocopheryl Polyethylene Glycol 1000 Succinate and a small-molecule Survivin suppressant synergistically induce apoptosis in SKBR3 breast cancer cells.
Literature	GENERIF_SUMMARY	Over-expression MiR-203 played the opposite role and attenuated the effects of AB209371 over-expression. Therefore, AB209371 may down-regulate miR-203 to up-regulate Survivin, thereby promoting ovarian carcinoma (OC) cell proliferation. Our study provided novel insights into the pathogenesis of OC.
Literature	GENERIF_SUMMARY	BIRC5/Survivin is a novel ATG12-ATG5 conjugate interactor and an autophagy-induced DNA damage suppressor in human cancer and mouse embryonic fibroblast cells.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Bone marrow stromal cells-induced resistance to drugs is associated with survivin upregulation which is a direct target of miR-101-3p.
Literature	GENERIF_SUMMARY	MicroRNA-214-3p modified tetrahedral framework nucleic acids target survivin to induce tumour cell apoptosis.
Literature	GENERIF_SUMMARY	dysregulation of miR-195-5p/-218-5p/BIRC5 axis predicts a poor prognosis in patients with gastric cancer
Literature	GENERIF_SUMMARY	miR-34a could modulate the susceptibility of gastric cancer to Platycodin D via targeting survivin
Literature	GENERIF_SUMMARY	Results demonstrate that Survivin is epigenetically modified in non-small cell lung cancer (NSCLC) tissues. A high expression of Survivin on the RNA level was negatively correlated with patients' survival in adenocarcinomas. Also, NSCLC smokers with low expression of Survivin show a better outcome.
Literature	GENERIF_SUMMARY	Survivin expression reflects aggressive histological and clinical behavior of HCC.
Literature	GENERIF_SUMMARY	we found that BIRC5 rs9904341, rs2071214 polymorphisms might cause the increased risk of urinary cancer, while rs17878467 reduces risk. An upgrade risk in rs9904341 of BIRC5 were associated with urinary cancer in allele contrast model (OR = 1.222, P = 0.012), homozygote contrast model (OR = 1.579, P = 0.0001) and recessive contrast model (OR = 1.433, P
Literature	GENERIF_SUMMARY	Functional Compartmentalization of HSP60-Survivin Interaction between Mitochondria and Cytosol in Cancer Cells.
Literature	GENERIF_SUMMARY	Loss of cytoplasmic survivin expression is an independent predictor of poor prognosis in radically operated prostate cancer patients.
Literature	GENERIF_SUMMARY	Study showed that the overexpression of survivin contributes significantly to aggressive tumor phenotypes in pancreatic ductal adenocarcinoma.
Literature	GENERIF_SUMMARY	CD44 mediates stem cell mobilization to damaged lung via its novel transcriptional targets, Cortactin and Survivin.
Literature	GENERIF_SUMMARY	Epithelial-mesenchymal transition of circulating tumor cells in prostate cancer is promoted by survivin.
Literature	GENERIF_SUMMARY	Targeting survivin sensitizes cervical cancer cells to radiation treatment.
Literature	GENERIF_SUMMARY	In this study, we designed and synthesized 7-19 residues of inner centromere protein (INCENP)-derived small peptides (INC peptides) as novel survivin-targeting agents.
Literature	GENERIF_SUMMARY	Besides the anti-apoptotic functions, recent discoveries have revealed that XIAP, survivin, and BRUCE also

Main Category	Category	Terms
		exhibit regulatory functions for autophagy in cells. As the role of autophagy in human diseases has already been discussed extensively in different reviews, this review will discuss the emerging autophagic role of XIAP, survivin, and BRUCE in cancer cells.
Literature	GENERIF_SUMMARY	Identification of prognostic significance of BIRC5 in breast cancer using integrative bioinformatics analysis.
Literature	GENERIF_SUMMARY	Subcellular Compartmentalization of Survivin is Associated with Biological Aggressiveness and Prognosis in Prostate Cancer.
Literature	GENERIF_SUMMARY	Unraveling survivin expression in chronic myeloid leukemia
Literature	GENERIF_SUMMARY	Replicative conditioning of Herpes simplex type 1 virus by Survivin promoter, combined to ERBB2 retargeting, improves tumour cell-restricted oncolysis.
Literature	GENERIF_SUMMARY	The 5' untranslated region of the anti-apoptotic protein Survivin contains an inhibitory upstream AUG codon.
Literature	GENERIF_SUMMARY	Increased Expression of BIRC2, BIRC3, and BIRC5 from the IAP Family in Mesenchymal Stem Cells of the Umbilical Cord Wharton's Jelly (WJSC) in Younger Women Giving Birth Naturally.
Literature	GENERIF_SUMMARY	Molecular characterisation of TP53 mutated squamous cell carcinomas of the lung to identify putative targets for therapy.
Literature	GENERIF_SUMMARY	Weight loss normalizes enhanced expression of the oncogene survivin in visceral adipose tissue and blood leukocytes from individuals with obesity.
Literature	GENERIF_SUMMARY	Dysregulation of microRNAs regulating survivin in CD4+ T cells in multiple sclerosis.
Literature	GENERIF_SUMMARY	Downregulation of serum survivin correlates with increased inflammation, enhanced disease severity and worse prognosis in sepsis patients.
Literature	GENERIF_SUMMARY	Survivin in breast cancer-derived exosomes activates fibroblasts by up-regulating SOD1, whose feedback promotes cancer proliferation and metastasis.
Literature	GENERIF_SUMMARY	Comparative evaluation of survivin expression in leukoplakia, lichen planus, and oral squamous cell carcinoma
Literature	GENERIF_SUMMARY	Differentially localized survivin and STAT3 as markers of gastric cancer progression
Literature	GENERIF_SUMMARY	Survivin modulation in the antimelanoma activity of prodiginines.
Literature	GENERIF_SUMMARY	Downregulation of TRIM27 suppresses gastric cancer cell proliferation via inhibition of the Hippo-BIRC5 pathway.
Literature	GENERIF_SUMMARY	Long noncoding RNA NR2F1AS1 facilitates the

Main Category	Category	Terms
		osteosarcoma cell malignant phenotype via the miR4855p/miR2185p/BIRC5 axis.
Literature	GENERIF_SUMMARY	Expression of survivin and p53 genes in patients with alopecia areata
Literature	GENERIF_SUMMARY	Identification of CDCA8, DSN1 and BIRC5 in Regulating Cell Cycle and Apoptosis in Osteosarcoma Using Bioinformatics and Cell Biology.
Literature	GENERIF_SUMMARY	Down-regulation of Survivin and Bcl-2 concomitant with the activation of caspase-3 as a mechanism of apoptotic death in KG1a and K562 cells upon exposure to a derivative from ciprofloxacin family.
Literature	GENERIF_SUMMARY	alpha5-nAChR and survivin
Literature	GENERIF_SUMMARY	Survivin expression and impact on head and neck cancer outcomes.
Literature	GENERIF_SUMMARY	Role of Napsin A and Survivin Immunohistochemical Expression in Bronchogenic Adenocarcinoma.
Literature	GENERIF_SUMMARY	BIRC5 is a prognostic biomarker associated with tumor immune cell infiltration.
Literature	GENERIF_SUMMARY	Survivin Expression Is Differentially Regulated by a Selective Cross-talk between RBM38 and miRNAs let-7b or miR-203a.
Literature	GENERIF_SUMMARY	Concentration of survivin in children with oligo- and polyarticular juvenile idiopathic arthritis (JIA)
Literature	GENERIF_SUMMARY	Natural Killer Cell Phenotype and Functionality Affected by Exposure to Extracellular Survivin and Lymphoma-Derived Exosomes.
Literature	GENERIF_SUMMARY	BIRC5, GAJ5, and lncRNA NPHP3-AS1 Are Correlated with the Development of Atrial Fibrillation-Valvular Heart Disease.
Literature	GENERIF_SUMMARY	Specific inhibition of the Survivin-CRM1 interaction by peptide-modified molecular tweezers.
Literature	GENERIF_SUMMARY	Survivin drives tumor-associated macrophage reprogramming
Literature	GENERIF_SUMMARY	Identification and Validation of a Prognostic Model Based on Three Autophagy-Related Genes in Hepatocellular Carcinoma.
Literature	GENERIF_SUMMARY	Therapeutic targeting of FOS in mutant TERT cancers through removing TERT suppression of apoptosis via regulating survivin and TRAIL-R2.
Literature	GENERIF_SUMMARY	CircANKRD52 Promotes the Tumorigenesis of Hepatocellular Carcinoma by Sponging miR-497-5p and Upregulating BIRC5 Expression.
Literature	GENERIF_SUMMARY	The prognostic significance of survivin expression in patients with HNSCC

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Literature	GENERIF_SUMMARY	Expression Profiles Reveal Involvement of VEGF, IGF1, BIRC5, and MMP1 in Vulvar Carcinogenesis.
Literature	GENERIF_SUMMARY	Lung development, repair and cancer
Literature	GENERIF_SUMMARY	Serum survivin is decreased in women with preeclampsia.
Literature	GENERIF_SUMMARY	Prognostic significance of survivin expression in pediatric ewing sarcoma.
Literature	GENERIF_SUMMARY	LncRNA PART1 alleviated myocardial ischemia/reperfusion injury via suppressing miR-503-5p/BIRC5 mediated mitochondrial apoptosis.
Literature	GENERIF_SUMMARY	Survivin promotes rheumatoid arthritis fibroblast-like synoviocyte cell proliferation, and the expression of angiogenesis-related proteins by activating the NOTCH pathway.
Literature	GENERIF_SUMMARY	Localization matters
Literature	GENERIF_SUMMARY	Identification of prognostic markers for hepatocellular carcinoma based on the epithelial-mesenchymal transition-related gene BIRC5.
Literature	GENERIF_SUMMARY	Diagnostic Utility of BAP1, EZH2 and Survivin in Differentiating Pleural Epithelioid Mesothelioma and Reactive Mesothelial Hyperplasia
Literature	GENERIF_SUMMARY	The Early Diagnostic and Prognostic Value of BIRC5 in Clear-Cell Renal Cell Carcinoma Based on the Cancer Genome Atlas Data.
Literature	GENERIF_SUMMARY	TMEM100 negatively regulated by microRNA106b facilitates cellular apoptosis by suppressing survivin expression in NSCLC.
Literature	GENERIF_SUMMARY	DEPDC1B regulates the progression of human chordoma through UBE2T-mediated ubiquitination of BIRC5.
Literature	GENERIF_SUMMARY	Regulation of survivin protein stability by USP35 is evolutionarily conserved.
Literature	GENERIF_SUMMARY	Upregulation of BIRC5 plays essential role in esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Expression levels of MiRNA-16, SURVIVIN and TP53 in Preeclamptic and Normotensive women.
Literature	GENERIF_SUMMARY	LINC00511 Knockdown Suppresses Resistance to Cisplatin in Lung Adenocarcinoma by Interacting with miR-182-3p and BIRC5.
Literature	GENERIF_SUMMARY	The transcriptional activity profile of inhibitor apoptosis protein encoding genes in colon cancer patients
Literature	GENERIF_SUMMARY	CircCAMSAP1 promotes non-small cell lung cancer proliferation and inhibits cell apoptosis by sponging miR-1182 and regulating BIRC5.

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Literature	GENERIF_SUMMARY	The Role of Survivin and Transcription Factor FOXP1 in Scarring After Glaucoma Surgery.
Literature	GENERIF_SUMMARY	BIRC5 promotes cancer progression and predicts prognosis in laryngeal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Polymorphism of Baculoviral Inhibitor of Apoptosis Repeat-Containing 5 (BIRC5) Can Be Associated with Clinical Outcome of Non-Small Cell Lung Cancer.
Literature	GENERIF_SUMMARY	Pan-cancer analysis identifies BIRC5 as a prognostic biomarker.
Literature	GENERIF_SUMMARY	Nuclear survivin is a prognosticator in gastroenteropancreatic neuroendocrine neoplasms
Literature	GENERIF_SUMMARY	BIRC5 regulates inflammatory tumor microenvironment-induced aggravation of penile cancer development in vitro and in vivo.
Literature	GENERIF_SUMMARY	miR-203 represses keratinocyte stemness by targeting survivin.
Literature	GENERIF_SUMMARY	Association of BIRC5 Gene Polymorphism with the Collateral Circulation and Severity of Large Artery Atherosclerotic Stroke.
Literature	GENERIF_SUMMARY	Fine particulate matter induces METTL3-mediated m(6)A modification of BIRC5 mRNA in bladder cancer.
Literature	GENERIF_SUMMARY	Survivin; a novel therapeutic target that correlates with survival of autoreactive T lymphocytes obtained from patients with ankylosing spondylitis.
Literature	GENERIF_SUMMARY	miR-92a-3p promotes breast cancer proliferation by regulating the KLF2/BIRC5 axis.
Literature	GENERIF_SUMMARY	Inhibition of USP1 enhances anticancer drugs-induced cancer cell death through downregulation of survivin and miR-216a-5p-mediated upregulation of DR5.
Literature	GENERIF_SUMMARY	Survivin in Breast Cancer
Literature	GENERIF_SUMMARY	Longitudinal associations of serum survivin with the severity and prognosis of community-acquired pneumonia patients.
Literature	GENERIF_SUMMARY	RUNX1-Survivin Axis Is a Novel Therapeutic Target for Malignant Rhabdoid Tumors.
Literature	GENERIF_SUMMARY	Oncogenic RAS promotes MYC protein stability by upregulating the expression of the inhibitor of apoptosis protein family member Survivin.
Literature	GENERIF_SUMMARY	The analysis of Survivin promoter (-31G/C) gene variation in oral squamous cell carcinoma risk and prognosis.
Literature	GENERIF_SUMMARY	Long non-coding RNA FGD5 antisense RNA 1 targets Baculovirus inhibitor 5 via microRNA-497-5p to alleviate calcific aortic valve disease.

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Literature	GENERIF_SUMMARY	TP53RK Drives the Progression of Chronic Kidney Disease by Phosphorylating Birc5.
Literature	GENERIF_SUMMARY	An Insight into Survivin in Relevance to Hematological, Biochemical and Genetic Characteristics in Tobacco Chewers with Oral Squamous Cell Carcinoma.
Literature	GENERIF_SUMMARY	Effect of RNAi-Mediated Survivin and Hypoxia-Inducible Factor 1alpha Gene Silencing on Proliferation, Invasion, Migration and Apoptosis of Gastric Cancer BGC-823 Cells.
Literature	GENERIF_SUMMARY	Establishing a prognostic model based on immune-related genes and identification of BIRC5 as a potential biomarker for lung adenocarcinoma patients.
Literature	GENERIF_SUMMARY	Impact of Survivin rs9904341 and rs17878467 Polymorphisms On Risk of Preeclampsia in Iran.
Literature	GENERIF_SUMMARY	Post-transcriptional regulation of BIRC5/survivin expression and induction of apoptosis in breast cancer cells by tristetraprolin.
Literature	GENERIF_SUMMARY	BIRC5 Gene Polymorphisms Are Associated with a Higher Stage of Local and Regional Disease in Oral and Oropharyngeal Squamous Cell Carcinomas.
Literature	GENERIF_SUMMARY	TRAF4 regulates ubiquitination-modulated survivin turnover and confers radioresistance.
Literature	GENERIF_SUMMARY	Downregulation of PTPRT elevates the expression of survivin and promotes the proliferation, migration, and invasion of lung adenocarcinoma.
Literature	GENERIF_SUMMARY	Survivin/BIRC5 as a novel molecular effector at the crossroads of glucose metabolism and radioresistance in head and neck squamous cell carcinoma.
Literature	GENERIF_SUMMARY	CYR61 confers chemoresistance by upregulating survivin expression in triple-negative breast cancer.
Literature	GENERIF_SUMMARY	USP36 inhibits apoptosis by deubiquitinating cIAP1 and survivin in colorectal cancer cells.
Literature	GENERIF_SUMMARY	CCT6A promotes cell proliferation in colon cancer by targeting BIRC5 associated with p53 status.
Pathways	KEGG_PATHWAY	Platinum drug resistance
Pathways	KEGG_PATHWAY	Apoptosis
Pathways	KEGG_PATHWAY	Apoptosis - multiple species
Pathways	KEGG_PATHWAY	Hippo signaling pathway
Pathways	KEGG_PATHWAY	Hepatitis B
Pathways	KEGG_PATHWAY	Pathways in cancer
Pathways	KEGG_PATHWAY	Chemical carcinogenesis - receptor activation
Pathways	KEGG_PATHWAY	Colorectal cancer

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Pathways	REACTOME_PATHWAY	Cytokine Signaling in Immune system
Pathways	REACTOME_PATHWAY	Amplification of signal from the kinetochores
Pathways	REACTOME_PATHWAY	Amplification of signal from unattached kinetochores via a MAD2 inhibitory signal
Pathways	REACTOME_PATHWAY	Signal Transduction
Pathways	REACTOME_PATHWAY	Cell Cycle
Pathways	REACTOME_PATHWAY	Immune System
Pathways	REACTOME_PATHWAY	Signaling by Rho GTPases
Pathways	REACTOME_PATHWAY	RHO GTPase Effectors
Pathways	REACTOME_PATHWAY	Generic Transcription Pathway
Pathways	REACTOME_PATHWAY	Separation of Sister Chromatids
Pathways	REACTOME_PATHWAY	Resolution of Sister Chromatid Cohesion
Pathways	REACTOME_PATHWAY	Mitotic Metaphase and Anaphase
Pathways	REACTOME_PATHWAY	SUMOylation
Pathways	REACTOME_PATHWAY	SUMO E3 ligases SUMOylate target proteins
Pathways	REACTOME_PATHWAY	Transcriptional Regulation by TP53
Pathways	REACTOME_PATHWAY	Metabolism of proteins
Pathways	REACTOME_PATHWAY	Signaling by Interleukins
Pathways	REACTOME_PATHWAY	SUMOylation of DNA replication proteins
Pathways	REACTOME_PATHWAY	TP53 Regulates Transcription of Cell Death Genes
Pathways	REACTOME_PATHWAY	RHO GTPases Activate Formins
Pathways	REACTOME_PATHWAY	Post-translational protein modification
Pathways	REACTOME_PATHWAY	Interleukin-4 and Interleukin-13 signaling
Pathways	REACTOME_PATHWAY	TP53 regulates transcription of several additional cell death genes whose specific roles in p53-dependent apoptosis remain uncertain
Pathways	REACTOME_PATHWAY	Mitotic Prometaphase
Pathways	REACTOME_PATHWAY	Mitotic Anaphase
Pathways	REACTOME_PATHWAY	M Phase
Pathways	REACTOME_PATHWAY	Cell Cycle, Mitotic
Pathways	REACTOME_PATHWAY	Mitotic Spindle Checkpoint
Pathways	REACTOME_PATHWAY	Cell Cycle Checkpoints
Pathways	REACTOME_PATHWAY	RNA Polymerase II Transcription
Pathways	REACTOME_PATHWAY	Gene expression (Transcription)

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	Neddylation
Pathways	REACTOME_PATHWAY	EML4 and NUDC in mitotic spindle formation
Pathways	REACTOME_PATHWAY	Signaling by Rho GTPases, Miro GTPases and RHOBTB3
Pathways	WIKIPATHWAYS	Apoptosis modulation and signaling
Pathways	WIKIPATHWAYS	IL11 signaling
Pathways	WIKIPATHWAYS	Apoptosis
Pathways	WIKIPATHWAYS	Apoptosis related network due to altered Notch3 in ovarian cancer
Pathways	WIKIPATHWAYS	RAC1 PAK1 p38 MMP2 pathway
Pathways	WIKIPATHWAYS	Photodynamic therapy induced HIF 1 survival signaling
Pathways	WIKIPATHWAYS	Photodynamic therapy induced NF kB survival signaling
Pathways	WIKIPATHWAYS	Hepatitis C and hepatocellular carcinoma
Pathways	WIKIPATHWAYS	VEGFA VEGFR2 signaling
Pathways	WIKIPATHWAYS	Chromosomal and microsatellite instability in colorectal cancer
Pathways	WIKIPATHWAYS	Extracellular vesicles in the crosstalk of cardiac cells
Pathways	WIKIPATHWAYS	Gastrin signaling
Pathways	WIKIPATHWAYS	Hepatitis B infection
Pathways	WIKIPATHWAYS	Network map of SARS CoV 2 signaling
Pathways	WIKIPATHWAYS	HDAC6 interactions in the central nervous system
Pathways	WIKIPATHWAYS	Cancer pathways
Protein Domains	INTERPRO	BIR_rpt
Protein Domains	INTERPRO	Baculoviral_IAP
Protein Domains	SMART	BIR
Protein Domains	UP_KW_DOMAIN	Coiled coil